

Verizon Communications Annual Report 2023

Form 10-K (NYSE:VZ)

Published: February 10th, 2023

PDF generated by stocklight.com



UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549
FORM 10-K

(Mark one)



ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d)
OF THE SECURITIES EXCHANGE ACT OF 1934

For the fiscal year ended December 31, 2022

OR



TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d)
OF THE SECURITIES EXCHANGE ACT OF 1934

For the transition period from to

Commission file number: 1-8606

Verizon Communications Inc.

(Exact name of registrant as specified in its charter)

Delaware

(State or other jurisdiction
of incorporation or organization)

1095 Avenue of the Americas

New York, New York

(Address of principal executive offices)

23-2259884

(I.R.S. Employer Identification No.)

10036

(Zip Code)

Registrant's telephone number, including area code: (212) 395-1000

Securities registered pursuant to Section 12(b) of the Act:

Title of Each Class	Trading Symbol(s)	Name of Each Exchange on Which Registered
Common Stock, par value \$0.10	VZ	New York Stock Exchange
Common Stock, par value \$0.10	VZ	The Nasdaq Global Select Market
1.625% Notes due 2024	VZ 24B	New York Stock Exchange
4.073% Notes due 2024	VZ 24C	New York Stock Exchange
0.875% Notes due 2025	VZ 25	New York Stock Exchange
3.25% Notes due 2026	VZ 26	New York Stock Exchange
1.375% Notes due 2026	VZ 26B	New York Stock Exchange
0.875% Notes due 2027	VZ 27E	New York Stock Exchange
1.375% Notes due 2028	VZ 28	New York Stock Exchange
1.125% Notes due 2028	VZ 28A	New York Stock Exchange
2.350% Fixed Rate Notes due 2028	VZ 28C	New York Stock Exchange
1.875% Notes due 2029	VZ 29B	New York Stock Exchange
0.375% Notes due 2029	VZ 29D	New York Stock Exchange
1.250% Notes due 2030	VZ 30	New York Stock Exchange
1.875% Notes due 2030	VZ 30A	New York Stock Exchange
4.250% Notes due 2030	VZ 30D	New York Stock Exchange
2.625% Notes due 2031	VZ 31	New York Stock Exchange
2.500% Notes due 2031	VZ 31A	New York Stock Exchange
3.000% Fixed Rate Notes due 2031	VZ 31D	New York Stock Exchange
0.875% Notes due 2032	VZ 32	New York Stock Exchange
0.750% Notes due 2032	VZ 32A	New York Stock Exchange
1.300% Notes due 2033	VZ 33B	New York Stock Exchange
4.75% Notes due 2034	VZ 34	New York Stock Exchange
4.750% Notes due 2034	VZ 34C	New York Stock Exchange
3.125% Notes due 2035	VZ 35	New York Stock Exchange
1.125% Notes due 2035	VZ 35A	New York Stock Exchange

Securities registered pursuant to Section 12(b) of the Act (continued):

Title of Each Class	Trading Symbol(s)	Name of Each Exchange on Which Registered
3.375% Notes due 2036	VZ 36A	New York Stock Exchange
2.875% Notes due 2038	VZ 38B	New York Stock Exchange
1.875% Notes due 2038	VZ 38C	New York Stock Exchange
1.500% Notes due 2039	VZ 39C	New York Stock Exchange
3.50% Fixed Rate Notes due 2039	VZ 39D	New York Stock Exchange
1.850% Notes due 2040	VZ 40	New York Stock Exchange
3.850% Fixed Rate Notes due 2041	VZ 41C	New York Stock Exchange

Securities registered pursuant to Section 12(g) of the Act: None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. ☒ Yes ☐ No

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. ☐ Yes ☒ No

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. ☒ Yes ☐ No

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). ☒ Yes ☐ No

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
		Emerging growth company	☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Act). ☐ Yes ☒ No

At June 30, 2022, the aggregate market value of the registrant's voting stock held by non-affiliates was approximately \$213,158,526,305.

At January 31, 2023, 4,199,882,937 shares of the registrant's common stock were outstanding, after deducting 91,550,709 shares held in treasury.

Documents Incorporated By Reference:

Portions of the registrant's definitive Proxy Statement to be delivered to shareholders in connection with the registrant's 2023 Annual Meeting of Shareholders (Part III).

TABLE OF CONTENTS

Item No.		Page
PART I		
Item 1.	<u>Business</u>	<u>4</u>
Item 1A.	<u>Risk Factors</u>	<u>14</u>
Item 1B.	<u>Unresolved Staff Comments</u>	<u>18</u>
Item 2.	<u>Properties</u>	<u>19</u>
Item 3.	<u>Legal Proceedings</u>	<u>19</u>
Item 4.	<u>Mine Safety Disclosures</u>	<u>19</u>
PART II		
Item 5.	<u>Market for Registrant’s Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities</u>	<u>20</u>
Item 6.	<u>[Reserved]</u>	<u>20</u>
Item 7.	<u>Management’s Discussion and Analysis of Financial Condition and Results of Operations</u>	<u>21</u>
Item 7A.	<u>Quantitative and Qualitative Disclosures About Market Risk</u>	<u>48</u>
Item 8.	<u>Financial Statements and Supplementary Data</u>	<u>51</u>
Item 9.	<u>Changes in and Disagreements with Accountants on Accounting and Financial Disclosure</u>	<u>107</u>
Item 9A.	<u>Controls and Procedures</u>	<u>107</u>
Item 9B.	<u>Other Information</u>	<u>107</u>
Item 9C.	<u>Disclosure Regarding Foreign Jurisdictions that Prevent Inspections</u>	<u>107</u>
PART III		
Item 10.	<u>Directors, Executive Officers and Corporate Governance</u>	<u>108</u>
Item 11.	<u>Executive Compensation</u>	<u>108</u>
Item 12.	<u>Security Ownership of Certain Beneficial Owners and Management and Related Stockholder Matters</u>	<u>108</u>
Item 13.	<u>Certain Relationships and Related Transactions, and Director Independence</u>	<u>109</u>
Item 14.	<u>Principal Accounting Fees and Services</u>	<u>109</u>
PART IV		
Item 15.	<u>Exhibits and Financial Statement Schedules</u>	<u>110</u>
Item 16.	<u>Form 10-K Summary</u>	<u>114</u>
	<u>Signatures</u>	<u>114</u>
	Certifications	

PART I

Item 1. Business

General

Verizon Communications Inc. (the Company) is a holding company that, acting through its subsidiaries (together with the Company, collectively, Verizon), is one of the world's leading providers of communications, technology, information and entertainment products and services to consumers, businesses and government entities. With a presence around the world, we offer data, video and voice services and solutions on our networks and platforms that are designed to meet customers' demand for mobility, reliable network connectivity, security and control.

Our principal executive offices are located at 1095 Avenue of the Americas, New York, New York 10036 (telephone number 212-395-1000).

We have two reportable segments that we operate and manage as strategic business units - Verizon Consumer Group (Consumer) and Verizon Business Group (Business).

Verizon Consumer Group

Our Consumer segment provides consumer-focused wireless and wireline communications services and products. Our wireless services are provided across one of the most extensive wireless networks in the United States (U.S.) under the Verizon brand, TracFone Wireless, Inc. (TracFone) brands and through wholesale and other arrangements. We also provide fixed wireless access (FWA) broadband through our wireless networks. Our wireline services are provided in nine states in the Mid-Atlantic and Northeastern U.S., as well as Washington D.C., over our 100% fiber-optic network through our Verizon Fios product portfolio and over a traditional copper-based network to customers who are not served by Fios.

In 2022, the Consumer segment's revenues were \$103.5 billion, representing approximately 76% of Verizon's consolidated revenues. As of December 31, 2022, Consumer had approximately 115 million wireless retail connections, of which 80% are postpaid connections (including FWA). In addition, at December 31, 2022 Consumer had approximately 8 million total broadband connections (which includes Fios internet, Digital Subscriber Line (DSL) and FWA connections), and approximately 3 million Fios video connections.

Verizon Business Group

Our Business segment provides wireless and wireline communications services and products, including data, video and conferencing services, corporate networking solutions, security and managed network services, local and long distance voice services and network access to deliver various Internet of Things (IoT) services and products. We also provide FWA broadband through our wireless networks. We provide these products and services to businesses, government customers and wireless and wireline carriers across the U.S. and select products and services to customers around the world.

In 2022, the Business segment's revenues were \$31.1 billion, representing approximately 23% of Verizon's consolidated revenues. As of December 31, 2022, Business had approximately 29 million wireless retail postpaid connections and approximately 1 million total broadband connections (which includes Fios internet, DSL and FWA connections).

Additional discussion of our reportable segments is included in Item 7. under the headings "Management's Discussion and Analysis of Financial Condition and Results of Operations - Overview" and - "Segment Results of Operations" and in Note 13 to the consolidated financial statements of Verizon Communications Inc. and Subsidiaries.

Service and Product Offerings

Our Consumer segment's wireless and wireline products and services are available to our retail customers, as well as resellers that purchase wireless network access from us on a wholesale basis. Our Business segment's wireless and wireline products and services are organized by the primary customer groups targeted by these offerings: Small and Medium Business, Global Enterprise, Public Sector and Other, and Wholesale.

Wireless

We offer wireless services and equipment to both Consumer customers and Business customers.

Wireless Services

Our Consumer and Business segments provide a wide variety of wireless services accessible on a broad range of devices. Customers can obtain our wireless services on a postpaid or prepaid basis. Retail (non-wholesale) postpaid accounts primarily represent retail customers that are directly served and managed by Verizon and use Verizon branded services. A single account may include monthly wireless services for a variety of connected devices. Our postpaid service is generally billed one month in

advance for a monthly access charge in return for access to and usage of network services. Our prepaid service is offered only to Consumer customers and enables individuals to obtain wireless services without credit verification by paying for all services in advance. As of December 31, 2022, we had 92 million postpaid connections, including FWA, and 23 million prepaid connections, representing approximately 80% and 20% of our Consumer wireless retail connections, respectively.

Access to the internet is available on all smartphones and nearly all basic phones. In addition, our customers can access the internet at broadband speeds on notebook computers and tablets that are either wireless-enabled or that are used in conjunction with separate dedicated devices that provide a mobile Wi-Fi connection.

Historically, customers on our fixed-term service plans paid higher access fees for their wireless service in exchange for the ability to purchase their wireless devices at subsidized prices. We no longer offer Consumer customers new fixed-term, subsidized service plans for devices; however, we continue to offer subsidized plans to our Business customers.

We also provide FWA broadband through our 5G or 4G LTE wireless networks to our Consumer and Business customers. FWA enables fixed broadband access using radio frequencies instead of cables and can be used to connect homes and businesses to the internet. As of December 31, 2022, we had approximately 1.5 million FWA broadband connections.

Wireless Equipment

Consumer and Business offer several categories of wireless equipment to customers, including a variety of smartphones and other handsets, wireless-enabled internet devices, such as tablets, and other wireless-enabled connected devices, such as smart watches. We permit customers to acquire equipment from us using device payment plans, which permit the customer to pay for the device in installments over time.

Verizon Consumer Group

In addition to the wireless services and equipment discussed above, the Consumer segment sells residential fixed connectivity solutions, including internet, video and voice services, and wireless network access to resellers on a wholesale basis. Consumer also provides non-connectivity services including device protection, cloud storage, and other products.

Residential Fixed Services. We provide residential fixed connectivity solutions to customers over our 100% fiber-optic network through our Verizon Fios product portfolio, and over a traditional copper-based network to customers who are not served by Fios. In addition, our Long-Term Evolution (LTE) Home fixed wireless access internet service is available in parts of all 50 states across the U.S.

We offer residential fixed services tailored to the needs of our customers. Depending on those needs at a particular time, our services may include features related to, among other things: internet access at different speed tiers using fiber-optic, copper or wireless technology; video services that may feature a variety of channel options, video on demand products, cloud-based services and digital video recording capabilities; over-the-top video services; voice services; and other home solutions.

Network Access Services. We sell network access to mobile virtual network operators (MVNOs) on a wholesale basis, who in turn resell wireless service under their own brand(s) to consumers.

Verizon Business Group

In addition to the wireless services and equipment discussed above, our Business segment provides wireless and wireline communications services and products, including data, video and conferencing services, corporate networking solutions, security and managed network services, local and long distance voice services and network access to deliver various IoT services and products.

Small and Medium Business

Small and Medium Business offers wireless services and equipment, conferencing services, tailored voice and networking products, Fios services, Internet Protocol (IP) networking, advanced voice solutions and security and managed information technology (IT) services to our U.S.-based small and medium businesses that do not meet the requirements to be categorized as Global Enterprise, as described below. In 2022, Small and Medium Business revenues were \$12.6 billion, representing approximately 41% of Business's total revenues.

In addition to the wireless services and equipment discussed above, Small and Medium Business provides fixed connectivity solutions comparable to the residential fixed services provided by Consumer, as well as business services and connectivity similar to the products and services offered by Global Enterprise, in each case with features and pricing designed to address the needs of small and medium businesses.

Global Enterprise

Global Enterprise offers services to large businesses, which are identified based on their size and volume of business with Verizon, as well as non-U.S. public sector customers. In 2022, Global Enterprise revenues were \$9.7 billion, representing approximately 31% of Business's total revenues.

Global Enterprise offers a broad portfolio of connectivity, security and professional services designed to enable our customers to optimize their business operations, mitigate business risks and capitalize on data. These services include the following:

- *Network Services.* We offer a portfolio of network connectivity products to help our customers connect with their employees, partners, vendors and customers. These products include private networking services, private cloud connectivity services, virtual and software defined networking services and internet access services.
- *Advanced Communications Services.* We offer a suite of services to our customers to help them communicate with their employees, partners, vendors, constituents and customers. These products include IP-based voice and video services, unified communications and collaboration tools and customer contact center solutions.
- *Security services.* We offer a suite of management and data security services that help our customers protect, detect and respond to security threats to their networks, data, applications and infrastructure.
- *Core services.* We provide a portfolio of domestic and global voice and data solutions utilizing traditional telecommunications technology, including voice calling, messaging services, conferencing, contact center solutions and private line and data access networks. Core services also include the provision of customer premises equipment, and installation, maintenance and site services.
- *IoT services.* We provide the network access required to deliver various IoT products and services. We work with companies that purchase network access from us to connect their devices, bundled together with their own solutions, which they sell to end users. We are building IoT capabilities by leveraging business models that monetize usage on our networks at the connectivity, platform and solution layers.

Public Sector and Other

Public Sector and Other offers wireless products and services as well as wireline connectivity and managed solutions to U.S. federal, state and local governments and educational institutions. These services include business services and connectivity similar to the products and services offered by Global Enterprise, in each case, with features and pricing designed to address the needs of governments and educational institutions. In 2022, Public Sector and Other revenues were \$6.1 billion, representing approximately 20% of Business's total revenues.

Public Sector and Other also includes solutions that support fleet tracking management, compliance management, field service management, asset tracking and other types of mobile resource management in the U.S. and around the world.

Wholesale

Wholesale offers wireline communications services including data, voice, local dial tone and broadband services primarily to local, long distance, and wireless carriers that use our facilities to provide services to their customers. In 2022, Wholesale revenues were \$2.6 billion, representing approximately 8% of Business's total revenues. A portion of Wholesale revenues are generated by a few large telecommunications companies, most of which compete directly with us. Wholesale's services include:

- *Data services.* We offer a portfolio of data services to enhance our Wholesale customers' networks and provide connections to their end users and subscribers.
- *Voice services.* We provide switched access services that allow carriers to complete their end-user calls that originate or terminate within our territory. In addition, we provide originating and terminating voice services throughout the U.S. and globally utilizing our time-division multiplexing and Voice over Internet Protocol (VoIP) networks.
- *Local services.* We offer an array of local dial tone and broadband services to competitive local exchange carriers, some of which are offered to comply with telecommunications regulations. In addition, we offer services such as colocation, resale and unbundled network elements in compliance with applicable regulations.

Distribution

We use a combination of direct, indirect and alternative distribution channels to market and distribute our products and services to Consumer and Business customers.

Our direct channel, including our company-operated stores, is a core component of our distribution strategy. Our sales and service centers and business direct sales teams also represent significant distribution channels for our services. In addition, we have a robust digital channel and omni-channel as a part of the customer experience in order to offer choice and convenience.

Our indirect channel includes agents that sell our wireless and wireline products and services at retail locations throughout the U.S., as well as through the internet. The majority of these sales are made under exclusive selling arrangements with us. We

also have relationships with high-profile national retailers that sell our wireless and wireline products and services, as well as convenience store chains that sell our wireless prepaid products and services.

In addition to our direct channel, our Business segment has additional distribution channels that include business solution fulfillment provided by resellers, non-stocked device fulfillment performed by distributors and integrated mobility services provided by system integrators and resellers.

Global Network and Technology

Our global network architecture is used by Consumer and Business. Our network technology platforms include both wireless and wireline technologies.

Network Evolution

We are transforming the architecture of our networks into our Intelligent Edge Network, providing improved efficiency and virtualization, increased automation and opportunities for edge computing services that will support our fiber-based and radio access network technologies. We expect that this new architecture will simplify operations by eliminating legacy network elements, speed the deployment of fifth-generation (5G) wireless technology and create new opportunities in the business market in a cost-efficient manner. As a part of our evolution, we retired our third-generation (3G) network in 2022. As a result, approximately 909 thousand wireless retail postpaid connections, including 392 thousand wireless postpaid phone connections, and 237 thousand retail prepaid connections, were removed from the customer base.

5G

Over the past several years, we have been leading the development of 5G wireless technology industry standards and the ecosystems for fixed and mobile 5G wireless services. 5G technology enables higher throughput and lower latency than the fourth-generation (4G) LTE technology and allows our networks to handle more traffic as the number of internet-connected devices grows.

We provide 5G wireless service utilizing millimeter wave spectrum and other spectrum bands, some of which use dynamic spectrum sharing (DSS) technology. In January 2022, we began deploying C-Band spectrum, which has been built out to cover approximately 189 million POPs in the U.S. as of December 31, 2022. We expect to continue deploying C-Band spectrum across the continental U.S. as more and more of the spectrum becomes available for our use. Our FWA broadband service also leverages the 5G network and our expanding C-Band footprint, in addition to our 4G FWA offering.

4G LTE

The wireless network technology platform that carries the majority of our wireless traffic is 4G LTE. As of December 31, 2022, our 4G LTE network covers approximately 328 million POPs, including those in areas served by our LTE in Rural America partners. Under this program, we have collaborated with wireless carriers in rural areas to build and operate a 4G LTE network using each carrier's network assets with our core 4G LTE equipment and 700 Megahertz (MHz) C Block and Advanced Wireless Services (AWS) spectrum. LTE Home Internet, our home broadband internet service leverages the Verizon 4G LTE network.

Wireless Network Reliability and Build-Out

We consider the reliability, speed, capacity and coverage of our wireless network to be key factors for our continued success. We believe that steady and consistent network and platform investments provide the foundation for innovative products and services. As we design and deploy our network, we focus on the number of successful data sessions the network enables, delivering on our advertised throughput speeds, and the number of calls that are connected on the first attempt and completed without being dropped. We utilize three strategies to maintain the quality of our network: increasing the density of our network elements, deploying new technologies as they are developed and putting additional wireless spectrum into service.

We have been densifying our network by utilizing small cell technology, in-building solutions and distributed antenna systems. Network densification enables us to add capacity to address increasing mobile video consumption and the growing demand for IoT products and services on our 4G LTE and 5G networks. We are also utilizing existing network capabilities to handle increased traffic without interrupting the quality of the customer experience. We continue to deploy advanced technologies to increase both network capacity and data rates.

In order to build and upgrade our existing 4G LTE network and deploy our 5G network, we must secure rights to a large number of sites and obtain zoning and other governmental approvals and fiber facilities for our macro and small cells, in-building systems and antennas and related radio equipment that comprise distributed antenna systems. We have relationships with a wide variety of vendors that supply various products and services that support our wireless network operations. We utilize tower site management firms as lessors or managers of a portion of our existing leased and owned tower sites.

Our networks in the U.S. include various elements of redundancy designed to enhance the reliability of the services provided to our customers. To mitigate the impact of power disruptions on our operations, we have battery backup at every switch and every macro cell. We also utilize backup generators at a majority of our macro cells and at every switch location. In addition, we have a

fleet of portable backup generators that can be deployed if needed. We further enhance reliability by using a fully redundant Multiprotocol Label Switching backbone network in critical locations.

In addition to our own network coverage, we have roaming agreements with a number of wireless service providers to enable our customers to receive wireless service in nearly all other areas in the U.S. where wireless service is available. We also offer a variety of international wireless voice and data services to our customers through roaming arrangements with wireless service providers outside the U.S.

Fios

Residential broadband service has seen significant growth in bandwidth demand over the past several years, and we believe that demand will continue to grow. We expect that the continued emergence of new video services, new data applications and the proliferation of IP devices in the home will continue to drive new network requirements for increased data speeds and throughput. We believe that the Passive Optical Network (PON) technology underpinning Fios positions us well to meet these demands in a cost-effective and efficient manner.

While deployed initially as a consumer broadband network, our PON infrastructure is also experiencing more widespread application in the Business segment, especially as businesses increasingly migrate to Ethernet-based access services.

Global IP

Verizon owns and operates one of the largest global fiber-optic networks in the world, providing connectivity to Business customers in more than 180 countries. Our global IP network includes long-haul, metro and submarine assets that enable and support international operations.

Global business is rapidly evolving to an "everything-as-a-service" model in which Business customers seek cloud-based, converged enterprise solutions delivered securely via managed and professional services. We are continuing to deploy packet optical transport technology in order to create a global network platform to meet this demand.

Spectrum

The spectrum licenses we hold can be used for mobile and fixed wireless voice, video and data communications services. We are licensed by the Federal Communications Commission (FCC) to provide these wireless services on portions of the 800 MHz band, also known as cellular spectrum, the 1800-1900 MHz band, also known as Personal Communication Services (PCS) spectrum, portions of the 700 MHz upper C block, AWS 1 and 3 spectrum in the 1700 and 2100 MHz bands in areas that collectively cover nearly all of the population of the U.S., and 3700-3900 MHz, also known as C-Band. We have also deployed 4G technologies in 3.5 Gigahertz (GHz) shared spectrum, using LTE/Citizens Broadband Radio Service, and in 5 GHz unlicensed spectrum, using LTE/Licensed Assisted Access. All of this spectrum is collectively called low and mid-band spectrum. We are using our low and mid-band spectrum to provide 4G LTE and 5G wireless services. As of December 31, 2022, we retired our 3G service. We are increasingly reallocating spectrum previously used for 3G service to provide 4G LTE service, and we completed that transition where needed at the end of 2022. We are also utilizing additional low and mid-band spectrum for 5G through DSS to complement both our C-Band and our spectrum licenses in the 28 and 39 GHz band, collectively called millimeter wave spectrum.

Millimeter wave spectrum is being used in conjunction with low and mid-band spectrum for our 5G technology deployment. We own millimeter wave spectrum predominantly in the 28 GHz and 37/39 GHz bands. Millimeter wave spectrum is currently being used to increase capacity for mobile and fixed wireless services in areas of high demand. We anticipate that demand will continue to increase over time, driven by growth in customer connections and the increased usage of wireless broadband services that use more bandwidth and require faster rates of speed, as well as the wider deployment of 5G mobile and fixed services. We expect to meet the demand for 4G and 5G spectrum needs with our existing spectrum assets. If demand continues to increase or if new spectrum is required for a future generation of technology, we can meet that demand by acquiring licenses or leasing spectrum from other licensees, or by acquiring new spectrum licenses from the FCC, if and when future FCC spectrum auctions occur.

We use low and mid-band spectrum and DSS technology, to allow 5G service to run simultaneously with 4G LTE on multiple spectrum bands. With DSS, whenever customers move outside Verizon's millimeter wave and C-Band coverage areas, their 5G-enabled devices will remain on 5G technology using the lower spectrum bands where this network is available. This allows us to more fully and effectively utilize our current spectrum resources to serve both 4G and 5G customers.

From time to time we have exchanged spectrum licenses with other wireless service providers through secondary market swap transactions. We expect to continue to pursue similar opportunities to trade spectrum licenses in order to meet capacity and expansion needs in the future. In certain cases, we have entered into intra-market spectrum swaps designed to increase the amount of contiguous spectrum within frequency bands in a specific market. Contiguous spectrum improves network performance and efficiency. These swaps, as well as any spectrum purchases, require us to obtain governmental approvals.

Information regarding spectrum license transactions is included in Note 3 to the consolidated financial statements of Verizon Communications Inc. and Subsidiaries.

Competition and Related Trends

The telecommunications industry is highly competitive. The rapid development of new technologies, services and products has eliminated many of the traditional distinctions among wireless, cable, internet and local and long distance communication services and brought new competitors to our markets. We expect competition to remain intense as traditional and non-traditional participants seek increased market share.

With respect to our wireless connectivity products and services, we compete against other national wireless service providers, including AT&T Inc. and T-Mobile US, Inc., as well as various regional wireless service providers. We also compete for retail activations with resellers that buy bulk wholesale service from wireless service providers, including Verizon, and resell it to their customers. Resellers include cable companies, such as Comcast Corporation and Charter Communications, Inc., and others. Competition remains intense as a result of various factors, including high rates of smartphone penetration in the wireless market, increased network investment by our competitors, the development and deployment of new technologies, including 5G, the introduction of new products and services, offerings that include additional bundled premium content, increased levels of promotions and service plan discounts, new market entrants, the availability of additional licensed and unlicensed spectrum and regulatory changes. In addition, increasing government incentives related to network deployment may enhance the ability of certain of our competitors to compete with us. Competition may also increase as smaller, stand-alone wireless service providers merge or transfer licenses to larger, better capitalized wireless service providers and as MVNOs resell wireless communication services. In addition, DISH Network Corporation has committed to deploy a facilities-based 5G broadband network in each of its license areas capable of serving at least 70% of the U.S. population by June 2023, which could result in additional competitive pressures in the U.S. wireless industry.

We also face competition from other communications and technology companies seeking to increase their brand recognition and capture customer revenue with respect to the provision of wireless products and services, in addition to non-traditional offerings in mobile data. For example, Microsoft Corporation, Alphabet Inc., Apple Inc., Meta Platforms, Inc. and others are offering alternative means for messaging and making wireless voice calls that, in certain cases, can be used in lieu of the wireless providers' voice service, as well as alternative means of accessing video content. In addition, we expect to see increasing competition in the provisioning of internet access by low Earth orbit satellite companies as well in the area of fixed wireless offerings that provide an alternative to traditional landline internet service providers.

With respect to our wireline connectivity services, we compete against cable companies, wireless service providers, domestic and foreign telecommunications providers, satellite television companies, internet service providers, over-the-top (OTT) providers and other companies that offer network services and managed enterprise solutions. We also face increasing competition from other providers of VoIP services as well as internet portal providers. Cable operators have increased the size and capacity of their networks in order to deliver digital products and services. Several major cable operators offer bundles with wireless services through strategic relationships. Traditional wireless carriers are also bundling broadband internet offerings with wireless services while increasing their broadband internet footprint. Customers have an increasing number of choices for obtaining video content from various online services. We expect the market will continue to shift from traditional linear video to OTT offerings. We also expect customer migration from traditional voice services to wireless services to continue as a growing number of customers place greater value on mobility and wireless companies position their services as a landline alternative.

We believe that the following are the most important competitive factors and trends in the telecommunications industry:

- *Network reliability, speed, capacity and coverage.* We consider networks that consistently provide high-quality, fast and reliable service to be a key differentiator in the market and driver of customer satisfaction. With the development of 5G technologies, wireless service providers are increasingly focusing their marketing efforts on network quality, speed and coverage, sometimes directly targeting customers of Verizon. In addition, lower prices, improved service quality and new service offerings, which in many cases include video content, have led to increased customer usage of connectivity services. We and other network-based providers must ensure that our networks can meet these increasing capacity usage requirements and offer highly reliable national coverage.
- *Pricing.* With respect to wireless services and equipment, pricing plays an increasingly important role in the wireless competitive landscape. As the demand for wireless services continues to grow, wireless service providers are offering a range of service plans at competitive prices. Many wireless service providers also bundle wireless service offerings with other content and offer promotional pricing and incentives, some of which may be targeted specifically to customers of Verizon. We and other wireless service providers, as well as equipment manufacturers, offer device payment options, which provide customers with the ability to pay for their device over a period of time, and some providers offer device leasing arrangements. In addition, aggressive device promotions have become more common in an effort to gain a greater share of subscribers interested in changing carriers as well as retaining existing customers. Pricing also plays an important role in the wireline competitive landscape, as traditional service providers compete aggressively in offerings such as IP Networking, Core Voice and other legacy products. In addition, as non-traditional modes of providing wireline communication services emerge, new entrants attempt to capture market share from incumbents

using competitive pricing. For example, VoIP and portal-based voice and video calling is often free or nearly free for customers and supported by advertising revenues.

- *Customer service.* We believe that high-quality customer service is a key factor in retaining customers and attracting new customers, including those of other providers. Our customer service, retention and satisfaction programs are based on providing customers with convenient and easy-to-use products and services and focusing on their needs in order to promote long-term relationships and minimize churn.

Customer service is highly valued by our Business customers. We provide Global Enterprise and Public Sector and Other customers with ready access to their system and performance information, and we conduct proactive testing of our networks to identify issues before they affect our customers. We service our Small and Medium Business customers through service representatives and online support, as well as through store-based representatives for small business customers. For Wholesale customers, we pursue service improvement through continued system automation initiatives.

- *Product differentiation.* Customer and revenue growth are increasingly dependent on the development of new and enhanced products and services, as the delivery of new and innovative products and services has been accelerating. Customers are shifting their focus from access to applications and are seeking ways to leverage their broadband, video and wireless connections. To compete effectively, providers need to continuously review, improve and refine their product portfolio and develop and rapidly deploy new products and services tailored to the needs of customers. We continue to pursue the development and rapid deployment of new and innovative products and services, both independently and in collaboration with application providers, content providers and device manufacturers. Features such as wireless and wireline inter-operability are becoming increasingly important, driven by both customer demand and technological advancement.
- *Sales and distribution.* A key to achieving sales success in the consumer and small and medium business sectors of the wireless industry is the reach and quality of sales channels and distribution points. We seek to optimally vary distribution channels among our company-operated stores selling wireless products and services, web-based sales and fulfillment capabilities, outside sales teams and telemarketing, our extensive indirect distribution network of retail outlets and our sale of wireless service to resellers, which resell wireless services to their end users.

In addition to these competitive factors and trends, our competitive position also depends on our ability to market our products and services effectively, anticipate and respond to various factors and trends affecting our industries, as well as the availability of capital resources.

Companies with a global presence are increasingly competing with us in our Business segment. A relatively small number of telecommunications and integrated service providers with global operations serve customers in the global enterprise market and, to a lesser extent, the global wholesale market. We compete with these providers for large contracts to provide integrated solutions to global enterprises and government customers. Many of these companies have strong market presence, brand recognition and existing customer relationships, all of which contribute to intensifying competition that may affect our future revenue growth.

In the Small and Medium Business market, customer purchasing behaviors and preferences continue to evolve. Solution speed and simplicity with user interfaces that have a consumer-like "look and feel" are becoming key differentiators for customers who are seeking full life-cycle offers that simplify the process of starting, running and growing their businesses. Several major cable operators also offer bundles with wireless services through strategic relationships.

In the Global Enterprise and Public Sector and Other markets, competition levels remain high, primarily as a result of increased industry focus on technology convergence. We compete in this area with system integrators, carriers, and hardware and software providers. In addition, some of the largest information technology services companies are making strategic acquisitions, divesting non-strategic assets and forging new alliances to improve their cost structure. Many new alliances and acquisitions have focused on emerging fields, such as cloud computing, software defined networking, communication applications and other computing tasks via networks, rather than by the use of in-house machines.

Our Wholesale business competes with traditional carriers for long-haul, voice and IP services. In addition, mobile video and data needs are driving a greater need for wireless backhaul. Network providers, cable companies and niche players are competitors for this business opportunity.

Regulatory Trends

Regulatory Landscape

Verizon operates in a regulated and highly competitive market, as described above. Some of our competitors are subject to fewer regulatory constraints than Verizon. For many services offered by Verizon, the FCC is our primary regulator. The FCC has jurisdiction over interstate telecommunications services and other matters under the Communications Act of 1934, as amended (Communications Act or Act). Other Verizon services are subject to state and local regulation.

Federal, State and Local Regulation

Wireless Services

The FCC regulates several aspects of our wireless operations. Generally, the FCC has jurisdiction over the construction, operation, acquisition and transfer of wireless communications systems. All wireless services require use of radio frequency spectrum, the assignment and distribution of which is subject to FCC oversight. If demand continues to increase or if new spectrum is required for a future generation of technology, we can meet our needs for licensed spectrum by purchasing licenses or leasing spectrum from others, or by participating in a competitive bidding process to acquire new spectrum from the FCC. Those processes are subject to certain reviews, approvals and potential conditions.

Today, Verizon holds FCC spectrum licenses that allow it to provide a wide range of mobile and fixed communications services, including both voice and data services. FCC spectrum licenses typically have a term of 10 years, at which time they are subject to renewal. While the FCC has routinely renewed all of Verizon's wireless licenses, challenges could be raised in the future. If a wireless license was revoked or not renewed, Verizon would not be permitted to provide services on the spectrum covered by that license. Some of our licenses require us to comply with so-called "open access" FCC regulations, which generally require licensees of particular spectrum to allow customers to use devices and applications of their choice, subject to certain technical limitations. The FCC has also imposed certain specific mandates on wireless carriers, including construction and geographic coverage requirements, technical operating standards, provision of enhanced 911 services, roaming obligations and requirements for wireless tower and antenna facilities.

The Act generally preempts regulation by state and local governments of the entry of, or the rates charged by, wireless carriers. The Act does not prohibit states from regulating the other "terms and conditions" of wireless service. For example, some states impose reporting and consumer protection requirements. Several states also have laws or regulations that address safety issues (for example, use of wireless handsets while driving), universal service funding, and taxation matters. Some states are also considering new network reliability or service quality requirements that may affect how and where we provide services if not preempted by federal law. In addition, wireless tower and antenna facilities are often subject to state and local zoning and land use regulation, and securing approvals for new or modified facilities is often a lengthy and expensive process.

Broadband

Verizon offers many different broadband services. The FCC currently recognizes broadband internet access services as "information services" subject to a "light touch" regulatory approach rather than to the traditional, utilities-style regulations. However, the FCC could return to a more utilities-style regulation of broadband. Additionally, several states have adopted or are considering adopting laws or executive orders that would regulate broadband. Although some of these have been challenged in court, the ultimate enforceability and effect of these state rules is uncertain. Regardless of regulation, Verizon remains committed to the open internet, which provides consumers with competitive choices and unblocked access to lawful websites and content. Our broadband commitment can be found on our website at <https://www.verizon.com/about/our-company/verizon-broadband-commitment>.

Wireline Voice

Verizon offers many different wireline voice services, including traditional telephone service and other services that rely on technologies such as VoIP. For regulatory purposes, legacy telephone services are generally considered to be "common carrier" services. Common carrier services are subject to heightened regulatory oversight with respect to rates, terms and conditions and other aspects of the services. The FCC has not decided the regulatory classification of VoIP but has said VoIP service providers must comply with certain rules, such as 911 capabilities and law enforcement assistance requirements.

State public utility commissions regulate Verizon's telephone operations with respect to certain telecommunications intrastate matters. Verizon operates as an "incumbent local exchange carrier" in nine states and the District of Columbia. These incumbent operations are subject to various levels of pricing flexibility and other state oversight and requirements. Verizon also has other wireline operations that are more lightly regulated. Some states, including California, impose reporting requirements and are considering new network reliability or service quality requirements for wireline voice services, including for VoIP.

Video

Verizon offers a multichannel video service that is regulated like traditional cable service. The FCC has a body of rules that apply to cable operators, and these rules also generally apply to Verizon. In areas where Verizon offers its facilities-based multichannel video services, Verizon has been required to obtain a cable franchise from local government entities, or in some cases a state-wide franchise, and comply with certain one-time and ongoing obligations as a result.

Privacy and Data Security

We are subject to local, state, federal, and international laws and regulations relating to privacy and data security that impact all parts of our business, including wireline, wireless, broadband and the development and roll out of new products, such as those in the artificial intelligence and IoT space. At the federal level, our business is governed by the FCC or the Federal Trade Commission (FTC), depending on the product or service. The California Consumer Privacy Act, Europe's General Data Protection Regulation and a number of other privacy laws recently adopted by other states and countries include significant

penalties for non-compliance. Generally, attention to privacy and data security requirements is increasing at all levels of government globally, and privacy-related legislation has been introduced or is under consideration in many locations. These regulations could have a significant impact on our businesses.

Public Safety and Cybersecurity

The FCC plays a role in addressing public safety concerns by regulating emergency communications services and mandating widespread availability of both media (broadcast/cable) and wireless emergency alerting services. In response to cyber attacks that have occurred or could occur in the future, however, the FCC or other regulators may attempt to increase regulation of the cybersecurity practices of providers. The FCC is also addressing the use by American companies of equipment produced by certain companies deemed to cause potential national security risks. Verizon does not currently use equipment in its networks from vendors under such restrictions. In addition, due to recent natural disasters, federal and state agencies may attempt to impose regulations to ensure continuity of service during disasters. For example, the FCC adopted rules requiring wireless providers to support roaming on each other's networks during disasters, and the California Public Utilities Commission has imposed regulations relating to back-up power for communications facilities.

Intercarrier Compensation and Network Access

The FCC regulates some of the rates that carriers pay each other for the exchange of voice traffic (particularly traditional wireline traffic) over different networks and other aspects of interconnection for some voice services. The FCC also regulates some of the rates and terms and conditions for certain wireline "business data services" and other services and network facilities. Verizon is both a seller and a buyer of these services, and both makes and receives interconnection payments.

Human Capital Resources

At Verizon, we know that our people are one of our most valuable assets. In order to realize our core business strategy, we have developed human capital programs and practices that support, develop and care for our employees from the time they join our team through the entirety of their careers with Verizon. These programs are centered on the following principles:

- Attract the right talent for our future and maintain a diverse workforce with high-value skills and expertise.
- Develop our employees to their full potential through best-in-class educational programs and exceptional development experiences and create a culture of continuous learning and engagement.
- Inspire individuals to build a career at Verizon by providing meaningful work and upskilling opportunities and establishing an inclusive work environment for all.

Verizon is committed to being an employer of choice. With approximately 117,100 employees on a full-time equivalent basis as of December 31, 2022, 89% of whom are based in the U.S., we know that we need employees with diverse backgrounds, experiences and perspectives to best meet the needs of the diverse customers and communities we serve. Our human capital programs and practices are designed to create a workplace where employees are empowered to share their authentic selves and feel seen and heard as vital contributors to Verizon's corporate purpose. In addition, Verizon has extensive on-the-job training opportunities, tuition reimbursement programs and career development support to enable our employees to maximize their potential and thrive professionally. Our long-standing commitment to diversity and inclusion has resulted in a strong representation of women and people of color. As of December 31, 2022, Verizon's global workforce was approximately 66.5% male, 33.3% female and 0.2% unknown or undeclared, and the race/ethnicity of our U.S. workforce was 52.5% White, 20.0% Black, 12.9% Hispanic, 8.2% Asian, 0.5% American Indian/Alaskan Native, 0.3% Native Hawaiian/Pacific Islander, 2.7% two or more races, and 2.9% unknown or undeclared. Women represented 36.4% of U.S. senior leadership (vice president level and above). People of color represented 32.5% of U.S. senior leadership.

Verizon respects our employees' rights to freedom of association and collective bargaining in compliance with applicable law, including the right to join or not join labor unions. We have a long history of working with the Communications Workers of America and the International Brotherhood of Electrical Workers—the two unions that in total represent approximately 22.5% of our employees as of December 31, 2022. The current collective bargaining agreements covering our union-represented employees who serve customers in our Mid-Atlantic and Northeast service areas extend through August 1, 2026. In addition, where applicable outside of the U.S., we engage with employee representative bodies such as works council. Verizon meets with U.S. national and local union leaders, as well as works council leaders outside the U.S., to talk about key business topics, including safety, customer service, plans to improve operational processes, our business performance and the impacts that changing technology and competition are having on our customers, employees and business strategy.

For a discussion of the oversight provided by the Company's Board of Directors over Verizon's human capital management practices, see the section entitled "Governance — Our Governance Framework — Oversight of human capital management" in our definitive Proxy Statement to be filed with the Securities and Exchange Commission and delivered to shareholders in connection with our 2023 Annual Meeting of Shareholders.

Patents, Trademarks and Licenses

We own or have licenses to various patents, copyrights, trademarks, domain names and other intellectual property rights necessary to conduct our business. We actively pursue the filing and registration of patents, copyrights, trademarks and domain names to protect our intellectual property rights within the United States and abroad. We also actively grant licenses, in exchange for appropriate fees or other consideration and subject to appropriate safeguards and restrictions, to other companies that enable them to utilize certain of our intellectual property rights and proprietary technology as part of their products and services. Such licenses enable the licensees to take advantage of Verizon's brands and the results of Verizon's research and development efforts. While these licenses result in valuable consideration for Verizon, we do not believe that the loss of such consideration, or the expiration of any of our intellectual property rights, would have a material effect on our results of operations.

We periodically receive offers from third parties to purchase or obtain licenses for patents and other intellectual property rights in exchange for royalties or other payments. We also periodically receive notices alleging that our products or services infringe on third-party patents or other intellectual property rights. These claims, whether against us directly or against third-party suppliers of products or services that we sell to our customers, if successful, could require us to pay damages or royalties, rebrand, or cease offering the relevant products or services.

Information About Our Executive Officers

See Part III, Item 10. "Directors, Executive Officers and Corporate Governance" of this Annual Report on Form 10-K for information about our executive officers.

Information on Our Internet Website

We make available, free of charge on our website, our annual reports on Form 10-K, quarterly reports on Form 10-Q, current reports on Form 8-K, and all amendments to those reports, at <https://www.verizon.com/about/investors> as soon as reasonably practicable after such reports are electronically filed with or furnished to the Securities and Exchange Commission (SEC). These reports and other information are also available on the SEC's website at <https://www.sec.gov>. We periodically provide other information for investors on our website, including news and announcements regarding our financial performance, information on environmental, social and corporate governance matters, and details related to our annual meeting of shareholders. We encourage investors, the media, our customers, business partners and other stakeholders to review the information we post on this channel. Website references in this report are provided as a convenience and do not constitute, and should not be viewed as, incorporation by reference of the information contained on, or available through, the websites. Therefore, such information should not be considered part of this report.

Cautionary Statement Concerning Forward-Looking Statements

In this report we have made forward-looking statements. These statements are based on our estimates and assumptions and are subject to risks and uncertainties. Forward-looking statements include the information concerning our possible or assumed future results of operations. Forward-looking statements also include those preceded or followed by the words "anticipates," "believes," "estimates," "expects," "forecasts," "hopes," "plans" or similar expressions. For those statements, we claim the protection of the safe harbor for forward-looking statements contained in the Private Securities Litigation Reform Act of 1995. We undertake no obligation to revise or publicly release the results of any revision to these forward-looking statements, except as required by law. Given these risks and uncertainties, readers are cautioned not to place undue reliance on such forward-looking statements.

The following important factors, along with those discussed elsewhere in this report and in other filings with the SEC, could affect future results and could cause those results to differ materially from those expressed in the forward-looking statements:

- the effects of competition in the markets in which we operate, including any inability to successfully respond to competitive factors such as prices, promotional incentives and evolving consumer preferences;
 - failure to take advantage of, or respond to competitors' use of, developments in technology and address changes in consumer demand;
 - performance issues or delays in the deployment of our 5G network resulting in significant costs or a reduction in the anticipated benefits of the enhancement to our networks;
 - the inability to implement our business strategy;
 - adverse conditions in the U.S. and international economies, including inflation in the markets in which we operate;
 - cyber attacks impacting our networks or systems and any resulting financial or reputational impact;
-

- damage to our infrastructure or disruption of our operations from natural disasters, extreme weather conditions, acts of war, terrorist attacks or other hostile acts and any resulting financial or reputational impact;
- the impact of public health crises on our operations, our employees and the ways in which our customers use our networks and other products and services
- disruption of our key suppliers' or vendors' provisioning of products or services, including as a result of geopolitical factors, public health crises or the potential impacts of global climate change;
- material adverse changes in labor matters and any resulting financial or operational impact;
- changes in the regulatory environment in which we operate, including any increase in restrictions on our ability to operate our networks or businesses;
- our high level of indebtedness;
- significant litigation and any resulting material expenses incurred in defending against lawsuits or paying awards or settlements;
- an adverse change in the ratings afforded our debt securities by nationally accredited ratings organizations or adverse conditions in the credit markets affecting the cost, including interest rates, and/or availability of further financing;
- significant increases in benefit plan costs or lower investment returns on plan assets;
- changes in tax laws or treaties, or in their interpretation; and
- changes in accounting assumptions that regulatory agencies, including the SEC, may require or that result from changes in the accounting rules or their application, which could result in an impact on earnings.

Item 1A. Risk Factors

The following discussion of "Risk Factors" identifies factors that may adversely affect our business, operations, financial condition or future performance. This information should be read in conjunction with "Management's Discussion and Analysis of Financial Condition and Result of Operations" and the consolidated financial statements and related notes. The following discussion of risks is not all-inclusive but is designed to highlight what we believe are the material factors to consider when evaluating our business and expectations. These factors could cause our future results to differ materially from our historical results and from expectations reflected in forward-looking statements.

Economic and Strategic Risks

We face significant competition that may reduce our profits.

We face significant competition in our industries. The rapid development of new technologies, services and products has eliminated many of the traditional distinctions among wireless, cable, internet and local and long distance communication services and brought new competitors to our markets, including other telecommunications companies, cable companies, wireless service providers, satellite providers, technology companies and application and device providers. While these changes have enabled us to offer new types of products and services, they have also allowed other providers to broaden the scope of their own competitive offerings. If we are unable to compete effectively, we could experience lower than expected revenues and earnings.

Wireless service providers are significantly altering the financial relationships with their customers through commercial offers that vary service and device pricing, promotions, incentives and levels of service provided – in some cases specifically targeting our customers and putting pressure on pricing and margins. In addition, the wireless industry's customer growth rate could moderate in comparison to historical growth rates, leading to increased competition for customers. Our ability to compete effectively will depend on, among other things, our network quality, capacity and coverage, the pricing of our products and services, the quality of our customer service, our development of new and enhanced products and services, the reach and quality of our sales and distribution channels, our ability to market our products and services effectively and our capital resources. It will also depend on how successfully we anticipate and respond to various factors affecting our industries, including new technologies and business models, changes in consumer preferences and demand for existing services, demographic trends and evolving economic conditions, including inflation. If we are not able to respond successfully to these competitive challenges, our results of operations and financial condition could be adversely impacted. In addition, a projected sustained decline in a reporting unit's revenues and earnings could have a significant negative impact on its fair value and result in an impairment charge, which could have an adverse impact on our results of operations.

If we are not able to take advantage of developments in technology and address changing consumer demand on a timely basis, or if the deployment of our 5G network is delayed or hindered for any reason, we

may experience a decline in the demand for our services, be unable to implement our business strategy and experience reduced profits.

Our industries are rapidly changing as new technologies are developed that offer consumers an array of choices for their communications needs and allow new entrants into the markets we serve. In order to grow and remain competitive, we will need to adapt to future changes in technology, enhance our existing offerings and introduce new offerings to address our customers' changing demands. If we are unable to meet future challenges from competing technologies on a timely basis or at an acceptable cost, we could lose customers to our competitors. We may not be able to accurately predict technological trends or the success of new services in the market. If our new services fail to gain acceptance in the marketplace, or if costs associated with the implementation and introduction of these services materially increase, our ability to retain and attract customers could be adversely affected.

In addition, the deployment of our 5G network is subject to a variety of risks, including those related to equipment and spectrum availability, unexpected costs, and regulatory matters that could cause deployment delays or network performance issues. These issues could result in significant costs, put us at a competitive disadvantage, or reduce the anticipated benefits of the enhancements to our networks.

As we introduce new offerings and technologies, such as 5G technology, we must phase out outdated and unprofitable technologies and services. If we are unable to do so on a cost-effective basis, we could experience reduced profits. In addition, there could be legal or regulatory restraints on our ability to phase out current services.

Adverse conditions in the U.S. and international economies could impact our results of operations and financial condition.

Unfavorable economic conditions, such as a recession or economic slowdown in the U.S. or elsewhere, or inflation in the markets in which we operate, could negatively affect the affordability of and demand for some of our products and services and our cost of doing business. In difficult economic conditions, consumers may seek to reduce discretionary spending by forgoing purchases of our products, electing to use fewer higher margin services, dropping down in price plans or obtaining lower-cost products and services offered by other companies. Similarly, under these conditions, the business customers that we serve may delay purchasing decisions, delay full implementation of service offerings or reduce their use of services. In addition, adverse economic conditions may lead to an increased number of our consumer and business customers that are unable to pay for services. Unfavorable economic conditions could also amplify other risk factors discussed in this section, including, but not limited to, our competitive position and margins. In 2022, as a result of the inflationary environment in the U.S., we experienced increases in our direct costs, including electricity and other energy-related costs for our network operations, and transportation and labor costs, as well as increased interest expenses related to rising interest rates. We believe that this inflationary environment and the resulting decline in real wages in the U.S. are altering consumer preferences and causing consumers to become more price conscious. These factors, along with impacts of the intense competition in our industries, resulted in increased costs and lower earnings per share during 2022, and caused us to lower our growth expectations and related financial guidance. We expect the inflationary environment and these other pressures to continue into 2023. Although we are taking various measures to improve efficiency and reduce costs, those measures may ultimately be ineffective or insufficient. If the economic slowdown in the U.S., inflation and other unfavorable economic conditions continue or worsen, they could have a material adverse effect on our results of operations and financial condition.

Operational Risks

Cyber attacks impacting our networks or systems could have an adverse effect on our business.

Cyber attacks, including through the use of malware, computer viruses, distributed denial of services attacks, ransomware attacks, credential harvesting, social engineering and other means for obtaining unauthorized access to or disrupting the operation of our networks and systems and those of our suppliers, vendors and other service providers, could have an adverse effect on our business. Cyber attacks may cause equipment failures, loss of information, including sensitive personal information of customers or employees or valuable technical and marketing information, as well as disruptions to our or our customers' operations. Cyber attacks against companies, including Verizon, have increased in frequency, scope and potential harm in recent years. They may occur alone or in conjunction with physical attacks, especially where disruption of service is an objective of the attacker. The development and maintenance of systems to prevent such attacks is costly and requires ongoing monitoring and updating to address their increasing prevalence and sophistication. While, to date, we have not been subject to cyber attacks that, individually or in the aggregate, have been material to Verizon's operations or financial condition, the preventive actions we take to reduce the risks associated with cyber attacks, including protection of our systems and networks, may be insufficient to repel or mitigate the effects of a major cyber attack in the future.

The inability to operate or use our networks and systems or those of our suppliers, vendors and other service providers as a result of cyber attacks, even for a limited period of time, may result in significant expenses to Verizon and/or a loss of market share to our competitors. The costs associated with a major cyber attack on Verizon could include expensive incentives offered to existing customers and business partners to retain their business, increased expenditures on cybersecurity measures and the use of alternate resources, lost revenues from business interruption and litigation. Further, certain of Verizon's businesses, such as those offering security solutions and infrastructure and cloud services to business customers, could be negatively affected if

our ability to protect our own networks and systems is called into question as a result of a cyber attack. Our presence in the IoT industry, which includes offerings of telematics products and services, could also increase our exposure to potential costs and expenses and reputational harm in the event of cyber attacks impacting these products or services. In addition, a compromise of security or a theft or other compromise of valuable information, such as financial data and sensitive or private personal information, could result in lawsuits and government claims, investigations or proceedings. Any of these occurrences could damage our reputation, adversely impact customer and investor confidence and result in a material adverse effect on Verizon's results of operation or financial condition.

Natural disasters, extreme weather conditions, acts of war, terrorist or other hostile acts could cause damage to our infrastructure and result in significant disruptions to our operations.

Our business operations are subject to interruption by power outages, acts of war, terrorist or other hostile acts, natural disasters or the potential impacts of climate change, including the increasing prevalence and intensity of hurricanes, wildfires, flooding, hail and storms. Such events could cause significant damage to our infrastructure upon which our business operations rely, resulting in degradation or disruption of service to our customers, as well as significant recovery time and expenditures to resume operations. Our system redundancy may be ineffective or inadequate to sustain our operations through all such events. We are implementing, and will continue to implement, measures to protect our infrastructure and operations from the impacts of these events in the future, but these measures and our overall disaster recovery planning may not be sufficient for all eventualities. These events could also damage the infrastructure of the suppliers that provide us with the equipment and services that we need to operate our business and provide products to our customers. These occurrences could result in lost revenues from business interruption, damage to our reputation and reduced profits.

Public health crises could materially adversely affect our business, financial condition and results of operations.

We are subject to risks related to public health crises, such as the COVID-19 pandemic, which had an adverse effect on our operating results in 2020. Our business is based on our ability to provide products and services to customers throughout the United States and around the world and the ability of those customers to use and pay for those products and services for their businesses and in their daily lives. As a result, our business, financial condition and results of operations could be materially adversely affected by a public health crisis that significantly impacts the way customers use and are able to pay for our products and services, the way our employees are able to provide services to our customers, and the ways that our partners and suppliers are able to provide products and services to us. For example, public and private sector policies and initiatives to reduce the transmission of COVID-19 and initiatives Verizon took in response to the health crisis to promote the health and safety of our employees and provide critical infrastructure and connectivity to our customers, along with the related global slowdown in economic activity, resulted in decreased revenues, increased costs and lower earnings per share during 2020. In addition, such a crisis could significantly increase the probability or consequences of the risks our business faces in ordinary circumstances, such as risks associated with our supplier and vendor relationships, risks of an economic slowdown, regulatory risks, and the costs and availability of financing.

We depend on key suppliers and vendors to provide equipment that we need to operate our business.

We depend on various key suppliers and vendors to provide us, directly or through other suppliers, with equipment and services, such as fiber, switch and network equipment, smartphones and other wireless devices that we need in order to operate our business and provide products and services to our customers. For example, our smartphone and other device suppliers often rely on one vendor for the manufacture and supply of critical components, such as chipsets, used in their devices, and there are a limited number of companies capable of supplying the network infrastructure equipment on which we depend. These suppliers or vendors could fail to provide equipment or service on a timely basis, or fail to meet our performance expectations, for a number of reasons, including, for example, disruption to the global supply chain as a result of geopolitical factors, public health crises, natural disasters or the potential impacts of global climate change. If such failures occur, we may be unable to provide products and services as and when requested by our customers, or we may be unable to continue to maintain or upgrade our networks. Due to the cost and time lag that can be associated with transitioning from one supplier to another, our business could be substantially disrupted if we were required to, or chose to, replace the products or services of one or more major suppliers with products or services from another source, especially if the replacement became necessary on short notice. Any such disruption could increase our costs, decrease our operating efficiencies and have a material adverse effect on our business, results of operations and financial condition.

The suppliers and vendors on which we rely may also be subject to litigation with respect to technology on which we depend, including litigation involving claims of patent infringement. Such claims are frequently made in the communications industry. We are unable to predict whether our business will be affected by any such litigation. We expect our dependence on key suppliers to continue as we develop and introduce more advanced generations of technology.

A significant portion of our workforce is represented by labor unions, and we could incur additional costs or experience work stoppages as a result of the renegotiation of our labor contracts.

As of December 31, 2022, approximately 22.5% of our workforce is represented by the Communications Workers of America or the International Brotherhood of Electrical Workers. While we have labor contracts in place with these unions, with subsequent negotiations we could incur additional costs and/or experience work stoppages, which could adversely affect our business

operations. In addition, while a small percentage of the workforce outside of our traditional wireline operations is represented by unions for bargaining, we cannot predict what impact increased union density in this workforce could have on our operations.

Regulatory and Legal Risks

Changes in the regulatory framework under which we operate could adversely affect our business prospects or results of operations.

Our domestic operations are subject to regulation by the FCC and other federal, state, and local agencies, and our international operations are regulated by various foreign governments and international bodies. These regulatory regimes frequently restrict or impose conditions on our ability to operate in designated areas and provide specified products or services. We are frequently required to maintain licenses for our operations and conduct our operations in accordance with prescribed standards. We are often involved in regulatory and other governmental proceedings or inquiries related to the application of these requirements. It is impossible to predict with any certainty the outcome of pending federal and state regulatory proceedings relating to our operations, or the reviews by federal or state courts of regulatory rulings. Without relief, existing laws and regulations may inhibit our ability to expand our business and introduce new products and services. Similarly, we cannot guarantee that we will be successful in obtaining the licenses needed to carry out our business plan or in maintaining our existing licenses. For example, the FCC grants wireless licenses for terms generally lasting 10 years, subject to renewal. The loss of, or a material limitation on, certain of our licenses could have a material adverse effect on our business, results of operations and financial condition.

New laws or regulations or changes to the existing regulatory framework at the federal, state, and local, or international level, such as those described below, those that incentivize business models or technologies different from ours or requirements limiting our ability to continue or discontinue service to customers could restrict the ways in which we manage our wireline and wireless networks and operate our businesses, impose additional costs, impair revenue opportunities, and potentially impede our ability to provide services in a manner that would be attractive to us and our customers.

- *Privacy and data protection* – We are subject to federal, state and international laws related to privacy and data protection. In particular, the California Consumer Privacy Act, Europe's General Data Protection Regulation and a number of other privacy laws recently adopted by other states and countries include significant penalties for non-compliance. Generally, attention to privacy and data security requirements is increasing at all levels of government globally, and privacy-related legislation has been introduced or is under consideration in many locations. These regulations could have a significant impact on our businesses.
- *Regulation of broadband internet access services*– In its 2015 Title II Order, the FCC nullified its longstanding "light touch" approach to regulating broadband internet access services and "reclassified" these services as telecommunications services subject to utilities-style common carriage regulation. The FCC repealed the 2015 Title II Order in December 2017, and returned to its traditional light-touch approach for these services. The 2017 order has been affirmed in part by the D.C. Circuit but may be revisited by the FCC or by Congress. Several states have also adopted or are considering adopting laws or executive orders that would impose net neutrality and other requirements on some of our services (in some cases different from the FCC's 2015 rules). Although some of these have been challenged in court, the ultimate enforceability and effect of these state rules is uncertain.
- *"Open Access"* – We hold certain wireless licenses that require us to comply with so-called "open access" FCC regulations, which generally require licensees of particular spectrum to allow customers to use devices and applications of their choice. Moreover, certain services could be subject to conflicting regulation by the FCC and/or various state and local authorities, which could significantly increase the cost of implementing and introducing new services.
- *Climate-Related Regulation and Policy*– Due to the nature of our operations, we have been, and expect to continue to be impacted by regulatory developments related to climate change, including, for example, the direct regulation of greenhouse gas emissions or carbon policies that could result in a tax on such emissions. In addition, policy-driven changes in the prices of fuel or energy in geographies in which we operate could make it more expensive for us to purchase energy to power our networks and data centers, and any increase in taxes on fuel could increase our costs associated with operating those vehicles in our fleet that are dependent on traditional fuels.

These developments and the further regulation of broadband, wireless, and our other activities and any related court decisions could result in significant increases in costs for us or restrict our ability to compete in the marketplace and limit the return we can expect to achieve on past and future investments in our networks.

We are subject to a substantial amount of litigation, which could require us to pay significant damages or settlements.

We are subject to a substantial amount of litigation and claims in arbitration, including, but not limited to, shareholder derivative suits, patent infringement lawsuits, wage and hour class actions, contract and commercial claims, personal injury claims, property claims, environmental claims, and lawsuits relating to our advertising, sales, billing and collection practices. In addition, our wireless business also faces personal injury and wrongful death lawsuits relating to alleged health effects of wireless phones

or radio frequency transmitters. We may incur significant expenses in defending these lawsuits. In addition, we may be required to pay significant awards or settlements.

Financial Risks

Verizon has significant debt, which could increase further if Verizon incurs additional debt in the future and does not retire existing debt.

As of December 31, 2022, Verizon had approximately \$130.6 billion of outstanding unsecured indebtedness, \$9.4 billion of unused borrowing capacity under our existing revolving credit facility and \$20.0 billion of outstanding secured indebtedness. Verizon's debt level and related debt service obligations could have negative consequences, including:

- requiring Verizon to dedicate significant cash flow from operations to the payment of principal, interest and other amounts payable on our debt, which would reduce the funds we have available for other purposes, such as working capital, capital expenditures, dividend payments and acquisitions;
- making it more difficult or expensive for Verizon to obtain any necessary future financing for working capital, capital expenditures, debt service requirements, debt refinancing, acquisitions or other purposes and limiting its ability to repurchase common stock;
- reducing Verizon's flexibility in planning for or reacting to changes in our industries and market conditions;
- making Verizon more vulnerable in the event of a downturn in our business; and
- exposing Verizon to increased interest rate risk to the extent that our debt obligations are subject to variable interest rates.

Adverse changes in the financial markets and other factors could increase our borrowing costs and the availability of financing.

We require a significant amount of capital to operate and grow our business. We fund our capital needs in part through borrowings in the public and private credit markets. Adverse changes in the financial markets, including increases in interest rates or changes in exchange rates, could increase our cost of borrowing, require us to post a significant amount of collateral, and/or make it more difficult for us to obtain financing for our operations or refinance existing indebtedness. In addition, our ability to obtain funding under asset-backed debt transactions is subject to our ability to continue to originate a sufficient amount of assets eligible to be securitized. Our borrowing costs also can be affected by short- and long-term debt ratings assigned by independent rating agencies, which are based, in significant part, on our performance as measured by customary credit metrics. A decrease in these ratings would likely increase our cost of borrowing and/or make it more difficult for us to obtain financing. A severe disruption in the global financial markets could impact some of the financial institutions with which we do business, and such instability could also affect our access to financing.

Increases in costs for pension benefits and active and retiree healthcare benefits may reduce our profitability and increase our funding commitments.

With approximately 117,100 employees and approximately 186,700 retirees as of December 31, 2022 eligible to participate in Verizon's benefit plans, the costs of pension benefits and active and retiree healthcare benefits have a significant impact on our profitability. Our costs of maintaining these plans, and the future funding requirements for these plans, are affected by several factors, including increases in healthcare costs, decreases in investment returns on funds held by our pension and other benefit plan trusts and changes in the discount rate and mortality assumptions used to calculate pension and other postretirement expenses. If we are unable to limit future increases in the costs of our benefit plans, those costs could reduce our profitability and increase our funding commitments.

Item 1B. Unresolved Staff Comments

None.

Item 2. Properties

Our principal properties do not lend themselves to simple description by character and location. Our total gross investment in property, plant and equipment was approximately \$308 billion at December 31, 2022 and \$290 billion at December 31, 2021, including the effect of retirements, but before deducting accumulated depreciation. Our gross investment in property, plant and equipment consisted of the following:

At December 31,	2022	2021
Network equipment	77.2 %	76.9 %
Land, buildings and building equipment	11.7 %	11.7 %
Furniture and other	11.1 %	11.4 %
	100.0 %	100.0 %

Network equipment consists primarily of cable (aerial, buried, underground or undersea) and the related support structures of poles and conduit, wireless plant, switching equipment, network software, transmission equipment and related facilities. Land, buildings and building equipment consists of land and land improvements, central office buildings or any other buildings that house network equipment, and buildings that are used for administrative and other purposes. Substantially all the switching centers are located on land and in buildings we own due to their critical role in the networks and high set-up and relocation costs. We also maintain facilities throughout the U.S. comprised of administrative and sales offices, customer care centers, retail sales locations, garage work centers, switching centers, cell sites and data centers. Furniture and other consists of telephone equipment, furniture, data processing equipment, office equipment, motor vehicles, construction in process, and leasehold improvements.

Item 3. Legal Proceedings

In the ordinary course of business, Verizon is involved in various litigation and regulatory proceedings at the state and federal level. As of the date of this report, we do not believe that any pending legal proceedings to which we or our subsidiaries are subject are required to be disclosed as material legal proceedings pursuant to this item. Verizon is not subject to any administrative or judicial proceeding arising under any federal, state or local provisions that have been enacted or adopted regulating the discharge of materials into the environment or primarily for the purpose of protecting the environment that is likely to result in monetary sanctions of \$1 million or more. For a discussion of our litigation risks, refer to Item 1A Risk Factors.

Item 4. Mine Safety Disclosures

None.

PART II

Item 5. Market for Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities

The principal market for trading in the common stock of the Company is the New York Stock Exchange under the symbol "VZ".

As of December 31, 2022, there were 462,587 shareholders of record. In addition, a significant number of holders of the Company's common stock are "street name" or beneficial holders, whose shares are held of record by banks, brokers, and other financial institutions.

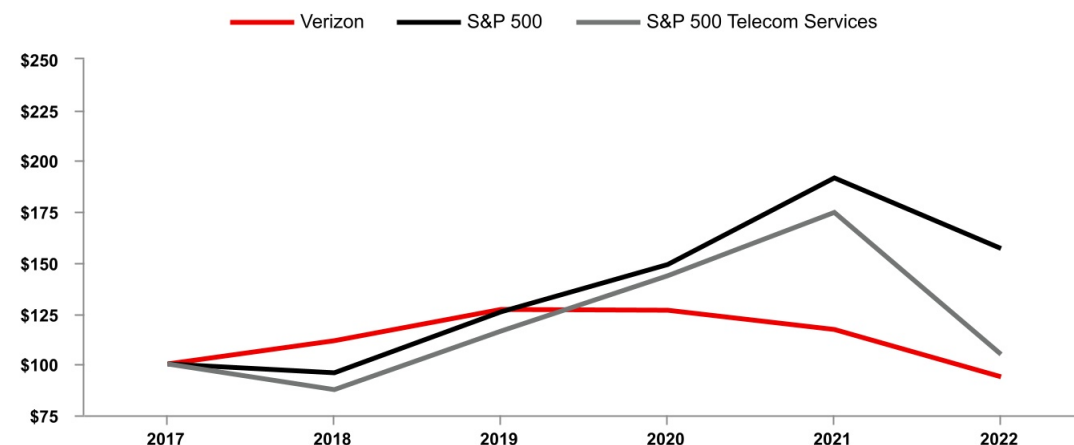
Stock Repurchases

In February 2020, the Board of Directors of the Company authorized a share buyback program to repurchase up to 100 million shares of our common stock. The program will terminate when the aggregate number of shares purchased reaches 100 million, or a new share repurchase plan superseding the current plan is authorized, whichever is sooner. Under the program, shares may be repurchased in privately negotiated transactions, on the open market, or otherwise, including through plans complying with Rule 10b5-1 under the Exchange Act. The timing and number of shares purchased under the program, if any, will depend on market conditions and our capital allocation priorities.

During the years ended December 31, 2022 and 2021, Verizon did not repurchase any shares of the Company's common stock under our authorized share buyback program. At December 31, 2022, the maximum number of shares that could be purchased by or on behalf of Verizon under our share buyback program was 100 million.

Stock Performance Graph

Comparison of Five-Year Total Return Among Verizon, S&P 500 and S&P 500 Telecommunications Services Index



	2017	2018	2019	2020	2021	2022
Verizon	\$ 100.0	\$ 111.3	\$ 126.7	\$ 126.5	\$ 117.0	\$ 93.6
S&P 500	100.0	95.6	125.7	148.8	191.5	156.8
S&P 500 Telecom Services	100.0	87.5	116.1	143.5	174.4	104.8

The graph compares the cumulative total returns of Verizon, the S&P 500 Stock Index and the S&P 500 Telecommunications Services Index over a five-year period. It assumes \$100 was invested on December 31, 2017 with dividends being reinvested.

Item 6. [Reserved]

Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations

Overview

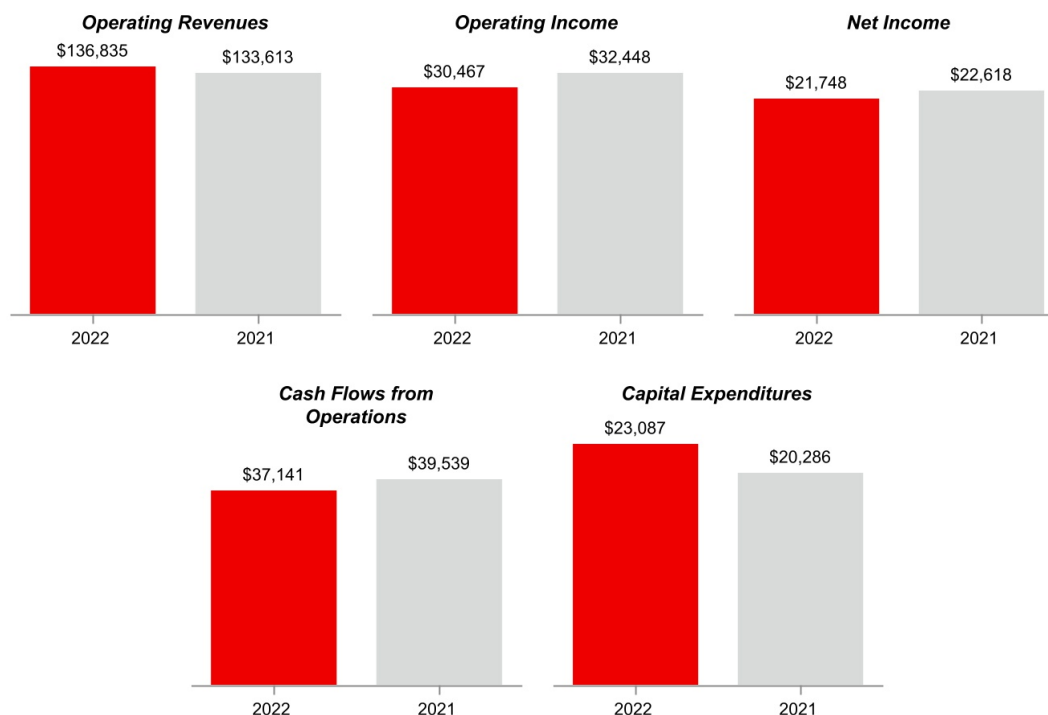
Verizon Communications Inc. (the Company) is a holding company that, acting through its subsidiaries (together with the Company, collectively, Verizon), is one of the world's leading providers of communications, technology, information and entertainment products and services to consumers, businesses and government entities. With a presence around the world, we offer data, video and voice services and solutions on our networks and platforms that are designed to meet customers' demand for mobility, reliable network connectivity, security and control.

To compete effectively in today's dynamic marketplace, we are focused on the capabilities of our high-performing networks to drive growth based on delivering what customers want and need in the digital world. In 2022, we focused on maintaining our network leadership, including by rapidly deploying C-Band spectrum; retaining and growing our high-quality customer base while balancing profitability in challenging market conditions; and driving monetization of our networks, platforms and solutions. We are creating business value by earning the trust of our stakeholders, limiting our environmental impact and supporting our customer base growth while creating social benefit through our products and services. Our strategy requires significant capital investments primarily to acquire wireless spectrum, put the spectrum into service, provide additional capacity for growth in our networks, invest in the fiber that supports our businesses, evolve and maintain our networks and develop and maintain significant advanced information technology systems and data system capabilities. 2022 was a peak year of capital investment for us as we rapidly deployed C-Band spectrum. We believe that this spectrum, together with our industry leading millimeter wave spectrum holding, 4G LTE network and fiber infrastructure, will drive innovative products and services and fuel our growth.

We are consistently deploying new network architecture and technologies to secure our leadership in both fourth-generation (4G) and fifth-generation (5G) wireless networks. We expect that our next-generation multi-use platform, which we call the Intelligent Edge Network, will simplify operations by eliminating legacy network elements, speed the deployment of 5G wireless technology and create new opportunities in the business market in a cost efficient manner. Our network leadership is the hallmark of our brand and the foundation for the connectivity, platforms and solutions upon which we build our competitive advantage.

Highlights of Our 2022 Financial Results

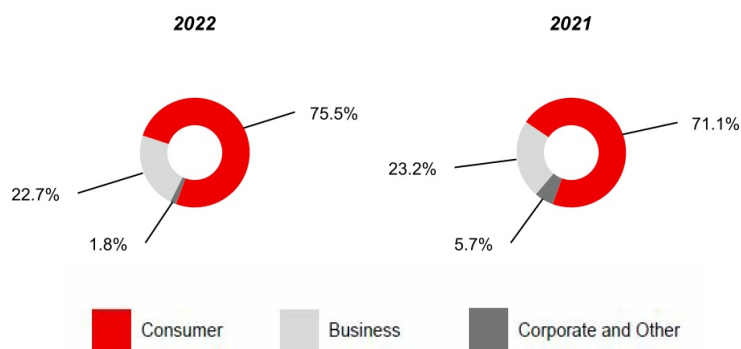
(dollars in millions)



Business Overview

We have two reportable segments that we operate and manage as strategic business units - Verizon Consumer Group (Consumer) and Verizon Business Group (Business).

Revenue by Segment



Note: Excludes eliminations.

Verizon Consumer Group

Our Consumer segment provides consumer-focused wireless and wireline communications services and products. Our wireless services are provided across one of the most extensive wireless networks in the United States (U.S.) under the Verizon brand, TracFone Wireless, Inc. (TracFone) brands and through wholesale and other arrangements. We also provide fixed wireless access (FWA) broadband through our wireless networks. Our wireline services are provided in nine states in the Mid-Atlantic and Northeastern U.S., as well as Washington D.C., over our 100% fiber-optic network through our Verizon Fios product portfolio and over a traditional copper-based network to customers who are not served by Fios. Our Consumer segment's wireless and wireline products and services are available to our retail customers, as well as resellers that purchase wireless network access from us on a wholesale basis.

Customers can obtain our wireless services on a postpaid or prepaid basis. Our postpaid service is generally billed one month in advance for a monthly access charge in return for access to and usage of network services. Our prepaid service is offered only to Consumer customers and enables individuals to obtain wireless services without credit verification by paying for all services in advance. The Consumer segment also offers several categories of wireless equipment to customers, including a variety of smartphones and other handsets, wireless-enabled internet devices, such as tablets, and other wireless-enabled connected devices, such as smart watches.

In addition to the wireless services and equipment discussed above, the Consumer segment sells residential fixed connectivity solutions, including internet, video and voice services, and wireless network access to resellers on a wholesale basis. The Consumer segment's operating revenues for the year ended December 31, 2022 totaled \$103.5 billion, an increase of \$8.2 billion, or 8.6%, compared to the year ended December 31, 2021. See "Segment Results of Operations" for additional information regarding our Consumer segment's operating performance and selected operating statistics.

Verizon Business Group

Our Business segment provides wireless and wireline communications services and products, including data, video and conferencing services, corporate networking solutions, security and managed network services, local and long distance voice services and network access to deliver various Internet of Things (IoT) services and products, including solutions that support fleet tracking management, compliance management, field service management, asset tracking and other types of mobile resource management. We also provide FWA broadband through our wireless networks. We provide these products and services to businesses, government customers and wireless and wireline carriers across the U.S. and select products and services to customers around the world. The Business segment's operating revenues for the year ended December 31, 2022 totaled \$31.1 billion and remained relatively flat compared to the year ended December 31, 2021. See "Segment Results of Operations" for additional information regarding our Business segment's operating performance and selected operating statistics.

Corporate and Other

Corporate and other primarily includes insurance captives, investments in unconsolidated businesses and development stage businesses that support our strategic initiatives, as well as unallocated corporate expenses, certain pension and other employee

benefit related costs and interest and financing expenses. Corporate and other also includes the historical results of divested businesses, including Verizon Media Group (Verizon Media) divested in September 2021, and other adjustments and gains and losses that are not allocated in assessing segment performance due to their nature. Although such transactions are excluded from the business segment results, they are included in reported consolidated earnings. Gains and losses from these transactions that are not individually significant are included in segment results as these items are included in the chief operating decision maker's assessment of segment performance. See "Consolidated Results of Operations" for additional information regarding Corporate and other results.

Capital Expenditures and Investments

We continue to invest in our wireless networks, high-speed fiber and other advanced technologies to position ourselves at the center of growth trends for the future. During the year ended December 31, 2022, these investments included \$23.1 billion for capital expenditures, inclusive of approximately \$6.2 billion in C-Band related capital expenditures. See "Cash Flows Used in Investing Activities" and "Liquidity and Capital Resources" for additional information. We believe that our investments aimed at expanding our portfolio of products and services will provide our customers with an efficient, reliable infrastructure for competing in the information economy.

Global Network and Technology

We are focusing our capital spending on adding capacity and density to our 4G Long-Term Evolution (LTE) network, while also building our next generation 5G network. We are densifying our networks by utilizing small cell technology, in-building solutions and distributed antenna systems. Network densification enables us to add capacity to address increasing mobile video consumption and the growing demand for IoT products and services on our 4G LTE and 5G networks. Over the past several years, we have been leading the development of 5G wireless technology industry standards and the ecosystems for fixed and mobile 5G wireless services. 5G technology enables higher throughput and lower latency than the 4G LTE technology and allows our networks to handle more traffic as the number of internet-connected devices grows. In January 2022, we began deploying C-Band spectrum, which has been built out to cover approximately 189 million POPs in the U.S. as of December 31, 2022. We expect to continue deploying C-Band spectrum across the continental U.S. as more and more of the spectrum becomes available for our use. We use low and mid-band spectrum and dynamic spectrum sharing (DSS) technology, to allow 5G service to run simultaneously with 4G LTE on multiple spectrum bands. With DSS, whenever customers move outside Verizon's millimeter wave and C-Band coverage areas, their 5G-enabled devices will remain on 5G technology using the lower spectrum bands where this network is available. This allows us to more fully and effectively utilize our current spectrum resources to serve both 4G and 5G customers.

To compensate for the shrinking market for traditional copper-based products, we continue to build fiber-based networks supporting data, video and advanced business services - areas where demand for reliable high-speed connections is growing. We are transforming the architecture of our networks into our Intelligent Edge Network, providing improved efficiency and virtualization, increased automation and opportunities for edge computing services that will support our fiber-based and radio access network technologies. We expect that this new architecture will simplify operations by eliminating legacy network elements, speed the deployment of 5G wireless technology and create new opportunities in the business market in a cost-efficient manner.

Consolidated Results of Operations

In this section, we discuss our overall results of operations and highlight special items that are not included in our segment results. In "Segment Results of Operations," we review the performance of our two reportable segments in more detail. A detailed discussion of 2020 items and year-over-year comparisons between 2021 and 2020 that are not included in this Form 10-K can be found in the "Management's Discussion and Analysis of Financial Condition and Results of Operations" in our Annual Report on Form 10-K for the year ended December 31, 2021.

Consolidated Operating Revenues

Years Ended December 31,	(dollars in millions)				
	2022	2021	Increase/(Decrease)		
			2022 vs. 2021		
Consumer	\$ 103,506	\$ 95,300	\$ 8,206	8.6	%
Business	31,072	31,042	30	0.1	
Corporate and other	2,510	7,722	(5,212)	(67.5)	
Eliminations	(253)	(451)	198	43.9	
Consolidated Operating Revenues	\$ 136,835	\$ 133,613	\$ 3,222	2.4	

Consolidated operating revenues increased during 2022 compared to 2021, due to increases in our Consumer and Business segments, partially offset by a decrease in Corporate and other.

Corporate and other revenues decreased during 2022 compared to 2021, primarily due to the sale of Verizon Media, which was completed in September 2021. Verizon Media's total operating revenues were approximately \$5.3 billion for the year ended December 31, 2021. See Note 3 to the consolidated financial statements for additional information on the sale of Verizon Media.

Revenues for our segments are discussed separately below under the heading "Segment Results of Operations."

Consolidated Operating Expenses

Years Ended December 31,			(dollars in millions)	
	2022	2021	Increase/(Decrease)	
Cost of services	\$ 28,637	\$ 31,234	\$ (2,597)	(8.3) %
Cost of wireless equipment	30,496	25,067	5,429	21.7
Selling, general and administrative expense	30,136	28,658	1,478	5.2
Depreciation and amortization expense	17,099	16,206	893	5.5
Consolidated Operating Expenses	\$ 106,368	\$ 101,165	\$ 5,203	5.1

Operating expenses for our segments are discussed separately below under the heading "Segment Results of Operations."

Cost of Services

Cost of services includes the following costs directly attributable to a service: salaries and wages, benefits, materials and supplies, content costs, contracted services, network access and transport costs, customer provisioning costs, computer systems support, costs to support our outsourcing contracts and technical facilities and traffic acquisition costs. Aggregate customer service costs, which include billing and service provisioning, are allocated between Cost of services and Selling, general and administrative expense.

Cost of services decreased during 2022 compared to 2021 primarily due to:

- a decrease in traffic acquisition costs of \$2.2 billion primarily related to the sale of Verizon Media;
- a decrease in personnel costs of \$745 million primarily related to the sale of Verizon Media;
- a decrease in digital content of \$335 million primarily driven by the sale of Verizon Media;
- a decrease in other direct costs of \$322 million primarily related to the sale of Verizon Media;
- a decrease in regulatory fees of \$201 million due to a lower Federal Universal Service Fund (FUSF) volume and rate;
- an increase in access costs of \$614 million primarily due to the inclusion of TracFone results, partially offset by a decline in voice services; and
- an increase in rent expense of \$212 million related to adding capacity to the networks to support demand and lease modifications for certain existing cell towers in April 2021 to support the build out of our 5G wireless network.

See Note 3 to the consolidated financial statements for additional information on the sale of Verizon Media and the acquisition of TracFone.

Cost of Wireless Equipment

Cost of wireless equipment increased during 2022 compared to 2021 primarily due to:

- an increase of \$2.2 billion due to the inclusion of TracFone results;
- an increase of \$2.1 billion driven by a higher volume of wireless devices sold primarily related to upgrades; and
- an increase of \$1.3 billion related to a shift to higher priced equipment in the mix of wireless devices sold.

Selling, General and Administrative Expense

Selling, general and administrative expense includes salaries and wages and benefits not directly attributable to a service or product, the provision for credit losses, taxes other than income taxes, advertising and sales commission costs, call center and information technology costs, regulatory fees, professional service fees and rent and utilities for administrative space. Also included is a portion of the aggregate customer care costs as discussed above in "Cost of Services."

Selling, general and administrative expense increased during 2022 compared to 2021 primarily due to:

- an increase in the provision for credit losses of \$822 million driven by increased device payment loan volume and an increase in the expected loss rate primarily as a result of the change in device payment plan terms to 36 months, as well as actions taken in prior years in response to the COVID-19 pandemic;
- the \$706 million net gain on the sale of Verizon Media in 2021;
- an increase in personnel costs of \$603 million primarily related to the inclusion of TracFone results and increases in third-party employee costs due to contracted services, partially offset by a decrease due to the sale of Verizon Media;
- a decrease of \$367 million in other general expenses driven by lease terminations primarily due to the sale of Verizon Media; and
- the \$223 million loss resulting from agreements entered into to sell certain wireless licenses in 2021.

See Note 3 to the consolidated financial statements for additional information on the sale of Verizon Media, the acquisition of TracFone and spectrum license transactions.

Depreciation and Amortization Expense

Depreciation and amortization expense increased during 2022 compared to 2021, primarily due to the change in the mix of net depreciable and amortizable assets, including acquisition-related intangible assets, as well as the deployment of C-Band spectrum.

Other Consolidated Results

Other Income (Expense), Net

Additional information relating to Other income (expense), net is as follows:

Years Ended December 31,	(dollars in millions)			
			Increase/(Decrease)	
	2022	2021	2022 vs. 2021	
Interest income	\$ 146	\$ 48	\$ 98	nm
Other components of net periodic benefit income	2,386	3,785	(1,399)	(37.0) %
Net debt extinguishment losses	(1,077)	(3,541)	2,464	69.6
Other, net	(82)	20	(102)	nm
Total	\$ 1,373	\$ 312	\$ 1,061	nm

nm - not meaningful

Other income (expense), net reflects certain items not directly related to our core operations, including interest income, gains and losses from non-operating asset dispositions, debt extinguishment costs, components of net periodic pension and postretirement benefit cost and income and certain foreign exchange gains and losses.

Other income (expense), net changed during 2022 compared to 2021 primarily due to:

- a net pension and postretirement benefits remeasurement gain of \$1.7 billion recorded during 2022, compared with a net pension and postretirement benefits remeasurement gain of \$2.4 billion recorded during 2021; and
- net debt extinguishment losses of \$1.1 billion primarily related to tender offers in 2022, compared with losses of \$3.5 billion in connection with tender offers, the redemptions of securities issued by Verizon and open market repurchases of various Company and subsidiary notes in 2021.

See Note 11 to the consolidated financial statements for additional information on the other components of net periodic pension and postretirement benefit income.

Interest Expense

Years Ended December 31,	(dollars in millions)			
			Increase/(Decrease)	
	2022	2021	2022 vs. 2021	
Total interest costs on debt balances	\$ 5,643	\$ 5,326	\$ 317	6.0 %
Less capitalized interest costs	2,030	1,841	189	10.3
Total	\$ 3,613	\$ 3,485	\$ 128	3.7
Average debt outstanding ^{(1) (3)}	\$ 151,226	\$ 147,035		
Effective interest rate ^{(2) (3)}	3.7 %	3.6 %		

⁽¹⁾ The average debt outstanding is a financial measure and is calculated by applying a simple average of prior thirteen-month end balances of total short-term and long-term debt, net of discounts, premiums and unamortized debt issuance costs.

⁽²⁾ The effective interest rate is the rate of actual interest incurred on debt. It is calculated by dividing the total interest costs on debt balances by the average debt outstanding.

⁽³⁾ We believe that this measure is useful to management, investors and other users of our financial information in evaluating our debt financing cost and trends in our debt leverage management.

Total interest expense increased during 2022 compared to 2021 primarily due to:

- an increase in interest costs due to a combination of higher average interest rates and higher average debt balances; partially offset by
- an increase in capitalized interest costs as a result of a full year of capitalized interest on C-Band licenses in 2022 compared to a partial year in 2021.

See Note 4 and Note 7 to the consolidated financial statements for additional information on spectrum licenses and debt transactions, respectively.

Provision for Income Taxes

Years Ended December 31,			(dollars in millions)	
	2022	2021	Increase/(Decrease)	
Provision for income taxes	\$ 6,523	\$ 6,802	\$ (279)	(4.1) %
Effective income tax rate	23.1 %	23.1 %		

The effective income tax rate is calculated by dividing the provision for income taxes by income before the provision for income taxes. The effective income tax rate for the year ended December 31, 2022 was comparable to the similar period in 2021. The decrease in the provision for income taxes was primarily due to the decrease in income before income taxes in 2022.

A reconciliation of the statutory federal income tax rate to the effective income tax rate for each period is included in Note 12 to the consolidated financial statements.

Consolidated Net Income, Consolidated EBITDA and Consolidated Adjusted EBITDA

Consolidated earnings before interest, taxes, depreciation and amortization expense (Consolidated EBITDA) and Consolidated Adjusted EBITDA, which are presented below, are non-generally accepted accounting principles (GAAP) measures that we believe are useful to management, investors and other users of our financial information in evaluating operating profitability on a more variable cost basis as they exclude the depreciation and amortization expense related primarily to capital expenditures and acquisitions that occurred in prior years, as well as in evaluating operating performance in relation to Verizon's competitors. Consolidated EBITDA is calculated by adding back interest, taxes, depreciation and amortization expense to net income.

Consolidated Adjusted EBITDA is calculated by excluding from Consolidated EBITDA the effect of the following non-operational items: equity in earnings of unconsolidated businesses and other income and expense, net, as well as the effect of certain special items. We believe that this measure is useful to management, investors and other users of our financial information in evaluating the effectiveness of our operations and underlying business trends in a manner that is consistent with management's evaluation of business performance. We believe that Consolidated Adjusted EBITDA is widely used by investors to compare a company's operating performance to its competitors by minimizing impacts caused by differences in capital structure, taxes, and depreciation and amortization policies. Further, the exclusion of non-operational items and special items enables comparability to prior period performance and trend analysis. See "Special Items" for additional information.

It is management's intent to provide non-GAAP financial information to enhance the understanding of Verizon's GAAP financial information, and it should be considered by the reader in addition to, but not instead of, the financial statements prepared in accordance with GAAP. Each non-GAAP financial measure is presented along with the corresponding GAAP measure so as not to imply that more emphasis should be placed on the non-GAAP measure. We believe that providing these non-GAAP measures in addition to the GAAP measures allows management, investors and other users of our financial information to more fully and accurately assess both consolidated and segment performance. The non-GAAP financial information presented may be determined or calculated differently by other companies and may not be directly comparable to that of other companies.

		(dollars in millions)	
Years Ended December 31,		2022	2021
Consolidated Net Income	\$	21,748	\$ 22,618
Add:			
Provision for income taxes		6,523	6,802
Interest expense		3,613	3,485
Depreciation and amortization expense ⁽¹⁾		17,099	16,206
Consolidated EBITDA	\$	48,983	\$ 49,111
Add (Less):			
Other (income) expense, net ^{(2) (3)}	\$	(1,373)	\$ (312)
Equity in earnings of unconsolidated businesses ⁽⁴⁾		(44)	(145)
Severance charges		304	209
Loss on spectrum licenses		—	223
Net gain from sale of Media		—	(706)
Consolidated Adjusted EBITDA	\$	47,870	\$ 48,380

⁽¹⁾ Includes Amortization of acquisition-related intangible assets, which were \$826 million and \$594 million during the years ended December 31, 2022 and 2021, respectively. See "Special Items" for additional information.

⁽²⁾ Includes Pension and benefits remeasurement credits of \$1.7 billion and \$2.4 billion during the years ended December 31, 2022 and 2021, respectively. See "Special Items" and "Other Income (Expense), Net" for additional information.

⁽³⁾ Includes Early debt redemption costs, which were \$1.2 billion and \$3.5 billion during the years ended December 31, 2022 and 2021, respectively. See "Special Items" for additional information.

⁽⁴⁾ Includes Net gain from disposition of asset, which was \$131 million during the year ended December 31, 2021. See "Special Items" for additional information.

The changes in Consolidated Net Income, Consolidated EBITDA and Consolidated Adjusted EBITDA in the table above during 2022 compared to 2021, were primarily a result of the factors described in connection with operating revenues and operating expenses.

Segment Results of Operations

We have two reportable segments that we operate and manage as strategic business units - Consumer and Business. We measure and evaluate our segments based on segment operating income. The use of segment operating income is consistent with the chief operating decision maker's assessment of segment performance.

To aid in the understanding of segment performance as it relates to segment operating income, management uses the following operating statistics to evaluate the overall effectiveness of our segments. We believe these operating statistics are useful to investors and other users of our financial information because they provide additional insight into drivers of our segments' operating results, key trends and performance relative to our peers. These operating statistics may be determined or calculated differently by other companies and may not be directly comparable to those statistics of other companies.

Wireless retail connections are retail customer device postpaid and prepaid connections as of the end of the period. Retail connections under an account may include those from smartphones and basic phones (collectively, phones), as well as tablets and other internet devices, including FWA, wearables and retail IoT devices. Wireless retail connections are calculated by adding total retail postpaid and prepaid new connections in the period to prior period retail connections, and subtracting total retail postpaid and prepaid disconnects in the period.

Wireless retail postpaid connections are retail postpaid customer device connections as of the end of the period. Retail postpaid connections under an account may include those from phones, as well as tablets and other internet devices, including postpaid FWA, wearables and retail IoT devices. Wireless retail postpaid connections are calculated by adding retail postpaid new connections in the period to prior period retail postpaid connections, and subtracting retail postpaid disconnects in the period.

Wireless retail prepaid connections are retail prepaid customer device connections as of the end of the period. Retail prepaid connections may include those from phones, as well as tablets and other internet devices, including prepaid FWA, and wearables. Wireless retail prepaid connections are calculated by adding retail prepaid new connections in the period to prior period retail prepaid connections, and subtracting retail prepaid disconnects in the period.

Fios internet connections are the total number of connections to the internet using Fios internet services as of the end of the period. Fios internet connections are calculated by adding Fios internet new connections in the period to prior period Fios internet connections, and subtracting Fios internet disconnects in the period.

Fios video connections are the total number of connections to traditional linear video programming using Fios video services as of the end of the period. Fios video connections are calculated by adding Fios video net additions in the period to prior period Fios video connections. Fios video net additions are calculated by subtracting the Fios video disconnects from the Fios video new connections.

Total broadband connections are the total number of connections to the internet using Fios internet services, Digital Subscriber Line (DSL), and postpaid, prepaid and IoT FWA as of the end of the period. Total broadband connections are calculated by adding total broadband connections, net additions in the period to prior period total broadband connections.

Wireless retail connections, net additions are the total number of additional retail customer device postpaid and prepaid connections, less the number of device disconnects in the period. Wireless retail connections, net additions in each period presented are calculated by subtracting the total retail postpaid and prepaid disconnects, net of certain adjustments, from the total retail postpaid and prepaid new connections in the period.

Wireless retail postpaid connections, net additions are the total number of additional retail customer device postpaid connections, less the number of device disconnects in the period. Wireless retail postpaid connections, net additions in each period presented are calculated by subtracting the retail postpaid disconnects, net of certain adjustments, from the retail postpaid new connections in the period.

Wireless retail prepaid connections, net additions are the total number of additional retail customer device prepaid connections, less the number of device disconnects in the period. Wireless retail prepaid connections, net additions in each period presented are calculated by subtracting the retail prepaid disconnects, net of certain adjustments, from the retail prepaid new connections in the period.

Wireless retail postpaid phone connections, net additions are the total number of additional retail customer postpaid phone connections, less the number of phone disconnects in the period. Wireless retail postpaid phone connections, net additions in each period presented are calculated by subtracting the retail postpaid phone disconnects, net of certain adjustments, from the retail postpaid phone new connections in the period.

Total broadband connections, net additions are the total number of additional total broadband connections, less the number of total broadband disconnects in the period. Total broadband connections, net additions in each period presented are calculated by subtracting the total broadband disconnects, net of certain adjustments, from the total broadband new connections in the period.

Wireless Churn is the rate at which service to retail, retail postpaid, or retail postpaid phone connections is terminated on average in the period. The churn rate in each period presented is calculated by dividing retail disconnects, retail postpaid disconnects, or retail postpaid phone disconnects by the average retail connections, average retail postpaid connections, or average retail postpaid phone connections, respectively, in the period.

Wireless retail postpaid ARPA is the calculated average retail postpaid service revenue per account (ARPA) from retail postpaid accounts in the period. Wireless retail postpaid service revenue does not include recurring device payment plan billings related to the Verizon device payment program, plan billings related to device warranty and insurance or regulatory fees. Wireless retail postpaid ARPA in each period presented is calculated by dividing retail postpaid service revenue by the average retail postpaid accounts in the period.

Wireless retail postpaid accounts are wireless retail customers that are directly served and managed under the Verizon brand and use its services as of the end of the period. Accounts include unlimited plans, shared data plans and corporate accounts, as well as legacy single connection plans and multi-connection family plans. A single account may include monthly wireless services for a variety of connected devices. Wireless retail postpaid accounts are calculated by adding retail postpaid new accounts to the prior period retail postpaid accounts.

Wireless retail postpaid connections per account is the calculated average number of retail postpaid connections per retail postpaid account as of the end of the period. Wireless retail postpaid connections per account is calculated by dividing the total number of retail postpaid connections by the number of retail postpaid accounts as of the end of the period.

Segment operating income margin reflects the profitability of the segment as a percentage of revenue. Segment operating income margin is calculated by dividing total segment operating income by total segment operating revenues.

Segment earnings before interest, taxes, depreciation and amortization (Segment EBITDA) which is presented below, is a non-GAAP measure and does not purport to be an alternative to operating income (loss) as a measure of operating performance. We believe this measure is useful to management, investors and other users of our financial information in evaluating operating profitability on a more variable cost basis as it excludes the depreciation and amortization expense related primarily to capital expenditures and acquisitions that occurred in prior years, as well as in evaluating operating performance in relation to our competitors. Segment EBITDA is calculated by adding back depreciation and amortization expense to segment operating income (loss). Segment EBITDA margin is calculated by dividing Segment EBITDA by total segment operating revenues. See Note 13 to the consolidated financial statements for additional information.

Verizon Consumer Group

Our Consumer segment provides consumer-focused wireless and wireline communications services and products. Our wireless services are provided across one of the most extensive wireless networks in the U.S. under the Verizon brand, TracFone brands and through wholesale and other arrangements. We also provide FWA broadband through our wireless networks. Our wireline services are provided in nine states in the Mid-Atlantic and Northeastern U.S., as well as Washington D.C., over our 100% fiber-optic network through our Verizon Fios product portfolio and over a traditional copper-based network to customers who are not served by Fios.

Operating Revenues and Selected Operating Statistics

		(dollars in millions, except ARPA)			
				Increase/(Decrease)	
Years Ended December 31,		2022	2021	2022 vs. 2021	
Service	\$	73,139	\$	67,733	\$ 5,406 8.0 %
Wireless equipment		23,168		19,781	3,387 17.1
Other		7,199		7,786	(587) (7.5)
Total Operating Revenues	\$	103,506	\$	95,300	\$ 8,206 8.6
Connections ('000):⁽¹⁾					
Wireless retail postpaid ⁽²⁾		91,856		91,543	313 0.3
Wireless retail prepaid ⁽²⁾⁽³⁾		22,664		23,852	(1,188) (5.0)
Total wireless retail		114,520		115,395	(875) (0.8)
Fios internet		6,740		6,541	199 3.0
Fios video		3,234		3,573	(339) (9.5)
Total broadband		7,900		6,989	911 13.0
Net Additions in Period ('000):⁽⁴⁾					
Wireless retail postpaid ⁽²⁾		965		1,114	(149) (13.4)
Wireless retail prepaid ⁽²⁾⁽³⁾		(445)		(52)	(393) nm
Total wireless retail		520		1,062	(542) (51.0)
Wireless retail postpaid phones ⁽²⁾		(655)		575	(1,230) nm
Total broadband		904		328	576 nm
Churn Rate:⁽²⁾					
Wireless retail ⁽⁴⁾		1.63 %		1.10 %	
Wireless retail postpaid		1.01 %		0.89 %	
Wireless retail postpaid phones		0.81 %		0.71 %	
Account Statistics:					
Wireless retail postpaid ARPA	\$	125.97	\$	122.30	\$ 3.67 3.0
Wireless retail postpaid accounts ('000) ⁽¹⁾		33,183		33,651	(468) (1.4)
Wireless retail postpaid connections per account ⁽¹⁾		2.77		2.72	0.05 1.8

⁽¹⁾ As of end of period

⁽²⁾ The number of wireless retail connections as of December 31, 2022 reflects a decline in our customer base related to the shutdown of our third-generation (3G) network in the fourth quarter of 2022 of approximately 576,000 wireless retail postpaid connections, including 180,000 wireless retail postpaid phone connections, and 237,000 wireless retail prepaid connections. In addition, the shutdown of our competitors' 3G network in the second and third quarter of 2022 resulted in a reduction to our customer base of 402,000 wireless retail prepaid connections and 102,000 wireless retail prepaid connections as of June 30, 2022 and September 30, 2022, respectively. The impact of the 3G network shutdowns has been excluded for purposes of calculating wireless retail net additions and wireless churn for the respective periods.

⁽³⁾ Acquisition of TracFone was completed on November 23, 2021. See Note 3 to the consolidated financial statements for additional information.

⁽⁴⁾ Includes certain adjustments

nm - not meaningful

Consumer's total operating revenues increased during 2022 compared to 2021, as a result of increases in Service and Wireless equipment revenues, partially offset by a decrease in Other revenue.

Service Revenue

Service revenue increased during 2022 compared to 2021, primarily driven by increases in wireless and Fios service revenues.

Wireless service revenue increased \$5.4 billion during 2022 compared to 2021 primarily due to:

- an increase of \$4.0 billion, representing the net impact of the acquisition of TracFone in the fourth quarter of 2021;
- an increase of \$769 million in access revenues related to our postpaid plans driven by mobile security products included in certain device protection packages such that a larger amount of the overall device protection revenue is recognized in Service revenue, pricing actions implemented in the second half of 2022, cloud services, and additional connections and migrations to higher priced plans, partially offset by related promotions as well as declines in our postpaid phone customer base;
- an increase of \$352 million in TravelPass revenues due to an increase in customer international travel; and
- an increase of \$253 million related to growth in non-retail service revenue.

For the year ended December 31, 2022, Fios service revenue totaled \$10.9 billion, representing an increase of \$127 million compared to 2021, primarily resulting from an increase in Fios internet connections, reflecting increased demand for higher broadband speeds, partially offset by a decrease in Fios video and voice revenues.

See Note 3 to the consolidated financial statements for additional information on the acquisition of TracFone.

Wireless Equipment Revenue

Wireless equipment revenue increased during 2022 compared to 2021 primarily due to:

- an increase of \$1.4 billion related to a shift to higher priced equipment in the mix of wireless devices sold;
- an increase of \$1.2 billion driven by a higher volume of wireless devices primarily related to a higher rate of upgrades partially offset by related promotions; and
- an increase of \$780 million due to the inclusion of TracFone results.

Other Revenue

Other revenue includes fees that partially recover the direct and indirect costs of complying with regulatory and industry obligations and programs, revenues associated with certain products included in our device protection offerings, leasing and interest when equipment is sold to the customer by an authorized agent under a device payment plan agreement.

Other revenue decreased during 2022 compared to 2021 primarily due to:

- a decrease of \$481 million that resulted from a change in the product mix within the device protection offering such that a smaller amount of the overall device protection revenue is recognized in Other revenue;
- a decrease of \$77 million related to interest recognition on equipment sold to the customer by an authorized agent under a device payment plan agreement, mainly due to an increase in contract terms to 36 months; and
- a decrease of \$51 million in other regulatory surcharges.

Operating Expenses

Years Ended December 31,	(dollars in millions)			
	2022	2021	Increase/(Decrease)	
Cost of services	\$ 17,746	\$ 16,581	\$ 1,165	7.0 %
Cost of wireless equipment	25,134	20,523	4,611	22.5
Selling, general and administrative expense	19,064	16,562	2,502	15.1
Depreciation and amortization expense	12,716	11,679	1,037	8.9
Total Operating Expenses	\$ 74,660	\$ 65,345	\$ 9,315	14.3

Cost of Services

Cost of services increased during 2022 compared to 2021 primarily due to:

- an increase in access costs of \$848 million primarily due to the inclusion of TracFone results; and
- an increase in rent expense of \$198 million related to adding capacity to the networks to support demand and lease modifications for certain existing cell towers in April 2021 in connection with the build out of our 5G wireless network.

Cost of Wireless Equipment

Cost of wireless equipment increased during 2022 compared to 2021 primarily due to:

- an increase of \$2.2 billion due to the inclusion of TracFone results;

- an increase of \$1.5 billion driven by a higher volume of wireless devices sold, primarily related to a higher rate of upgrades; and
- an increase of \$1.0 billion related to a shift to higher priced equipment in the mix of wireless devices sold.

Selling, General and Administrative Expense

Selling, general and administrative expense increased during 2022 compared to 2021 primarily due to:

- an increase in personnel costs of \$1.3 billion primarily due to the inclusion of TracFone results and increased third-party employee costs due to contracted services;
- an increase in provision for credit losses of \$745 million driven by increased device payment loan volume and an increase in the expected loss rate primarily as a result of the change in device payment plan terms to 36 months, as well as actions taken in prior years in response to the COVID-19 pandemic; and
- an increase in advertising expense of \$454 million primarily due to the inclusion of TracFone results and brand marketing, including the launch of the 5G Ultra campaign in early 2022.

Depreciation and Amortization Expense

Depreciation and amortization expense increased during 2022 compared to 2021, driven by the change in the mix of total Verizon depreciable and amortizable assets and Consumer's usage of those assets.

Segment Operating Income and EBITDA

Years Ended December 31,			(dollars in millions)	
	2022	2021	Increase/(Decrease)	
Segment Operating Income	\$ 28,846	\$ 29,955	\$ (1,109)	(3.7) %
Add Depreciation and amortization expense	12,716	11,679	1,037	8.9
Segment EBITDA	\$ 41,562	\$ 41,634	\$ (72)	(0.2)
Segment operating income margin	27.9 %	31.4 %		
Segment EBITDA margin	40.2 %	43.7 %		

The changes in the table above during the periods presented were primarily a result of the factors described in connection with operating revenues and operating expenses.

Verizon Business Group

Our Business segment provides wireless and wireline communications services and products, including data, video and conferencing services, corporate networking solutions, security and managed network services, local and long distance voice services and network access to deliver various IoT services and products. We also provide FWA broadband through our wireless networks. We provide these products and services to businesses, government customers and wireless and wireline carriers across the U.S. and select products and services to customers around the world. The Business segment is organized in four customer groups: Small and Medium Business, Global Enterprise, Public Sector and Other, and Wholesale.

Operating Revenues and Selected Operating Statistics

Years Ended December 31,	(dollars in millions)			
			Increase/(Decrease)	
	2022	2021	2022 vs. 2021	
Small and Medium Business	\$ 12,613	\$ 11,774	\$ 839	7.1 %
Global Enterprise	9,734	10,224	(490)	(4.8)
Public Sector and Other	6,118	6,324	(206)	(3.3)
Wholesale	2,607	2,720	(113)	(4.2)
Total Operating Revenues⁽¹⁾	\$ 31,072	\$ 31,042	\$ 30	0.1
Connections ('000):⁽²⁾				
Wireless retail postpaid ⁽³⁾	28,733	27,411	1,322	4.8
Fios internet	373	356	17	4.8
Fios video	67	71	(4)	(5.6)
Total broadband	1,036	599	437	73.0
Net Additions in Period ('000):⁽⁴⁾				
Wireless retail postpaid ⁽³⁾	1,640	1,001	639	63.8
Wireless retail postpaid phones ⁽³⁾	856	509	347	68.2
Total broadband	386	81	305	nm
Churn Rate:⁽³⁾				
Wireless retail postpaid	1.38 %	1.27 %		
Wireless retail postpaid phones	1.07 %	1.03 %		

⁽¹⁾ Service and other revenues included in our Business segment amounted to approximately \$27.0 billion and \$27.7 billion for the years ended December 31, 2022 and 2021, respectively. Wireless equipment revenues included in our Business segment amounted to approximately \$4.0 billion and \$3.4 billion for the years ended December 31, 2022 and 2021, respectively.

⁽²⁾ As of end of period

⁽³⁾ The number of wireless retail connections as of December 31, 2022 reflects a decline in our customer base related to the shutdown of our 3G network in the fourth quarter of 2022 of approximately 333,000 wireless retail postpaid connections, including 212,000 wireless retail postpaid phone connections. The impact of the 3G network shutdown has been excluded for purposes of calculating wireless retail net additions and wireless churn for the respective periods.

⁽⁴⁾ Includes certain adjustments

nm - not meaningful

Business's total operating revenues increased during 2022 compared to 2021, as a result of an increase in Small and Medium Business revenue, partially offset by decreases in Global Enterprise, Public Sector and Other and Wholesale revenues.

Small and Medium Business

Small and Medium Business offers wireless services and equipment, conferencing services, tailored voice and networking products, Fios services, Internet Protocol (IP) networking, advanced voice solutions and security and managed information technology services to our U.S.-based small and medium businesses that do not meet the requirements to be categorized as Global Enterprise, as described below.

Small and Medium Business revenues increased during 2022 compared to 2021 primarily due to:

- an increase in wireless equipment revenue of \$507 million driven by a higher volume of devices sold and a shift to higher priced equipment in the mix of devices sold, partially offset by an increase in promotions;
- an increase in wireless service revenue of \$395 million primarily driven by an increase in the amount of wireless retail postpaid connections as well as the economic adjustment charge that took effect in the second quarter of 2022; and
- a decrease of \$85 million related to the loss of voice and DSL service connections.

Fios revenues totaled \$1.0 billion, which represents an increase of \$41 million during 2022 compared to 2021. The increase was primarily related to increases in total connections, as well as increased demand for higher broadband speeds.

Global Enterprise

Global Enterprise offers services to large businesses, which are identified based on their size and volume of business with Verizon, as well as non-U.S. public sector customers.

Global Enterprise revenues decreased during 2022 compared to 2021 primarily due to:

- a decrease of \$544 million in networking revenue and traditional data and voice communication services related to secular pressures in the marketplace;
- a decrease of \$118 million due to lower FUSF volume and rate along with resulting surcharges;
- an increase of \$117 million in wireless equipment revenue driven by a shift to higher priced equipment in the mix of devices sold and higher volume of devices sold;
- an increase of \$89 million in wireless service revenue primarily driven by an increase in the amount of wireless retail postpaid connections as well as the economic adjustment charge that took effect in the second quarter of 2022; and
- an increase of \$34 million in customer premise equipment primarily due to higher volumes.

Public Sector and Other

Public Sector and Other offers wireless products and services as well as wireline connectivity and managed solutions to U.S. federal, state and local governments and educational institutions. These services include business services and connectivity similar to the products and services offered by Global Enterprise, in each case, with features and pricing designed to address the needs of governments and educational institutions.

Public Sector and Other revenues decreased during 2022 compared to 2021 primarily due to:

- a decrease of \$169 million in networking revenue and traditional data and voice communication services related to secular pressures in the marketplace; and
- a decrease of \$44 million related to lower FUSF volume and rate along with resulting surcharges.

Wholesale

Wholesale offers wireline communications services including data, voice, local dial tone and broadband services primarily to local, long distance, and wireless carriers that use our facilities to provide services to their customers.

Wholesale revenues decreased during 2022 compared to 2021 primarily due to:

- a decrease of \$113 million related to declines in traditional voice communication and network connectivity as a result of technology substitution and rationalization of international traffic, as well as a decrease in core data.

Operating Expenses

Years Ended December 31,			(dollars in millions)	
	2022	2021	Increase/(Decrease)	
			2022 vs. 2021	
Cost of services	\$ 10,483	\$ 10,653	\$ (170)	(1.6) %
Cost of wireless equipment	5,362	4,544	818	18.0
Selling, general and administrative expense	8,284	8,324	(40)	(0.5)
Depreciation and amortization expense	4,312	4,084	228	5.6
Total Operating Expenses	\$ 28,441	\$ 27,605	\$ 836	3.0

Cost of Services

Cost of services decreased during 2022 compared to 2021 primarily due to:

- a decrease in regulatory fees of \$226 million primarily related to lower FUSF volume and rate;
- a decrease in access costs of \$219 million resulting from a decline in voice services;
- an increase in network and equipment maintenance of \$66 million primarily related to fleet maintenance;
- an increase of \$65 million in customer premise equipment primarily due to higher volumes; and
- an increase in personnel expense of \$54 million related to rate and actuarial assumption changes in connection with certain post-employment benefits.

Cost of Wireless Equipment

Cost of wireless equipment increased during 2022 compared to 2021 primarily due to:

- an increase of \$527 million driven by a higher volume of wireless devices sold; and
- an increase of \$360 million related to a shift to higher priced equipment in the mix of wireless devices sold.

Selling, General and Administrative Expense

Selling, general and administrative expense decreased during 2022 compared to 2021 primarily due to:

- a decrease of \$88 million primarily due to gains recognized in connection with a non-strategic divestiture and other activity;
- a decrease of \$78 million primarily due to taxes other than income mainly driven by a one-time international tax benefit;
- a decrease in other general expenses of \$72 million partly related to rent and lease expenses;
- an increase in personnel expense of \$148 million primarily related to increased third-party employee costs due to contracting services; and

- an increase of \$73 million in provision for credit losses driven by increased device payment loan volume and an increase in the expected loss rate primarily as a result of the change in device payment plan terms to 36 months.

Depreciation and Amortization Expense

Depreciation and amortization expense increased during 2022 compared to 2021, driven by the change in the mix of total Verizon depreciable and amortizable assets and Business usage of those assets.

Segment Operating Income and EBITDA

Years Ended December 31,	(dollars in millions)			
			Increase/(Decrease)	
	2022	2021	2022 vs. 2021	
Segment Operating Income	\$ 2,631	\$ 3,437	\$ (806)	(23.5) %
Add Depreciation and amortization expense	4,312	4,084	228	5.6
Segment EBITDA	\$ 6,943	\$ 7,521	\$ (578)	(7.7)
Segment operating income margin	8.5 %	11.1 %		
Segment EBITDA margin	22.3 %	24.2 %		

The changes in the table above during the periods presented were primarily a result of the factors described in connection with operating revenues and operating expenses.

Special Items

Special items included in Income Before Provision For Income Taxes were as follows:

Years Ended December 31,	(dollars in millions)	
	2022	2021
Amortization of acquisition-related intangible assets^{(1) (2)}		
Depreciation and amortization expense	\$ 826	\$ 594
Severance, pension and benefits charges (credits)		
Selling, general and administrative expense	304	209
Other (income) expense, net	(1,675)	(2,379)
Early debt redemption costs		
Other (income) expense, net	1,241	3,541
Net gain from disposition of asset and business		
Selling, general and administrative expense	—	(706)
Equity in (earnings) losses of unconsolidated businesses	—	(131)
Loss on spectrum licenses		
Selling, general and administrative expense	—	223
Total	\$ 696	\$ 1,351

⁽¹⁾ Certain amounts have been reclassified to conform to the current period presentation.

⁽²⁾ Amounts are included in segment results of operations.

The Consolidated Adjusted EBITDA non-GAAP measure presented in the Consolidated Net Income, Consolidated EBITDA and Consolidated Adjusted EBITDA discussion (see "Consolidated Results of Operations") excludes all of the amounts included above, as described below.

The income and expenses related to special items included in our consolidated results of operations were as follows:

Years Ended December 31,	(dollars in millions)	
	2022	2021
Within Total Operating Expenses⁽¹⁾	\$ 1,130	\$ 320
Within Other (income) expense, net	(434)	1,162
Within Equity in (earnings) losses of unconsolidated businesses	—	(131)
Total	\$ 696	\$ 1,351

⁽¹⁾ Certain amounts have been reclassified to conform to the current period presentation.

Amortization of Acquisition-Related Intangible Assets

During 2022 and 2021, we recorded pre-tax amortization expense of \$826 million and \$594 million, respectively, related to acquired intangible assets.

Severance, Pension and Benefits Charges (Credits)

During 2022, in accordance with our accounting policy to recognize actuarial gains and losses in the period in which they occur, we recorded net pre-tax pension and benefits credits of \$1.7 billion in our pension and postretirement benefit plans. The credits were recorded in Other income (expense), net in our consolidated statement of income and were primarily driven by a credit of \$7.0 billion due to an increase in our discount rate assumption used to determine the current year liabilities of our pension plans (\$4.1 billion) and postretirement benefit plans (\$2.9 billion) from a weighted-average of 2.9% at December 31, 2021 to a weighted-average of 5.2% at December 31, 2022, a charge of \$5.5 billion due to the difference between our estimated and our actual return on assets and a credit of \$206 million due to other actuarial assumption adjustments. During 2022, we also recorded net pre-tax severance charges of \$304 million, related to involuntary separations under our existing plans, in Selling, general and administrative expense in our consolidated statement of income.

During 2021, in accordance with our accounting policy to recognize actuarial gains and losses in the period in which they occur, we recorded net pre-tax pension and benefits credits of \$2.4 billion in our pension and postretirement benefit plans. The credits were recorded in Other income (expense), net in our consolidated statement of income and were primarily driven by a credit of \$1.1 billion due to an increase in our discount rate assumption used to determine the current year liabilities of our pension plans and postretirement benefit plans from a weighted-average of 2.6% at December 31, 2020 to a weighted-average of 2.9% at December 31, 2021, a credit of \$847 million due to the difference between our estimated and our actual return on assets and a credit of \$453 million due to other actuarial assumption adjustments. During 2021, we also recorded net pre-tax severance charges of \$209 million related to voluntary separations under our existing plans in Selling, general and administrative expense in our consolidated statement of income.

Due to the presentation of the other components of net periodic benefit cost, we recognize a portion of the pension and benefits charges (credits) in Other income (expense), net in our consolidated statements of income.

See Note 11 to the consolidated financial statements for additional information related to severance, pension and benefits charges (credits).

Early Debt Redemption Costs

During 2022, we recorded pre-tax early debt redemption costs of \$1.2 billion primarily in connection with tender offers.

During 2021, we recorded pre-tax early debt redemption costs of \$3.5 billion in connection with tender offers, the redemptions of securities issued by Verizon and open market repurchases of various Company and subsidiary notes.

See Note 7 to the consolidated financial statements for additional information related to our early debt redemptions.

Net Gain from Disposition of Asset and Business

During 2021, we recorded a pre-tax net gain of \$837 million, in connection with the sale of an investment and the sale of Verizon Media.

See Note 3 to the consolidated financial statements for additional information.

Loss on Spectrum Licenses

During 2021, we recognized a pre-tax loss of \$223 million as a result of signing two agreements to sell certain wireless licenses.

See Note 3 to the consolidated financial statements for additional information.

Operating Environment and Trends

The telecommunications industry is highly competitive. The rapid development of new technologies, services and products has eliminated many of the traditional distinctions among wireless, cable, internet and local and long distance communication services and brought new competitors to our markets. We expect competition to remain intense as traditional and non-traditional participants seek increased market share. Our high-quality customer base and networks differentiate us from our competitors and give us the ability to plan and manage through changing economic and competitive conditions. We remain focused on executing on the fundamentals of the business: maintaining a high-quality customer base, delivering strong financial and operating results and strengthening our balance sheet. We continue to lead in 4G LTE performance while aggressively building out our 5G network.

The U.S. wireless market has achieved a high penetration of smartphones, which reduces the opportunity for new phone connection growth for the industry. We expect future revenue growth in the industry to be driven by expanding existing customer

relationships, increasing the number of ways customers can connect with wireless networks and services and increasing the penetration of FWA and connected devices including wearables, tablets and IoT devices. Although certain use cases for 5G technologies and related ecosystems are in early development stages, we expect this technology will provide a significant opportunity for growth in the coming years. With respect to our wireless connectivity products and services, we compete against other national wireless service providers, including AT&T Inc. and T-Mobile US, Inc., as well as various regional wireless service providers. We also compete for retail activations with resellers that buy bulk wholesale service from wireless service providers, including Verizon, and resell it to their customers. Resellers include cable companies, such as Comcast Corporation and Charter Communications, Inc., and others. We face competition from other communications and technology companies seeking to increase their brand recognition and capture customer revenue with respect to the provision of wireless products and services, in addition to non-traditional offerings in mobile data. For example, Microsoft Corporation, Alphabet Inc., Apple Inc., Meta Platforms, Inc. and others are offering alternative means for messaging and making wireless voice calls that, in certain cases, can be used in lieu of the wireless providers' voice service, as well as alternative means of accessing video content. In addition, we expect to see increasing competition in the provisioning of internet access by low Earth orbit satellite companies as well in the area of fixed wireless offerings that provide an alternative to traditional landline internet service providers.

With respect to wireless services and equipment, pricing plays an increasingly important role in the wireless competitive landscape. We compete in this area by offering our customers services and devices that we believe they will regard as the best available value for the price. As the demand for wireless services continues to grow, wireless service providers are offering a range of service plans at competitive prices. These service offerings will vary from time to time based on customer needs, technology changes and market conditions and may be provided as standard plans or as part of limited time promotional offers. In addition, aggressive device promotions have become more common in an effort to gain a greater share of subscribers interested in changing carriers as well as retaining existing customers.

We expect future service revenue growth opportunities to arise from increased access revenue as customer demand for 5G connectivity continues to expand and customers shift to higher access plans, driven in part by attractive bundled content with premium brands, as well as from increased connections per account. Future service revenue growth opportunities will be dependent on expanding the penetration of our services, increasing the number of ways that our customers can connect with our networks and services and the development of new ecosystems. We and other wireless service providers, as well as equipment manufacturers, offer device payment options, which provide customers with the ability to pay for their device over a period of time, and some providers offer device leasing arrangements.

Current and potential competitors in the wireline service market include cable companies, wireless service providers, domestic and foreign telecommunications providers, satellite television companies, internet service providers, over-the-top providers and other companies that offer network services and managed enterprise solutions. We also face increasing competition from other providers of VoIP services as well as internet portal providers.

In addition, companies with a global presence are increasingly competing with us in our wireline services. A relatively small number of telecommunications and integrated service providers with global operations serve customers in the global enterprise market and, to a lesser extent, the global wholesale market. We compete with these providers for large contracts to provide integrated solutions to global enterprises and government customers. Many of these companies have strong market presence, brand recognition and existing customer relationships, all of which contribute to intensifying competition that may affect our future revenue growth.

Despite this challenging environment, we expect that we will be able to grow key aspects of our wireline services. We continue to provide network reliability and offer products, which include fiber-optic internet access, several video services, and voice services. Further, we will continue to offer our business and government customers a converged-enterprise environment for voice and data services as well as more robust IP products and services, and advance our IoT strategies by leveraging business models that monetize usage on our networks at the connectivity, platform and solution layers.

We will also continue to focus on cost efficiencies to allow us to have flexibility to adjust to changes in the competitive and economic environments and increase shareholder value.

Macroeconomic Environment

In 2022, as a result of the inflationary environment in the U.S., we experienced increases in our direct costs, including electricity and other energy-related costs for our network operations, and transportation and labor costs, as well as increased interest expenses related to rising interest rates. We believe that this inflationary environment and the resulting decline in real wages in the U.S. are altering consumer preferences, and causing certain consumers to become more price conscious. We expect the inflationary environment and the resulting pressures to continue in 2023. For a discussion of the risks relating to unfavorable economic conditions and inflation to our business, refer to Item 1A Risk Factors.

2023 Connection Trends

In our Consumer segment, we expect to continue to attract new customers and maintain high-quality retail postpaid customers, capitalizing on demand for data services and providing our customers new ways of using wireless services in their daily lives. We expect that future connection growth will be driven by FWA, as well as smartphones, tablets and other connected devices such as wearables. We believe the combination of our wireless network performance and service offerings provides a superior

customer experience, supporting increased penetration of data services and the continued attraction and retention of higher valued retail postpaid connections. For our prepaid business, we expect to continue to operate in a competitive environment while making investments to achieve long-term success in this space.

We expect to continue to grow our Fios internet connections as we seek to increase our penetration rates within our Fios service areas, further supported by the demand for higher speed internet connections. At the same time, we expect accelerating fixed wireless access connections to complement strong Fios results as demand for broadband services continues to grow. In Fios video, the business continues to face ongoing pressure as observed throughout the linear television market. Given these trends, we provide standalone Fios internet plans in addition to the choice of Fios video or a streaming service provided by a partner. We have experienced continuing access line and DSL losses as customers have disconnected both primary and secondary lines and switched to alternative technologies such as wireless, Voice over Internet Protocol, and cable for voice and data services.

In our Business segment, we offer wireless products and services to business and government customers across the U.S. We continue to grow our retail connections while operating in a highly competitive environment. We expect that this connection growth, combined with our industry-leading network assets, will provide additional opportunities to sell solutions, such as those around security, advanced communications and professional services. We also expect to expand our existing services offered to business customers through our Intelligent Edge Network, our multi-use platform.

In addition, in both our Consumer and our Business segments, we expect to support connection growth in part by adding capacity and further expanding our wireless coverage, and by continuing the build-out of our 5G network.

2023 Operating Revenue Trends

In our Consumer segment, we expect to see a continuation of service revenue growth in 2023 as customers shift to higher access plans with additional services and increase the number of devices they connect with our networks and services. We expect continued growth in wireless service revenue, driven by migrations to higher priced plans, increases in fixed wireless access connections, and the pricing actions implemented in the second quarter of 2022. We expect Fios revenue to benefit in 2023 as growth in our broadband customer base and increases in demand for higher speed internet connections offset the impact of the shift from the bundling of wireline services to standalone Internet service.

In our Business segment, we expect wireless revenue to expand, driven by growth from increases in wireless volumes and fixed wireless access contributions. We expect that Fios, through increased penetration, will also contribute to revenue growth and legacy traditional wireline services will continue to face secular pressures.

2023 Operating Expense and Cash Flow from Operations Trends

We expect our consolidated operating income margin and adjusted consolidated EBITDA margin to remain strong as we continue to drive revenue growth and undertake initiatives to reduce our overall cost structure. We expect that our ability to generate cash flows will benefit from our anticipated reduction in capital expenditures as we reach the end of our dedicated C-Band spending program, as well as our actions to better monetize our assets which increase our return on invested capital. We created the Verizon Global Services group, an organization that is expected to drive efficiencies by centralizing key functions and strategically delivering key services across Verizon.

We create value for our shareholders by investing the cash flows generated by our business in opportunities and transactions that support continued profitable growth, thereby increasing customer satisfaction and usage of our products and services. In addition, we have used our cash flows to maintain and grow our dividend payout to shareholders. Our Board of Directors increased the quarterly dividend by 2.0% during 2022, making this the sixteenth consecutive year in which we have raised our dividend.

Our goal is to use our cash to create long-term value for our shareholders. We will continue to look for investment opportunities that will help us to grow the business, strengthen our balance sheet, acquire spectrum licenses (see "Cash Flows from Investing Activities"), pay dividends to our shareholders and, when appropriate, buy back shares of our outstanding common stock (see "Cash Flows from Financing Activities").

Liquidity and Capital Resources

We use the net cash generated from our operations to fund expansion and modernization of our networks, service and repay external financing, pay dividends, invest in new businesses and spectrum and, when appropriate, buy back shares of our outstanding common stock. Our sources of funds, primarily from operations and, to the extent necessary, from external financing arrangements, are sufficient to meet ongoing operating and investing requirements over the next 12 months and beyond.

Our cash and cash equivalents balance is \$2.6 billion as of December 31, 2022. Our cash and cash equivalents are held both domestically and internationally, and are invested to maintain principal and provide liquidity. See "Market Risk" for additional information regarding our foreign currency risk management strategies.

We expect that our capital spending requirements will continue to be financed primarily through internally generated funds. Debt or equity financing may be needed to fund additional investments or development activities, or to maintain an appropriate capital structure to ensure our financial flexibility. Our external financing arrangements include credit facilities and other bank lines of credit, an active commercial paper program, vendor financing arrangements, issuances of registered debt or equity securities, U.S. retail medium-term notes and other securities that are privately-placed or offered overseas. In addition, we monetize certain receivables through asset-backed debt transactions.

Capital Expenditures

Our 2023 capital program includes capital to fund advanced networks and services, including expanding our core networks, adding capacity and density to our 5G network in order to stay ahead of our customers' increasing data demands and deploying C-Band, transforming our structure to deploy the Intelligent Edge Network while reducing the cost to deliver services to our customers, and pursuing other opportunities to drive operating efficiencies. We expect that our network architecture will simplify operations by eliminating legacy network elements, improve our 4G LTE coverage, speed the deployment of 5G technology, and create new enterprise opportunities in the business market. We anticipate cash requirements for our 2023 capital program to be between \$18.25 billion and \$19.25 billion, inclusive of the final \$1.75 billion of the incremental \$10 billion related to our accelerated C-Band deployment program.

Contractual Obligations and Commitments

We have various contractual obligations and commitments. The following represent our anticipated material cash requirements from known contractual and other obligations as of December 31, 2022:

- Long-term debt, including current maturities, commitments of \$149.4 billion, of which \$9.3 billion are expected to be due within the next twelve months. Related interest payments are \$73.6 billion, of which \$5.6 billion, are expected to be due within the next twelve months. Items included in long-term debt with variable coupon rates exclude unamortized debt issuance costs, and are described in Note 7 to the consolidated financial statements.
- Operating lease obligations of \$29.8 billion and Finance lease obligations of \$1.8 billion, of which \$4.6 billion and \$586 million, respectively, are expected to be due within the next twelve months. In addition, Verizon has an obligation of \$675 million representing future minimum payments under the sublease arrangement for our cell towers, of which \$297 million is expected to be due within the next twelve months. See Note 6 to the consolidated financial statements for additional information.
- Unconditional purchase obligations, with terms in excess of one year, amount to \$24.5 billion, of which \$9.4 billion is expected to be due within the next twelve months. Items included in unconditional purchase obligations are primarily commitments to purchase network equipment, software and services, content, marketing services and other items which will be used or sold in the ordinary course of business. These amounts do not represent our entire anticipated purchases in the future, but represent only those items that are the subject of contractual obligations. We also purchase products and services as needed with no firm commitment. See Note 16 to the consolidated financial statements for additional information.
- Estimated commitments for our allocable share of clearing costs incurred by, and incentive payments due to, the incumbent license holders associated with C-Band wireless spectrum acquired under Auction 107. The remaining commitment is estimated to be \$4.8 billion, all of which is expected to be due within the next twelve months.
- Other long-term liabilities, including current maturities, of \$4.1 billion, of which \$827 million is expected to be due within the next twelve months. Other long-term liabilities represent estimated postretirement benefit and qualified pension plan contributions. Qualified pension plan contributions include estimated minimum funding contributions. We expect that there will be no required pension funding through 2026, subject to changes in market conditions. Postretirement benefit payments include future postretirement benefit payments. These estimated amounts: (1) are subject to change based on changes to assumptions and future plan performance, which could impact the timing and/or amounts of these payments; and (2) exclude expectations beyond 5 years due to uncertainty of the timing and amounts.
- We are not able to make a reasonable estimate of when the unrecognized tax benefits balance of \$2.8 billion and related interest and penalties will be settled with the respective taxing authorities until the related tax audits are further developed or resolved. See Note 12 to the consolidated financial statements for additional information.

Consolidated Financial Condition

		(dollars in millions)	
Years Ended December 31,		2022	2021
Cash Flows Provided By (Used In)			
Operating activities	\$	37,141	\$ 39,539
Investing activities		(28,662)	(67,153)
Financing activities		(8,529)	8,277
Decrease in cash, cash equivalents and restricted cash	\$	(50)	\$ (19,337)

Cash Flows Provided By Operating Activities

Our primary source of funds continues to be cash generated from operations. Net cash provided by operating activities decreased by \$2.4 billion during 2022 compared to 2021, primarily driven by higher device payment receivables as our device payment portfolio increased over the comparative period, which was partially due to the change in device payment plan terms to 36 months. This decrease was further driven by a decrease in earnings. We expect to make approximately \$200 million of discretionary pension contributions in 2023. Additionally, we expect that there will be no required pension funding through 2026, subject to changes in market conditions.

Cash Flows Used In Investing Activities

Capital Expenditures

Capital expenditures continue to relate primarily to the use of capital resources to increase the operating efficiency and productivity of our networks, maintain our existing infrastructure, facilitate the introduction of new products and services and enhance responsiveness to competitive challenges.

Capital expenditures, including capitalized software, were \$23.1 billion and \$20.3 billion for 2022 and 2021, respectively. Capital expenditures increased approximately \$2.8 billion during 2022, compared to 2021, primarily due to accelerating our 5G technology deployment. See "Global Network and Technology" for more details.

Acquisitions of Wireless Licenses

In February 2021, the Federal Communications Commission (FCC) completed an auction, Auction 107, for mid-band spectrum known as C-Band. During 2021, we paid approximately \$45.9 billion for spectrum licenses in connection with this auction and related costs, of which \$1.3 billion was primarily paid for certain obligations related to projected clearing costs. During 2022, we made additional payments of \$1.6 billion for obligations related to accelerated clearing incentives and clearing costs associated with the auction.

In March 2022, Verizon signed agreements with satellite operators in which operators agreed to clear C-Band spectrum in certain markets and frequencies ahead of the previously expected December 2023 timeframe. During 2022, Verizon made payments of approximately \$310 million associated with these agreements.

During 2022 and 2021, we recorded capitalized interest related to wireless licenses of \$1.7 billion and \$1.6 billion, respectively.

During 2022 and 2021, we entered into and completed various other wireless license acquisitions for cash consideration of an insignificant amount and \$95 million, respectively.

Cash Received (Paid) Related to Acquisitions of Businesses, Net of Cash Acquired

In September 2020, we entered into a purchase agreement to acquire TracFone, a provider of prepaid and value mobile services in the U.S. The transaction closed on November 23, 2021 (the Acquisition Date). The aggregate cash consideration paid by Verizon at the closing of the transaction was approximately \$3.6 billion, net of cash acquired, approximately 57.6 million shares of our common stock valued at approximately \$3.0 billion, and up to an additional \$650 million in future cash contingent consideration. During 2022, Verizon received net cash proceeds of \$248 million for the final settlement of working capital, which was included in our consideration as of the Acquisition Date. See Note 3 to the consolidated financial statements for additional information.

In October 2020, we entered into a definitive agreement to acquire certain assets of Bluegrass Cellular (Bluegrass), a rural wireless operator serving central Kentucky. The transaction closed in March 2021. The aggregate cash consideration paid by Verizon at the closing of the transaction was approximately \$412 million, net of cash acquired.

During 2022 and 2021, we completed various other acquisitions for cash consideration of an insignificant amount and approximately \$51 million, respectively.

See "Acquisitions and Divestitures" for information on our acquisitions.

Collateral Payments Related to Derivative Contracts, Net of Repayments

During 2022 and 2021, we made payments of \$2.3 billion and an insignificant amount, respectively, of derivative collateral, net of repayments. See Note 9 to the consolidated financial statements for additional information.

Proceeds from Disposition of Business

During 2021, we received cash proceeds of \$4.1 billion, net of cash transferred, in connection with the sale of Verizon Media. See Note 3 to the consolidated financial statements for additional information.

Other, Net

During 2021, we received cash of \$321 million in connection with the settlement of a note receivable related to TracFone and net cash proceeds of \$98 million in connection with the sale of an investment.

Cash Flows Provided By (Used In) Financing Activities

We seek to maintain a mix of fixed and variable rate debt to lower borrowing costs within reasonable risk parameters and to protect against earnings and cash flow volatility resulting from changes in market conditions. During 2022, net cash used in financing activities was \$8.5 billion. During 2021, net cash provided by financing activities was \$8.3 billion.

2022

During 2022, our net cash used in financing activities of \$8.5 billion was primarily driven by \$13.6 billion used for repayments, redemptions and repurchases of long-term borrowings (secured and unsecured) as well as finance lease obligations, \$10.8 billion used for dividend payments and \$2.1 billion used for other financing activities. These cash flows used in financing activities were partially offset by \$17.8 billion provided by proceeds from long-term borrowings, which included \$10.7 billion of proceeds from our asset-backed debt transactions.

Proceeds from and Repayments, Redemptions, and Repurchases of Long-Term Borrowings

At December 31, 2022, our total debt decreased to \$150.6 billion compared to \$150.9 billion at December 31, 2021. Our effective interest rate was 3.7% and 3.6% during the years ended December 31, 2022 and 2021, respectively. We have entered into interest rate swaps to achieve a targeted mix of fixed and variable rate debt, managing our exposure to changes in interest rates. See also "Market Risk" and Note 7 to the consolidated financial statements for additional information.

At December 31, 2022, approximately \$34.0 billion, or 22.5%, of the aggregate principal amount of our total debt portfolio consisted of foreign denominated debt, primarily Euro and British Pound Sterling. We have entered into cross currency swaps on our foreign denominated debt in order to fix our future interest and principal payments in U.S. dollars and mitigate the impact of foreign currency transaction gains or losses. See "Market Risk" for additional information.

Verizon may acquire debt securities issued by Verizon and its affiliates through open market purchases, redemptions, privately negotiated transactions, tender offers, exchange offers, or otherwise, upon such terms and at such prices as Verizon may from time to time determine, for cash or other consideration.

Other, Net

Other, net financing activities during 2022 includes the cash consideration payments to acquire additional interests in certain controlled wireless partnerships and early debt redemption costs. See Note 14 to the consolidated financial statements for additional information on noncontrolling interests. See Note 15 to the consolidated financial statements for additional information on the early debt redemption costs.

Dividends

The Board of Directors of the Company assesses the level of our dividend payments on a periodic basis taking into account such factors as long-term growth opportunities, internal cash requirements and the expectations of our shareholders. During the third quarter of 2022, our Board of Directors increased our quarterly dividend payment by 2.0% to \$0.6525 from \$0.6400 per share in the preceding quarter. This is the sixteenth consecutive year that Company's Board of Directors has approved a quarterly dividend increase.

As in prior periods, dividend payments were a significant use of capital resources. During 2022, we paid \$10.8 billion in dividends.

2021

During 2021, our net cash provided by financing activities of \$8.3 billion was primarily driven by \$41.4 billion provided by proceeds from long-term borrowings, which included \$8.4 billion of proceeds from our asset-backed debt transactions. These cash flows provided by financing activities were partially offset by \$18.9 billion used for repayments, redemptions and repurchases of long-term borrowings (secured and unsecured) as well as finance lease obligations, \$10.4 billion used for dividend payments and \$3.8 billion used for other financing activities.

Proceeds from and Repayments, Redemptions, and Repurchases of Long-Term Borrowings

At December 31, 2021, our total debt was \$150.9 billion, and during the year ended December 31, 2021, our effective interest rate was 3.6%. The substantial majority of our total debt portfolio consisted of fixed rate indebtedness, therefore, changes in interest rates did not have a material effect on our interest payments. See "Market Risk" and Note 7 to the consolidated financial statements for additional information.

At December 31, 2021, approximately \$33.5 billion, or 22.2%, of the aggregate principal amount of our total debt portfolio consisted of foreign denominated debt, primarily Euro and British Pound Sterling. We have entered into cross currency swaps on our foreign denominated debt in order to fix our future interest and principal payments in U.S. dollars and mitigate the impact of foreign currency transaction gains or losses. See "Market Risk" for additional information.

Other, Net

Other, net financing activities during 2021 includes \$320 million in payments related to vendor financing arrangements, \$161 million in postings of derivative collateral and early debt redemption costs. See Note 15 to the consolidated financial statements for additional information on the early debt redemption costs.

Dividends

During the third quarter of 2021, our Board of Directors increased our quarterly dividend payment by 2.0% to \$0.6400 per share.

As in prior periods, dividend payments were a significant use of capital resources. During 2021, we paid \$10.4 billion in dividends.

Asset-Backed Debt

As of December 31, 2022, the carrying value of our asset-backed debt was \$20.0 billion. Our asset-backed debt includes Asset-Backed Notes (ABS Notes) issued to third-party investors (Investors) and loans (ABS Financing Facilities) received from banks and their conduit facilities (collectively, the Banks). Our consolidated asset-backed debt bankruptcy remote legal entities (each, an ABS Entity or collectively, the ABS Entities) issue the debt or are otherwise party to the transaction documentation in connection with our asset-backed debt transactions. Under the terms of our asset-backed debt, Celco Partnership (Celco), a wholly-owned subsidiary of the Company, and certain other Company affiliates (collectively, the Originators) transfer device payment plan agreement receivables and certain other receivables (collectively referred to as certain receivables) to one of the ABS Entities, which in turn transfers such receivables to another ABS Entity that issues the debt. Verizon entities retain the equity interests and residual interests, as applicable, in the ABS Entities, which represent the rights to all funds not needed to make required payments on the asset-backed debt and other related payments and expenses.

Our asset-backed debt is secured by the transferred receivables and future collections on such receivables. These receivables transferred to the ABS Entities and related assets, consisting primarily of restricted cash, will only be available for payment of asset-backed debt and expenses related thereto, payments to the Originators in respect of additional transfers of certain receivables, and other obligations arising from our asset-backed debt transactions, and will not be available to pay other obligations or claims of Verizon's creditors until the associated asset-backed debt and other obligations are satisfied. The Investors or Banks, as applicable, which hold our asset-backed debt have legal recourse to the assets securing the debt, but do not have any recourse to Verizon with respect to the payment of principal and interest on the debt. Under a parent support agreement, the Company has agreed to guarantee certain of the payment obligations of Celco and the Originators to the ABS Entities.

Cash collections on the receivables collateralizing our asset-backed debt securities are required at certain specified times to be placed into segregated accounts. Deposits to the segregated accounts are considered restricted cash and are included in Prepaid expenses and other and Other assets in our consolidated balance sheets.

Proceeds from our asset-backed debt transactions are reflected in Cash flows from financing activities in our consolidated statements of cash flows. The asset-backed debt issued is included in Debt maturing within one year and Long-term debt in our consolidated balance sheets.

See Note 7 to the consolidated financial statements for additional information.

Long-Term Credit Facilities

(dollars in millions)	Maturities	Facility Capacity	At December 31, 2022		
			Unused Capacity	Principal Amount Outstanding	
Verizon revolving credit facility ⁽¹⁾	2026	\$ 9,500	\$ 9,436	\$	—
Various export credit facilities ⁽²⁾	2024 - 2031	11,000	1,000		6,735
Total		\$ 20,500	\$ 10,436	\$	6,735

⁽¹⁾ The revolving credit facility does not require us to comply with financial covenants or maintain specified credit ratings, and it permits us to borrow even if our business has incurred a material adverse change. The revolving credit facility provides for the issuance of letters of credit. As of December 31, 2022, there have been no drawings against the \$9.5 billion revolving credit facility since its inception.

⁽²⁾ During 2022 and 2021, we drew down \$3.0 billion and \$1.0 billion, respectively, from these facilities. Borrowings under certain of these facilities are amortized semi-annually in equal installments up to the applicable maturity dates. Maturities reflect

maturity dates of principal amounts outstanding. Any amounts borrowed under these facilities and subsequently repaid cannot be reborrowed.

Common Stock

Common stock has been used from time to time to satisfy some of the funding requirements of employee and shareholder plans. During both the years ended December 31, 2022 and 2021, we issued 2.1 million shares of common stock from treasury stock, which had aggregate values of \$91 million.

In connection with our acquisition of TracFone in November 2021, we issued approximately 57.6 million shares of common stock from treasury stock valued at approximately \$3.0 billion. See Note 3 to the consolidated financial statements for additional information.

In February 2020, the Board of Directors of the Company authorized a share buyback program to repurchase up to 100 million shares of our common stock. The program will terminate when the aggregate number of shares purchased reaches 100 million, or a new share repurchase plan superseding the current plan is authorized, whichever is sooner. The program permits Verizon to repurchase shares over time, with the amount and timing of repurchases depending on market conditions and corporate needs. There were no repurchases of common stock during 2022 and 2021 under our authorized share buyback program.

Credit Ratings

Verizon's credit ratings did not change in 2022 or 2021.

Securities ratings assigned by rating organizations are expressions of opinion and are not recommendations to buy, sell or hold securities. A securities rating is subject to revision or withdrawal at any time by the assigning rating organization. Each rating should be evaluated independently of any other rating.

Covenants

Our credit agreements contain covenants that are typical for large, investment grade companies. These covenants include requirements to pay interest and principal in a timely fashion, pay taxes, maintain insurance with responsible and reputable insurance companies, preserve our corporate existence, keep appropriate books and records of financial transactions, maintain our properties, provide financial and other reports to our lenders, limit pledging and disposition of assets and mergers and consolidations, and other similar covenants.

We and our consolidated subsidiaries are in compliance with all of our restrictive covenants in our debt agreements.

Change In Cash, Cash Equivalents and Restricted Cash

Our Cash and cash equivalents at December 31, 2022 totaled \$2.6 billion, a \$316 million decrease compared to December 31, 2021, primarily as a result of the factors discussed above.

Restricted cash at December 31, 2022 totaled \$1.5 billion, a \$266 million increase compared to restricted cash at December 31, 2021, primarily related to cash collections on certain receivables that are required at certain specified times to be placed into segregated accounts. The increase was primarily driven by the completion of asset-backed debt transactions during the year ended December 31, 2022.

Free Cash Flow

Free cash flow is a non-GAAP financial measure that reflects an additional way of viewing our liquidity that, we believe, when viewed with our GAAP results, provides management, investors and other users of our financial information with a more complete understanding of factors and trends affecting our cash flows. Free cash flow is calculated by subtracting capital expenditures (including capitalized software) from net cash provided by operating activities. We believe it is a more conservative measure of cash flow since capital expenditures are necessary for ongoing operations. Free cash flow has limitations due to the fact that it does not represent the residual cash flow available for discretionary expenditures. For example, free cash flow does not incorporate payments made on finance lease obligations or cash payments for business acquisitions or wireless licenses. Therefore, we believe it is important to view free cash flow as a complement to our entire consolidated statements of cash flows.

The following table reconciles net cash provided by operating activities to free cash flow:

		(dollars in millions)	
Years Ended December 31,		2022	2021
Net cash provided by operating activities	\$	37,141	\$ 39,539
Less Capital expenditures (including capitalized software)		23,087	20,286
Free cash flow	\$	14,054	\$ 19,253

The decrease in free cash flow during 2022 is a reflection of the decrease in operating cash flows, as well as the increase in capital expenditures, both of which are discussed above.

Employee Benefit Plans Funded Status and Contributions

Employer Contributions

We operate numerous qualified and nonqualified pension plans and other postretirement benefit plans. These plans primarily relate to our domestic business units. We made no discretionary contribution to our qualified pension plan in either 2022 or 2021. During 2022 and 2021, we made contributions of \$53 million and \$58 million to our nonqualified pension plans, respectively.

Our overall investment strategy is to achieve a mix of assets that allows us to meet projected benefit payments while taking into consideration risk and return. In an effort to reduce the risk of our portfolio strategy and better align assets with liabilities, we have adopted a liability driven pension strategy that seeks to better match the interest rate sensitivity of the liability hedging assets with the interest rate sensitivity of the liability. We expect that the strategy will reduce the likelihood that assets will decline at a time when liabilities increase (referred to as liability hedging), with the goal to reduce the risk of underfunding to the plan and its participants and beneficiaries. Over time, as the asset allocation shifts to more liability hedging assets, this strategy will generally result in lower expected asset returns. In 2023, we expect discretionary qualified pension plan contributions to be approximately \$200 million and nonqualified pension plan contributions to be insignificant.

Contributions to our other postretirement benefit plans generally relate to payments for benefits on an as-incurred basis since these other postretirement benefit plans do not have funding requirements similar to the pension plans. We contributed \$692 million and \$885 million to our other postretirement benefit plans in 2022 and 2021, respectively. Contributions to our other postretirement benefit plans are estimated to be approximately \$820 million in 2023.

Leasing Arrangements

See Note 6 to the consolidated financial statements for additional information related to leasing arrangements.

Guarantees

We guarantee the debentures of our operating telephone company subsidiaries. See Note 7 to the consolidated financial statements for additional information.

In connection with the execution of agreements for the sale of businesses and investments, Verizon ordinarily provides representations and warranties to the purchasers pertaining to a variety of nonfinancial matters, such as ownership of the securities being sold, as well as financial losses. See Note 16 to the consolidated financial statements for additional information.

As of December 31, 2022, letters of credit totaling approximately \$834 million, which were executed in the normal course of business and support several financing arrangements and payment obligations to third parties, were outstanding. See Note 16 to the consolidated financial statements for additional information.

Other Future Obligations

During 2022, Verizon entered into four renewable energy purchase agreements (REPAs) with third parties, in addition to 20 signed in previous years. See Note 16 to the consolidated financial statements for additional information. Under the REPAs, we plan to purchase up to an aggregate of approximately 3.0 gigawatts of capacity across multiple states, including Arizona, Illinois, Indiana, Iowa, Maryland, Nebraska, New York, North Carolina, Ohio, Oklahoma, Pennsylvania, Texas and West Virginia.

Critical Accounting Estimates and Recently Issued Accounting Standards

Critical Accounting Estimates

A summary of the critical accounting estimates used in preparing our financial statements are as follows:

Wireless Licenses and Goodwill

Wireless licenses and Goodwill are a significant component of our consolidated assets. Both our wireless licenses and goodwill are treated as indefinite-lived intangible assets and, therefore are not amortized, but rather are tested for impairment annually in the fourth fiscal quarter, unless there are events requiring an earlier assessment or changes in circumstances during an interim period suggesting impairment indicators are present. We believe our estimates and assumptions are reasonable and represent appropriate marketplace considerations as of the valuation date. Although we use consistent methodologies in developing the assumptions and estimates underlying the fair value calculations used in our impairment tests, these estimates and assumptions are uncertain by nature, may change over time and can vary from actual results. It is possible that in the future there may be changes in our estimates and assumptions, including the timing and amount of future cash flows, margins, growth rates, market participant assumptions, comparable benchmark companies and related multiples and discount rates, which could result in different fair value estimates. Significant and adverse changes to any one or more of the above-noted estimates and assumptions could result in an impairment to our wireless licenses and goodwill impairment for one or more of our reporting units.

Wireless Licenses

The carrying value of our wireless licenses was approximately \$149.8 billion as of December 31, 2022. We aggregate our wireless licenses into one single unit of accounting, as we utilize our wireless licenses on an integrated basis as part of our nationwide wireless network. Our wireless licenses provide us with the exclusive right to utilize certain radio frequency spectrum to provide wireless communication services. There are currently no legal, regulatory, contractual, competitive, economic or other factors that limit the useful life of our wireless licenses.

We test our wireless licenses for potential impairment annually or more frequently if impairment indicators are present. We have the option to first perform a qualitative assessment to determine whether it is necessary to perform a quantitative impairment test. However, we may elect to bypass the qualitative assessment in any period and proceed directly to performing the quantitative impairment test. It is our policy to perform quantitative impairment assessment at least every three years.

During the fourth quarter of 2022, we performed a qualitative impairment assessment as our annual impairment test to determine whether it is more likely than not that the fair value of our wireless licenses was less than the carrying amount. As part of our qualitative assessment we considered several factors including the business enterprise value of our combined wireless business, macroeconomic conditions (including changes in interest rates and discount rates), industry and market considerations (including industry revenue and EBITDA margin results, projections and recent merger and acquisition activity), the recent and projected financial performance of our combined wireless business as a whole, as well as other factors including the result of our last quantitative assessment. Our annual impairment test in 2022 indicated that it is more likely than not that the fair value of our wireless licenses remained above their carrying value and, therefore, did not result in an impairment.

During the fourth quarter of 2021, we performed a quantitative impairment assessment, which indicated that the fair value of our wireless licenses is substantially in excess of their carrying value and, therefore, did not result in an impairment. Our quantitative impairment assessment consisted of comparing the estimated fair value of our aggregate wireless licenses to the aggregated carrying amount as of the test date. Under our quantitative assessment, we estimated the fair value of our wireless licenses using the Greenfield approach. The Greenfield approach is an income based valuation approach that values the wireless licenses by calculating the cash flow generating potential of a hypothetical start-up company that goes into business with no assets except the wireless licenses to be valued. A discounted cash flow analysis is used to estimate what a marketplace participant would be willing to pay to purchase the aggregated wireless licenses as of the valuation date. As a result, we were required to make significant estimates about future cash flows and profitability specifically associated with our wireless licenses, an appropriate discount rate based on the risk associated with those estimated cash flows and assumed terminal value and growth rates. We considered current and expected future economic conditions, current and expected availability of wireless network technology and infrastructure and related equipment and the costs thereof as well as other relevant factors in estimating future cash flows and profitability. The discount rate represented our estimate of the weighted-average cost of capital (WACC), or expected return, that a marketplace participant would have required as of the valuation date. We developed the discount rate based on our consideration of the cost of debt and equity of a group of guideline companies as of the valuation date. Accordingly, our discount rate incorporated our estimate of the expected return a marketplace participant would have required as of the valuation date, including the risk premium associated with the current and expected economic conditions as of the valuation date. The terminal value growth rate represented our estimate of the marketplace's long-term growth rate.

Goodwill

At December 31, 2022, the balance of our goodwill was approximately \$28.7 billion, of which \$21.1 billion was in our Consumer reporting unit and \$7.5 billion was in our Business reporting unit.

To determine if goodwill is potentially impaired, we have the option to perform a qualitative assessment to determine whether it is more likely than not that the fair value of a reporting unit is less than its carrying value. If we elect not to conduct the qualitative assessment or if indications of a potential impairment exist, the determination of whether an impairment has occurred requires the fair value of each reporting unit to be assessed. It is our policy to perform quantitative impairment assessments at least every three years.

Under the qualitative assessment, we consider several factors, including the business enterprise value of the reporting unit from the last quantitative test and the excess of fair value over carrying value from this test, macroeconomic conditions (including changes in interest rates and discount rates), industry and market considerations (including industry revenue and EBITDA margin results, projections and recent merger and acquisition activity), the recent and projected financial performance of the reporting unit, as well as other factors.

Under the quantitative assessment, the fair value of the reporting unit is calculated using a market approach and a discounted cash flow method, as a form of the income approach. The market approach includes the use of comparative multiples to complement discounted cash flow results. The discounted cash flow method is based on the present value of two components-projected cash flows and a terminal value. The terminal value represents the expected normalized future cash flows of the reporting unit beyond the cash flows from the discrete projection period. The fair value of the reporting unit is calculated based on the sum of the present value of the cash flows from the discrete period and the present value of the terminal value. The discount rate represented our estimate of the WACC, or expected return, that a marketplace participant would have required as of the valuation date. The application of our goodwill impairment test required key assumptions underlying our valuation model.

The discounted cash flow analysis factored in assumptions on discount rates and terminal growth rates to reflect risk profiles of key strategic revenue and cost initiatives, as well as revenue and EBITDA growth relative to history and market trends and expectations. The market multiples approach incorporated significant judgment involved in the selection of comparable public company multiples and benchmarks. The selection of companies and multiples was influenced by differences in growth and profitability, and volatility in market prices of peer companies. These valuation inputs are inherently judgmental, and an adverse change in one or a combination of these inputs could trigger a goodwill impairment loss in the future.

During the fourth quarter of 2022, we performed a qualitative impairment assessment for our Consumer reporting unit. Our qualitative assessment indicated that it was more likely than not that the fair value of our Consumer reporting unit exceeded its carrying value and, therefore, did not result in an impairment.

During the fourth quarter of 2022, we performed a quantitative impairment assessment for our Business reporting unit given it has continued to experience increased market pressures resulting in lower than expected revenues and earnings that may continue to persist over the near term. At the goodwill impairment measurement date of October 31, 2022, our Business reporting unit had a fair value that exceeded its carrying amount by approximately 8% and, therefore, did not result in an impairment. We do not anticipate reasonable changes in significant estimates to change the outcome of the quantitative impairment assessment. For instance, if either the terminal value growth rate declined by 50 basis points, or if the discount rate increased by 50 basis points, the fair value of our Business reporting unit would still exceed its carrying value. However, as a result of our 2022 goodwill assessment, management believes there is an increasing risk that our Business reporting unit may be required to recognize an impairment charge in the future. See Note 4 to the consolidated financial statements for additional information.

A projected sustained decline in a reporting unit's revenues and earnings could have a significant negative impact on its fair value and may result in impairment charges. Such a decline could be driven by, among other things: (1) decreases in sales volumes or long-term growth rate as a result of competitive pressures or other factors; or (2) the inability to achieve or delays in achieving the goals in strategic initiatives. Also, adverse changes to macroeconomic factors, such as increases in long-term interest rates, would also negatively impact the fair value of the reporting unit.

At December 31, 2021, the balance of our goodwill was approximately \$28.6 billion, of which \$21.0 billion was in our Consumer reporting unit and \$7.5 billion was in our Business reporting unit. We performed quantitative impairment assessments for both our Consumer and Business reporting units during the fourth quarter of 2021. At the goodwill impairment measurement date of October 31, 2021, our quantitative assessments indicated that the fair values for our Consumer and Business reporting units were substantially in excess of their carrying values and, therefore, did not result in impairment.

Pension and Other Postretirement Benefit Plans

We maintain benefit plans for most of our employees, including, for certain employees, pension and other postretirement benefit plans. Benefit plan assumptions, including the discount rate used, the long-term rate of return on plan assets, the determination of the substantive plan and health care trend rates are periodically updated and impact the amount of benefit plan income, expense, assets and obligations. Changes to one or more of these assumptions could significantly impact our accounting for pension and other postretirement benefits.

In determining pension and other postretirement obligations, the weighted-average discount rate was selected to approximate the composite interest rates available on a selection of high-quality bonds available in the market at December 31, 2022. The bonds selected had maturities that coincided with the time periods during which benefit payments are expected to occur, were non-callable and available in sufficient quantities to ensure marketability (at least \$300 million par outstanding). Bond yields are subject to uncertainty for a number of reasons including corporate performance, credit rating downgrades and upgrades, government fiscal policy decisions, and general market volatility. The expected long-term rates of return on plan assets used in determining Verizon's pension and other postretirement obligations are based on expectations for future investment returns for the plans' asset allocation. The rates are subject to uncertainty for a number of reasons including corporate performance, credit ratings, monetary policy, inflation, exchange rates, investor behavior and general market volatility.

A sensitivity analysis of the impact of changes in the discount rate and the long-term rate of return on plan assets on the benefit obligations and expense (income) recorded, as well as an increase or a decrease in the actual versus expected return on plan assets as of December 31, 2022 and for the year then ended pertaining to Verizon's pension and postretirement benefit plans, is provided in the table below. The amounts in the table below related to discount rate changes are gross impacts on benefit obligations and expense, and do not reflect changes in asset values as a result of interest rate changes, for which our pension plan is highly hedged.

(dollars in millions)	Percentage point change	Increase/(decrease) at December 31, 2022
Pension plans discount rate	+0.50 \$	(676)
	-0.50	740
Rate of return on pension plan assets	+1.00	(168)
	-1.00	168
Postretirement plans discount rate	+0.50	(507)
	-0.50	552
Rate of return on postretirement plan assets	+1.00	(5)
	-1.00	5

In addition to our liability hedging assets, we also employ an interest rate hedging strategy to further minimize the impact of discount rate changes on the funded ratio of the pension plan. While the target hedge ratio varies depending on the funded status of the plan and the level of interest rates, the target hedge ratio was greater than 80% at December 31, 2022, significantly limiting volatility.

The annual measurement date for both our pension and other postretirement benefits is December 31. We use the full yield curve approach to estimate the interest cost component of net periodic benefit cost for pension and other postretirement benefits. The full yield curve approach refines our estimate of interest cost by applying the individual spot rates from a yield curve composed of the rates of return on several hundred high-quality fixed income corporate bonds available at the measurement date. These individual spot rates align with the timing of each future cash outflow for benefit payments and therefore provide a more precise estimate of interest cost.

See Note 11 to the consolidated financial statements for additional information.

Income Taxes

Our current and deferred income taxes and associated valuation allowances are impacted by events and transactions arising in the normal course of business as well as in connection with the adoption of new accounting standards, changes in tax laws and rates, acquisitions and dispositions of businesses and non-recurring items. As a global commercial enterprise, our income tax rate and the classification of income taxes can be affected by many factors, including estimates of the timing and realization of deferred income tax assets and the timing and amount of income tax payments. We account for tax benefits taken or expected to be taken in our tax returns in accordance with the accounting standard relating to the uncertainty in income taxes, which requires the use of a two-step approach for recognizing and measuring tax benefits taken or expected to be taken in a tax return. We review and adjust our liability for unrecognized tax benefits based on our best judgment given the facts, circumstances and information available at each reporting date. To the extent that the final outcome of these tax positions is different than the amounts recorded, such differences may impact income tax expense and actual tax payments. We recognize any interest and penalties accrued related to unrecognized tax benefits in income tax expense. Actual tax payments may materially differ from estimated liabilities as a result of changes in tax laws as well as unanticipated transactions impacting related income tax balances. See Note 12 to the consolidated financial statements for additional information.

Property, Plant and Equipment

Our Property, plant and equipment balance represents a significant component of our consolidated assets. We record property, plant and equipment at cost. We depreciate property, plant and equipment on a straight-line basis over the estimated useful life of the assets. The estimated useful life is subject to change due to a variety of factors such as change in asset capacity or performance, technical obsolescence, market expectations and competition impacts. In connection with our ongoing review of the estimated useful lives of property, plant and equipment during 2022, we determined that the estimated useful life of our property, plant and equipment would remain unchanged. We expect that a one year increase in estimated useful lives of our property, plant and equipment would result in a decrease to our 2022 depreciation expense of \$2.2 billion and that a one year decrease would result in an increase of approximately \$3.5 billion in our 2022 depreciation expense.

Accounts Receivable

Accounts receivable are recorded at amortized cost less an allowance for credit losses that are not expected to be recovered. The gross amount of accounts receivable and corresponding allowance for credit losses are presented separately in the consolidated balance sheets. We maintain allowances for credit losses resulting from the expected failure or inability of our customers to make required payments. We recognize the allowance for credit losses at inception and reassess quarterly based on management's expectation of the asset's collectability. The allowance is based on multiple factors including historical experience with bad debts, the credit quality of the customer base, the aging of such receivables and current macroeconomic conditions, such as the COVID-19 pandemic, as well as management's expectations of conditions in the future, as applicable. The impact of these factors on the allowance involves significant level of estimation and is subject to uncertainty. Our allowance for credit losses is based on management's assessment of the collectability of assets pooled together with similar risk characteristics.

We record an allowance to reduce the receivables to the amount that is expected to be collectible. For device payment plan agreement receivables, we record bad debt expense based on a default and loss calculation using our proprietary loss model. The expected loss rate is determined based on customer credit scores and other qualitative factors as noted above. The loss rate is assigned individually on a customer by customer basis and the custom credit scores are then aggregated by vintage and used in our proprietary loss model to calculate the weighted-average loss rate used for determining the allowance balance. The weighted-average expected loss rate increased 0.44% at December 31, 2022 as compared to at December 31, 2021. We expect that an increase or decrease of 0.25% in the weighted-average loss rate would result in a change of \$111 million in the allowance.

We monitor the collectability of our wireless service receivables as one overall pool. Wireline service receivables are disaggregated and pooled by the following customer groups: consumer, small and medium business, global enterprise, public sector and wholesale. For wireless service receivables and wireline consumer and small and medium business receivables, the allowance is calculated based on a 12 month rolling average write-off balance multiplied by the average life-cycle of an account from billing to write-off. The risk of loss is assessed over the contractual life of the receivables and we adjust the historical loss amounts for current and future conditions based on management's qualitative considerations. For global enterprise, public sector and wholesale wireline receivables, the allowance for credit losses is based on historical write-off experience and individual customer credit risk, as applicable. We consider multiple factors in determining the allowance as discussed above.

If there is a deterioration of our customers' financial condition or if future actual default rates on receivables in general differ from those currently anticipated, we may have to adjust our allowance for credit losses, which would affect earnings in the period the adjustments are made. See Note 8 to the consolidated financial statements for additional information.

Acquisitions and Divestitures

Spectrum License Transactions

From time to time, we enter into agreements to buy, sell or exchange spectrum licenses. We believe these spectrum license transactions have allowed us to continue to enhance the reliability of our wireless network while also resulting in a more efficient use of spectrum.

In March 2020, the FCC's incentive auction, Auction 103, for spectrum licenses in the upper 37 GHz, 39 GHz, and 47 GHz bands concluded. Verizon participated in this incentive auction and was the high bidder on 4,940 licenses, which primarily consisted of 37 GHz and, to a lesser extent, 39 GHz spectrum. As an incumbent licensee, our 39 GHz licenses provided us with incentive payments that were applied towards the purchase price of spectrum in the auction. The value of the licenses won by Verizon amounted to \$3.4 billion, of which \$1.8 billion was settled with the relinquished 39 GHz licenses.

In February 2021, the FCC concluded Auction 107 for C-Band wireless spectrum. Verizon paid \$45.5 billion for the licenses it won, of which \$44.6 billion was paid in the first quarter of 2021. In accordance with the rules applicable to the auction, Verizon is required to make additional payments to acquire the licenses. The payments are for our allocable share of clearing costs incurred by, and incentive payments due to, the incumbent license holders associated with the auction, which are estimated to be \$7.7 billion. During 2021, we made payments of \$1.3 billion primarily related to certain obligations for projected clearing costs. During 2022, we made additional payments of \$1.6 billion for obligations related to accelerated clearing incentives and clearing costs. We expect to continue to make payments related to clearing cost and incentive payment obligations through 2024, which we expect to be \$4.8 billion. These payments are dependent on the incumbent license holders accelerated clearing of the spectrum for Verizon's use and, therefore, the final timing and amounts could differ based on the incumbent holders' execution of their clearing process. In accordance with the FCC order, the clearing must be completed by December 2025. The carrying value of the wireless spectrum won in Auction 107 consists of all payments required to participate and purchase licenses in the auction, including Verizon's allocable share of clearing costs incurred by, and incentive payments due to, the incumbent license holders associated with the auction that we are obligated to pay in order to acquire the licenses, as well as capitalized interest to the extent qualifying activities have occurred.

In March 2022, Verizon signed agreements with satellite operators in which operators agreed to clear C-Band spectrum in certain markets and frequencies ahead of the previously expected December 2023 timeframe. During 2022, Verizon incurred costs associated with these agreements of approximately \$340 million, of which \$310 million was paid as of the year ended December 31, 2022. This early clearance accelerated Verizon's access to more spectrum in a number of key markets to support its 5G technology deployment.

See Note 3 to the consolidated financial statements for additional information regarding our spectrum license transactions.

TracFone Wireless, Inc.

In September 2020, we entered into a purchase agreement (TracFone Purchase Agreement) with América Móvil to acquire TracFone, a leading provider of prepaid and value mobile services in the U.S. The transaction closed on November 23, 2021. In accordance with the terms of the TracFone Purchase Agreement, Verizon acquired all of TracFone's outstanding stock in exchange for approximately \$3.5 billion in cash, net of cash acquired and working capital and other adjustments, 57,596,544 shares of our common stock valued at approximately \$3.0 billion, and up to an additional \$650 million in future cash contingent

consideration related to the achievement of certain performance measures and other commercial arrangements. The fair value of the common stock was determined on the basis of its closing market price on the Acquisition Date. The estimated fair value of the contingent consideration as of the Acquisition Date was approximately \$560 million calculated using a probability-weighted discounted cash flow model and significant unobservable inputs, thus representing a Level 3 measurement. The contingent consideration payable is based on the achievement of certain revenue and operational targets, measured over a two-year earn out period, as defined in the TracFone Purchase Agreement. During 2022, Verizon made payments of \$188 million related to the contingent consideration, which is reflected in Cash flows from financing activities in our consolidated statement of cash flows for the twelve months ended December 31, 2022. Payments related to the contingent consideration are expected to continue through 2024. As of December 31, 2022, the fair value of the contingent consideration was \$317 million. During 2022, we recorded gains of \$57 million related to fair value adjustments for the contingent consideration within Selling, general and administrative expense in our consolidated statement of income. See Note 3 and Note 9 to the consolidated financial statements for additional information.

Bluegrass Cellular

In October 2020, we entered into a definitive agreement to acquire certain assets of Bluegrass, a rural wireless operator serving central Kentucky. The transaction closed in March 2021. The aggregate cash consideration paid by Verizon at the closing of the transaction was approximately \$412 million, net of cash acquired. See Note 3 to the consolidated financial statements for additional information.

Verizon Media Divestiture

On May 2, 2021, Verizon entered into a definitive agreement with an affiliate of Apollo Global Management Inc. (the Apollo Affiliate) pursuant to which we agreed to sell Verizon Media in return for consideration of \$4.3 billion in cash, subject to customary adjustments, \$750 million in non-convertible preferred limited partnership units of the Apollo Affiliate, and 10% of the fully-diluted common limited partnership units of the Apollo Affiliate.

On September 1, 2021, we completed the sale of Verizon Media. As of the close of the transaction, cash proceeds, the fair value of the non-convertible preferred limited partnership units of the Apollo Affiliate, and the fair value of 10% of the fully-diluted common limited partnership units of the Apollo Affiliate were \$4.3 billion, \$496 million, and \$124 million, respectively. We recorded a pre-tax gain on sale of approximately \$1.0 billion (after-tax \$1.0 billion) in Selling general and administrative expense in our consolidated statement of income for the year ended December 31, 2021. In addition, we incurred \$346 million of various costs associated with this disposition which are primarily recorded in Selling general and administrative expense in our consolidated statement of income for the year ended December 31, 2021. See Note 3 to the consolidated financial statements for additional information.

Other

From time to time, we enter into strategic agreements to acquire various other businesses and investments. See Note 3 to the consolidated financial statements for additional information.

In connection with a non-strategic divestiture and other activity, we recorded a pre-tax net gain of an insignificant amount in our consolidated statement of income for the year ended December 31, 2022.

In December 2021, we completed the sale of an investment. In connection with this transaction, we recorded a pre-tax gain of \$131 million in Equity in earnings (losses) of unconsolidated businesses in our consolidated statement of income for the year ended December 31, 2021.

Item 7A. Quantitative and Qualitative Disclosures About Market Risk

We are exposed to various types of market risk in the normal course of business, including the impact of interest rate changes, foreign currency exchange rate fluctuations, changes in investment, equity and commodity prices and changes in corporate tax rates. We employ risk management strategies, which may include the use of a variety of derivatives including cross currency swaps, forward starting interest rate swaps, interest rate swaps, interest rate caps, treasury rate locks and foreign exchange forwards. We do not hold derivatives for trading purposes.

It is our general policy to enter into interest rate, foreign currency and other derivative transactions only to the extent necessary to achieve our desired objectives in optimizing exposure to various market risks. Our objectives include maintaining a mix of fixed and variable rate debt to lower borrowing costs within reasonable risk parameters and to protect against earnings and cash flow volatility resulting from changes in market conditions. We do not hedge our market risk exposure in a manner that would completely eliminate the effect of changes in interest rates and foreign exchange rates on our earnings.

Counterparties to our derivative contracts are major financial institutions with whom we have negotiated derivatives agreements (ISDA master agreements) and credit support annex (CSA) agreements which provide rules for collateral exchange. The CSA agreements contain rating based thresholds such that we or our counterparties may be required to hold or post collateral based upon changes in outstanding positions as compared to established thresholds and changes in credit ratings. We do not offset fair value amounts recognized for derivative instruments and fair value amounts recognized for the right to reclaim cash collateral or

the obligation to return cash collateral arising from derivative instruments recognized at fair value. At December 31, 2022, we did not hold any collateral. At December 31, 2022, we posted \$2.3 billion of collateral related to derivative contracts under collateral exchange agreements, which was recorded as Prepaid expenses and other in our consolidated balance sheet. At December 31, 2021, we held and posted \$0.1 billion and an insignificant amount, respectively, of collateral related to derivative contracts under collateral exchange arrangements, which were recorded as Other current liabilities and Prepaid expenses and other, respectively, in our consolidated balance sheet. While we may be exposed to credit losses due to the nonperformance of our counterparties, we consider the risk remote and do not expect that any such nonperformance would result in a significant effect on our results of operations or financial condition due to our diversified pool of counterparties. See Note 9 to the consolidated financial statements for additional information regarding the derivative portfolio.

Interest Rate Risk

We are exposed to changes in interest rates, primarily on our short-term debt and the portion of long-term debt that carries floating interest rates. As of December 31, 2022, approximately 74% of the aggregate principal amount of our total debt portfolio consisted of fixed-rate indebtedness, including the effect of interest rate swap agreements designated as hedges. The impact of a 100-basis-point change in interest rates affecting our floating rate debt would result in a change in annual interest expense, including our interest rate swap agreements that are designated as hedges, of approximately \$420 million. The interest rates on our existing long-term debt obligations are unaffected by changes to our credit ratings.

Certain of our floating rate debt and certain of our interest rate derivative transactions utilize interest rates that are linked to the London Inter-Bank Offered Rate (LIBOR) as the benchmark rate. The U.S. dollar LIBOR rates utilized in these transactions will cease publication after June 30, 2023, in accordance with recent U.S. and international regulatory guidance for reform. The consequences of these developments cannot be entirely predicted but could include an increase in the cost of our floating rate debt or exposure under our interest rate derivative transactions. We do not anticipate a significant impact to our financial position given our current mix of variable and fixed-rate debt and taking into account the impact of our interest rate hedging. In 2021, we began utilizing interest rates that are linked to the Secured Overnight Financing Rate as the benchmark rate for various floating rate transactions.

The table that follows summarizes the fair values of our long-term debt, including current maturities, and interest rate swap derivatives as of December 31, 2022 and 2021. The table also provides a sensitivity analysis of the estimated fair values of these financial instruments assuming 100-basis-point upward and downward shifts in the yield curve. Our sensitivity analysis does not include the fair values of our commercial paper and bank loans, if any, because they are not significantly affected by changes in market interest rates.

		(dollars in millions)		
Long-term debt and related derivatives		Fair Value	Fair Value assuming + 100 basis point shift	Fair Value assuming - 100 basis point shift
At December 31, 2022	\$	143,648	\$	152,427
At December 31, 2021		169,179	156,078	184,496

Interest Rate Swaps

We enter into interest rate swaps to achieve a targeted mix of fixed and variable rate debt. We principally receive fixed rates and pay variable rates, resulting in a net increase or decrease to Interest expense. These swaps are designated as fair value hedges and hedge against interest rate risk exposure of designated debt issuances. At December 31, 2022, there was no amount related to the fair value of the asset and the fair value of the liability of these contracts was \$4.6 billion. At December 31, 2021, the fair value of the asset and liability of these contracts was \$473 million and \$666 million, respectively. The increase in the fair value of the liability of these contracts is primarily due to interest rate changes during the year ended December 31, 2022. At December 31, 2022 and 2021, the total notional amount of the interest rate swaps was \$26.1 billion and \$19.8 billion, respectively.

Forward Starting Interest Rate Swaps

We have entered into forward starting interest rate swaps designated as cash flow hedges in order to manage our exposure to interest rate changes on future forecasted transactions. At December 31, 2022, there was no amount related to the fair value of the asset or liability of these contracts. At December 31, 2021, there was no amount related to the fair value of the asset and the fair value of the liability of these contracts was \$302 million. At December 31, 2022, there was no outstanding notional amount for forward starting interest rate swaps. At December 31, 2021, the total notional amount of the forward starting interest rate swaps was \$1.0 billion.

Treasury Rate Locks

From time to time we enter into treasury rate locks to mitigate our interest rate risk. There was no outstanding notional amount for treasury rate locks at December 31, 2022 or 2021.

Foreign Currency Translation

The functional currency for our foreign operations is primarily the local currency. The translation of income statement and balance sheet amounts of our foreign operations into U.S. dollars is recorded as cumulative translation adjustments, which are included in Accumulated other comprehensive loss in our consolidated balance sheets. Gains and losses on foreign currency transactions are recorded in the consolidated statements of income. At December 31, 2022, our primary translation exposure was to the British Pound Sterling, Euro and Australian Dollar.

Cross Currency Swaps

We have entered into cross currency swaps to exchange our British Pound Sterling, Euro, Swiss Franc, Canadian Dollar and Australian Dollar-denominated cash flows into U.S. dollars and to fix our cash payments in U.S. dollars, as well as to mitigate the impact of foreign currency transaction gains or losses. On March 31, 2022, we voluntarily elected to de-designate our cross currency swaps previously designated as cash flow hedges and re-designated the swaps as fair value hedges. Subsequently executed cross currency swaps are also designated as fair value hedges. The fair value of the asset of these contracts was \$305 million and \$589 million at December 31, 2022 and 2021, respectively. At December 31, 2022 and 2021, the fair value of the liability of these contracts was \$3.6 billion and \$1.6 billion, respectively. At December 31, 2022 and 2021, the total notional amount of the cross currency swaps was \$35.0 billion and \$32.5 billion, respectively.

Foreign Exchange Forwards

We also have foreign exchange forwards which we use as economic hedges but for which we have elected not to apply hedge accounting. We enter into British Pound Sterling and Euro foreign exchange forwards to mitigate our foreign exchange rate risk related to non-functional currency denominated monetary assets and liabilities of international subsidiaries.

At December 31, 2022, the fair value of the asset and liability of these contracts was insignificant. At December 31, 2021, the fair value of the asset of these contracts was insignificant and there was no amount related to the liability of these contracts. At December 31, 2022 and 2021, the total notional amount of the foreign exchange forwards was \$920 million and \$932 million, respectively.

Item 8. Financial Statements and Supplementary Data

Report of Independent Registered Public Accounting Firm

To the Shareholders and the Board of Directors of Verizon Communications Inc.:

Opinion on Internal Control Over Financial Reporting

We have audited Verizon Communications Inc. and subsidiaries' (Verizon) internal control over financial reporting as of December 31, 2022, based on criteria established in Internal Control – Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework) (the COSO criteria). In our opinion, Verizon maintained, in all material respects, effective internal control over financial reporting as of December 31, 2022, based on the COSO criteria.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the consolidated balance sheets of Verizon as of December 31, 2022 and 2021, the related consolidated statements of income, comprehensive income, cash flows, and changes in equity for each of the three years in the period ended December 31, 2022, and the related notes and the financial statement schedule listed in the Index at Item 15(a) and our report dated February 10, 2023 expressed an unqualified opinion thereon.

Basis for Opinion

Verizon's management is responsible for maintaining effective internal control over financial reporting and for its assessment of the effectiveness of internal control over financial reporting included in the accompanying Management's Annual Report on Internal Control over Financial Reporting. Our responsibility is to express an opinion on Verizon's internal control over financial reporting based on our audit. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to Verizon in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects.

Our audit included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, testing and evaluating the design and operating effectiveness of internal control based on the assessed risk, and performing such other procedures as we considered necessary in the circumstances. We believe that our audit provides a reasonable basis for our opinion.

Definition and Limitations of Internal Control Over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

/s/ **Ernst & Young LLP**
Ernst & Young LLP
New York, New York

February 10, 2023

Report of Independent Registered Public Accounting Firm

To the Shareholders and the Board of Directors of Verizon Communications Inc.:

Opinion on the Financial Statements

We have audited the accompanying consolidated balance sheets of Verizon Communications Inc. and subsidiaries (Verizon or the Company) as of December 31, 2022 and 2021, the related consolidated statements of income, comprehensive income, cash flows, and changes in equity for each of the three years in the period ended December 31, 2022, and the related notes and the financial statement schedule listed in the Index at Item 15(a) (collectively referred to as the "consolidated financial statements"). In our opinion, the consolidated financial statements present fairly, in all material respects, the financial position of Verizon at December 31, 2022 and 2021, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2022, in conformity with U.S. generally accepted accounting principles.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), Verizon's internal control over financial reporting as of December 31, 2022, based on criteria established in Internal Control-Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework) and our report dated February 10, 2023 expressed an unqualified opinion thereon.

Basis for Opinion

These financial statements are the responsibility of Verizon's management. Our responsibility is to express an opinion on Verizon's financial statements based on our audits. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to Verizon in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the financial statements. We believe that our audits provide a reasonable basis for our opinion.

Critical Audit Matters

The critical audit matters communicated below are matters arising from the current period audit of the financial statements that were communicated or required to be communicated to the audit committee and that: (1) relate to accounts or disclosures that are material to the financial statements and (2) involved our especially challenging, subjective or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matters below, providing separate opinions on the critical audit matters or on the accounts or disclosures to which they relate.

<i>Description of the Matter</i>	Valuation of Employee Benefit Obligations The Company sponsors several pension plans and other post-employment benefit plans. At December 31, 2022, the Company's aggregate defined benefit pension obligation was \$15.3 billion and exceeded the fair value of pension plan assets of \$13.7 billion, resulting in an unfunded defined benefit pension obligation of \$1.6 billion. Also, at December 31, 2022, the other postretirement benefits obligation was approximately \$11.1 billion. As explained in Note 11 of the consolidated financial statements, the Company updates the estimates used to measure employee benefit obligations and plan assets in the fourth quarter and upon a remeasurement event to reflect the actual return on plan assets and updated actuarial assumptions. Auditing the employee benefit obligations was complex due to the highly judgmental nature of the actuarial assumptions (e.g., discount rate, health care cost trends and per capita claims cost trends) used in the measurement process. These assumptions had a significant effect on the projected benefit obligation.
----------------------------------	---

How We Addressed the Matter in Our Audit We obtained an understanding, evaluated the design and tested the operating effectiveness of controls over the employee benefits obligation valuation process. For example, we tested controls over management's review of the employee benefit obligation calculations, the significant actuarial assumptions and the data inputs provided to the actuary.

To test the employee benefit obligations, our audit procedures included, among others, evaluating the methodologies used, the significant actuarial assumptions discussed above and the underlying data used by the Company. We compared the actuarial assumptions used by management to historical trends, current economic factors and evaluated the change in the employee benefit obligations from prior year due to the change in service cost, interest cost, actuarial gains and losses, benefit payments, contributions and other activities. In addition, we involved an actuarial specialist to assist in evaluating management's methodology for determining the discount rate that reflects the maturity and duration of the benefit payments and is used to measure the employee benefit obligations. As part of this assessment, we compared the projected cash flows to prior year projections and compared the current year benefits paid to the prior year projected cash flows. To evaluate the health care cost trends and per capita claims cost trends, we involved an actuarial specialist to assist in evaluating the assumptions and assessed whether the information was consistent with publicly available information, and whether any market data adjusted for entity-specific adjustments were applied. We also tested the completeness and accuracy of the underlying data, including the participant data provided to management's actuarial specialists.

Impairment Evaluation for Verizon Business Group Goodwill

Description of the Matter At December 31, 2022, the Company's goodwill related to its Verizon Business Group (Business) reporting unit was \$7.5 billion and represented 2.0% of total assets. As discussed in Notes 1 and 4 of the consolidated financial statements, goodwill is not amortized but rather is tested for impairment at the reporting unit level at least annually, or more frequently if impairment indicators are present. The impairment test compares the fair value of the reporting unit (calculated using a combination of a market approach and an income approach) to its carrying amount.

Auditing management's goodwill impairment test was complex and highly judgmental due to the inherent subjectivity of developing an estimate of the fair value of the reporting unit, which is based on assumptions about future conditions, transactions, or events whose outcome is uncertain and will therefore be subject to change over time. In particular, the fair value estimate was sensitive to significant assumptions such as the discount rate, revenue growth rate and earnings before interest, taxes, depreciation and amortization (EBITDA) margins, which are affected by expected future market or economic conditions.

How We Addressed the Matter in Our Audit We obtained an understanding, evaluated the design and tested the operating effectiveness of controls over the Company's goodwill impairment review process. For example, we tested controls over management's review of the valuation models and the significant assumptions described above.

To test the estimated fair value of the Company's Business reporting unit, our audit procedures included, among others, assessing the suitability and application of the valuation methodologies selected and evaluating the significant assumptions discussed above and underlying data used by the Company in its analysis. We compared the significant assumptions used by management to current industry and economic trends, market information, and other relevant factors. We performed sensitivity analyses of significant assumptions to determine what changes in assumptions are particularly sensitive when assessing the likelihood of impairment, or when calculating the amount of an impairment. In addition, we involved a valuation specialist to assist in the evaluation of the assumptions and other relevant information that are most significant to the fair value estimate. We also assessed the historical accuracy of management's forecasts of financial results used in developing prior fair value estimates to assist in evaluating the reliability of the current forecasts.

/s/ **Ernst & Young LLP**
Ernst & Young LLP
We have served as Verizon's auditor since 2000.
New York, New York

February 10, 2023

Consolidated Statements of Income

Verizon Communications Inc. and Subsidiaries

Years Ended December 31,	(dollars in millions, except per share amounts)		
	2022	2021	2020
Operating Revenues			
Service revenues and other	\$ 109,625	\$ 110,449	\$ 109,872
Wireless equipment revenues	27,210	23,164	18,420
Total Operating Revenues	136,835	133,613	128,292
Operating Expenses			
Cost of services (exclusive of items shown below)	28,637	31,234	31,401
Cost of wireless equipment	30,496	25,067	19,800
Selling, general and administrative expense	30,136	28,658	31,573
Depreciation and amortization expense	17,099	16,206	16,720
Total Operating Expenses	106,368	101,165	99,494
Operating Income	30,467	32,448	28,798
Equity in earnings (losses) of unconsolidated businesses	44	145	(45)
Other income (expense), net	1,373	312	(539)
Interest expense	(3,613)	(3,485)	(4,247)
Income Before Provision For Income Taxes	28,271	29,420	23,967
Provision for income taxes	(6,523)	(6,802)	(5,619)
Net Income	\$ 21,748	\$ 22,618	\$ 18,348
Net income attributable to noncontrolling interests	\$ 492	\$ 553	\$ 547
Net income attributable to Verizon	21,256	22,065	17,801
Net Income	\$ 21,748	\$ 22,618	\$ 18,348
Basic Earnings Per Common Share			
Net income attributable to Verizon	\$ 5.06	\$ 5.32	\$ 4.30
Weighted-average shares outstanding (in millions)	4,202	4,148	4,140
Diluted Earnings Per Common Share			
Net income attributable to Verizon	\$ 5.06	\$ 5.32	\$ 4.30
Weighted-average shares outstanding (in millions)	4,204	4,150	4,142

See Notes to Consolidated Financial Statements

Consolidated Statements of Comprehensive Income

Verizon Communications Inc. and Subsidiaries

Years Ended December 31,	(dollars in millions)		
	2022	2021	2020
Net Income	\$ 21,748	\$ 22,618	\$ 18,348
Other Comprehensive Loss, Net of Tax (Expense) Benefit			
Foreign currency translation adjustments, net of tax of \$(13), \$(17) and \$19	(153)	(141)	180
Unrealized gain (loss) on cash flow hedges, net of tax of \$(11), \$30 and \$197	322	(85)	(571)
Unrealized loss on fair value hedges, net of tax of \$148, \$0 and \$0	(431)	—	—
Unrealized loss on marketable securities, net of tax of \$3, \$3 and \$(2)	(25)	(9)	(2)
Defined benefit pension and postretirement plans, net of tax of \$221, \$205 and \$221	(651)	(621)	(676)
Other comprehensive loss attributable to Verizon	(938)	(856)	(1,069)
Total Comprehensive Income	\$ 20,810	\$ 21,762	\$ 17,279
Comprehensive income attributable to noncontrolling interests	\$ 492	\$ 553	\$ 547
Comprehensive income attributable to Verizon	20,318	21,209	16,732
Total Comprehensive Income	\$ 20,810	\$ 21,762	\$ 17,279

See Notes to Consolidated Financial Statements

Consolidated Balance Sheets

Verizon Communications Inc. and Subsidiaries

		(dollars in millions, except per share amounts)	
At December 31,		2022	2021
Assets			
Current assets			
Cash and cash equivalents	\$	2,605	\$ 2,921
Accounts receivable		25,332	24,742
Less Allowance for credit losses		826	896
Accounts receivable, net		24,506	23,846
Inventories		2,388	3,055
Prepaid expenses and other		8,358	6,906
Total current assets		37,857	36,728
Property, plant and equipment		307,689	289,897
Less Accumulated depreciation		200,255	190,201
Property, plant and equipment, net		107,434	99,696
Investments in unconsolidated businesses		1,071	1,061
Wireless licenses		149,796	147,619
Goodwill		28,671	28,603
Other intangible assets, net		11,461	11,677
Operating lease right-of-use assets		26,130	27,883
Other assets		17,260	13,329
Total assets	\$	379,680	\$ 366,596
Liabilities and Equity			
Current liabilities			
Debt maturing within one year	\$	9,963	\$ 7,443
Accounts payable and accrued liabilities		23,977	24,833
Current operating lease liabilities		4,134	3,859
Other current liabilities		12,097	11,025
Total current liabilities		50,171	47,160
Long-term debt		140,676	143,425
Employee benefit obligations		12,974	15,410
Deferred income taxes		43,441	40,685
Non-current operating lease liabilities		21,558	23,203
Other liabilities		18,397	13,513
Total long-term liabilities		237,046	236,236
Commitments and Contingencies (Note 16)			
Equity			
Series preferred stock (\$0.10 par value; 250,000,000 shares authorized; none issued)		—	—
Common stock (\$0.10 par value; 6,250,000,000 shares authorized in each period; 4,291,433,646 shares issued in each period)		429	429
Additional paid in capital		13,420	13,861
Retained earnings		82,380	71,993
Accumulated other comprehensive loss		(1,865)	(927)
Common stock in treasury, at cost (91,572,258 and 93,634,725 shares outstanding)		(4,013)	(4,104)
Deferred compensation – employee stock ownership plans (ESOPs) and other		793	538
Noncontrolling interests		1,319	1,410
Total equity		92,463	83,200
Total liabilities and equity	\$	379,680	\$ 366,596

See Notes to Consolidated Financial Statements

Consolidated Statements of Cash Flows

Verizon Communications Inc. and Subsidiaries

Years Ended December 31,	(dollars in millions)		
	2022	2021	2020
Cash Flows from Operating Activities			
Net Income	\$ 21,748	\$ 22,618	\$ 18,348
Adjustments to reconcile net income to net cash provided by operating activities:			
Depreciation and amortization expense	17,099	16,206	16,720
Employee retirement benefits	(2,046)	(3,391)	840
Deferred income taxes	2,973	4,264	1,553
Provision for expected credit losses	1,611	789	1,380
Equity in losses (earnings) of unconsolidated businesses, net of dividends received	(10)	36	91
Changes in current assets and liabilities, net of effects from acquisition/disposition of businesses:			
Accounts receivable	(1,978)	(1,592)	189
Inventories	627	(905)	(369)
Prepaid expenses and other	928	150	1,202
Accounts payable and accrued liabilities and Other current liabilities	(33)	1,457	(966)
Other, net	(3,778)	(93)	2,780
Net cash provided by operating activities	37,141	39,539	41,768
Cash Flows from Investing Activities			
Capital expenditures (including capitalized software)	(23,087)	(20,286)	(18,192)
Cash received (paid) related to acquisitions of businesses, net of cash acquired	248	(4,065)	(520)
Acquisitions of wireless licenses	(3,653)	(47,596)	(3,896)
Collateral payments related to derivative contracts, net of repayments	(2,265)	(21)	—
Proceeds from disposition of business	33	4,122	—
Other, net	62	693	(904)
Net cash used in investing activities	(28,662)	(67,153)	(23,512)
Cash Flows from Financing Activities			
Proceeds from long-term borrowings	7,074	33,034	25,822
Proceeds from asset-backed long-term borrowings	10,732	8,383	5,635
Net proceeds from (repayments of) short-term commercial paper	106	—	(9)
Repayments of long-term borrowings and finance lease obligations	(8,616)	(14,063)	(9,775)
Repayments of asset-backed long-term borrowings	(4,948)	(4,800)	(7,413)
Dividends paid	(10,805)	(10,445)	(10,232)
Other, net	(2,072)	(3,832)	(2,703)
Net cash provided by (used in) financing activities	(8,529)	8,277	1,325
Increase (decrease) in cash, cash equivalents and restricted cash	(50)	(19,337)	19,581
Cash, cash equivalents and restricted cash, beginning of period	4,161	23,498	3,917
Cash, cash equivalents and restricted cash, end of period (Note 1)	\$ 4,111	\$ 4,161	\$ 23,498

See Notes to Consolidated Financial Statements

Consolidated Statements of Changes in Equity

Verizon Communications Inc. and Subsidiaries

Years Ended December 31,	(dollars in millions, except per share amounts, and shares in thousands)					
	2022		2021		2020	
	Shares	Amount	Shares	Amount	Shares	Amount
Common Stock						
Balance at beginning of year	4,291,434	\$ 429	4,291,434	\$ 429	4,291,434	\$ 429
Balance at end of year	4,291,434	429	4,291,434	429	4,291,434	429
Additional Paid In Capital						
Balance at beginning of year		13,861		13,404		13,419
Other (Note 14)		(441)		457		(15)
Balance at end of year		13,420		13,861		13,404
Retained Earnings						
Balance at beginning of year		71,993		60,464		53,147
Opening balance sheet adjustment (Note 1)		—		—		(200)
Adjusted opening balance		71,993		60,464		52,947
Net income attributable to Verizon		21,256		22,065		17,801
Dividends declared (\$2.585, \$2.535, \$2.485 per share)		(10,860)		(10,532)		(10,284)
Other		(9)		(4)		—
Balance at end of year		82,380		71,993		60,464
Accumulated Other Comprehensive Income (Loss)						
Balance at beginning of year attributable to Verizon		(927)		(71)		998
Foreign currency translation adjustments		(153)		(141)		180
Unrealized gain (loss) on cash flow hedges		322		(85)		(571)
Unrealized loss on fair value hedges		(431)		—		—
Unrealized loss on marketable securities		(25)		(9)		(2)
Defined benefit pension and postretirement plans		(651)		(621)		(676)
Other comprehensive loss		(938)		(856)		(1,069)
Balance at end of year attributable to Verizon		(1,865)		(927)		(71)
Treasury Stock						
Balance at beginning of year	(93,635)	(4,104)	(153,304)	(6,719)	(155,606)	(6,820)
Employee plans (Note 14)	2,048	90	2,057	90	2,298	101
Shareholder plans (Note 14)	15	1	15	1	4	—
Acquisitions (Note 3)	—	—	57,597	2,524	—	—
Balance at end of year	(91,572)	(4,013)	(93,635)	(4,104)	(153,304)	(6,719)
Deferred Compensation-ESOPs and Other						
Balance at beginning of year		538		335		222
Restricted stock equity grant		423		369		275
Amortization		(168)		(166)		(162)
Balance at end of year		793		538		335
Noncontrolling Interests						
Balance at beginning of year		1,410		1,430		1,440
Total comprehensive income		492		553		547
Distributions and other (Note 14)		(583)		(573)		(557)
Balance at end of year		1,319		1,410		1,430
Total Equity		\$ 92,463		\$ 83,200		\$ 69,272

See Notes to Consolidated Financial Statements

Notes to Consolidated Financial Statements

Verizon Communications Inc. and Subsidiaries

Note 1. Description of Business and Summary of Significant Accounting Policies

Description of Business

Verizon Communications Inc. (the Company) is a holding company that, acting through its subsidiaries (together with the Company, collectively, Verizon), is one of the world's leading providers of communications, technology, information and entertainment products and services to consumers, businesses and government entities. With a presence around the world, we offer data, video and voice services and solutions on our networks and platforms that are designed to meet customers' demand for mobility, reliable network connectivity, security and control.

We have two reportable segments that we operate and manage as strategic business units - Verizon Consumer Group (Consumer) and Verizon Business Group (Business).

Our Consumer segment provides consumer-focused wireless and wireline communications services and products. Our wireless services are provided across one of the most extensive wireless networks in the United States (U.S.) under the Verizon brand, TracFone Wireless, Inc. (TracFone) brands and through wholesale and other arrangements. We also provide fixed wireless access (FWA) broadband through our wireless networks. Our wireline services are provided in nine states in the Mid-Atlantic and Northeastern U.S., as well as Washington D.C., over our 100% fiber-optic network through our Verizon Fios product portfolio and over a traditional copper-based network to customers who are not served by Fios. Our Consumer segment's wireless and wireline products and services are available to our retail customers, as well as resellers that purchase wireless network access from us on a wholesale basis.

Our Business segment provides wireless and wireline communications services and products, including data, video and conferencing services, corporate networking solutions, security and managed network services, local and long distance voice services and network access to deliver various Internet of Things (IoT) services and products. We also provide FWA broadband through our wireless networks. We provide these products and services to businesses, government customers and wireless and wireline carriers across the U.S. and select products and services to customers around the world.

Consolidation

The method of accounting applied to investments, whether consolidated or equity, involves an evaluation of all significant terms of the investments that explicitly grant or suggest evidence of control or influence over the operations of the investee. The consolidated financial statements include our controlled subsidiaries, as well as variable interest entities (VIE) where we are deemed to be the primary beneficiary. For controlled subsidiaries that are not wholly-owned, the noncontrolling interests are included in Net income and Total equity. Investments in businesses that we do not control, but have the ability to exercise significant influence over operating and financial policies, are accounted for using the equity method. Equity method investments are included in Investments in unconsolidated businesses in our consolidated balance sheets. All significant intercompany accounts and transactions have been eliminated.

Basis of Presentation

We have reclassified certain prior year amounts to conform to the current year presentation.

Use of Estimates

We prepare our financial statements using U.S. generally accepted accounting principles (GAAP), which requires management to make estimates and assumptions that affect reported amounts and disclosures. These estimates and assumptions take into account historical and forward-looking factors that the Company believes are reasonable, including but not limited to public health crises, such as the COVID-19 pandemic, and related economic implications. Actual results could differ significantly from those estimates.

Examples of significant estimates include the allowance for credit losses, the recoverability of intangible assets, property, plant and equipment, and other long-lived assets, the incremental borrowing rate for the lease liability, fair value measurements, including those related to financial instruments, goodwill, spectrum licenses and intangible assets, unrecognized tax benefits, valuation allowances on tax assets, pension and postretirement benefit obligations, contingencies and the identification and valuation of assets acquired and liabilities assumed in connection with business combinations.

Revenue Recognition

We earn revenue from contracts with customers, primarily through the provision of telecommunications and other services and through the sale of wireless equipment. These services include a variety of communication and connectivity services for our Consumer and Business customers including other carriers that use our facilities to provide services to their customers, as well as professional and integrated managed services for our large enterprises and government customers. We account for these revenues under Accounting Standards Update (ASU) 2014-09, "Revenue from Contracts with Customers" (Topic 606).

We also earn revenues that are not accounted for under Topic 606 from leasing arrangements (such as those for towers and equipment), captive reinsurance arrangements primarily related to wireless device insurance and the interest on equipment financed under a device payment plan agreement when sold to the customer by an authorized agent.

Nature of Products and Services

Telecommunications

Service

We offer wireless services through a variety of plans on a postpaid or prepaid basis. For wireless service, we recognize revenue using an output method, either as the service allowance units are used or as time elapses, because it reflects the pattern by which we satisfy our performance obligation through the transfer of service to the customer. Monthly service is generally billed in advance, which results in a contract liability. See Note 2 for additional information. For postpaid plans, where monthly usage exceeds the allowance, the overage usage represents options held by the customer for incremental services and the usage-based fee is recognized when the customer exercises the option (typically on a month-to-month basis).

For our contracts related to wireline communication and connectivity services, in general, fixed monthly fees for service are billed one month in advance, which results in a contract liability, and service revenue is recognized over the enforceable contract term as the service is rendered, as the customer simultaneously receives and consumes the benefits of the services through network access and usage. While substantially all of our wireline service revenue contracts are the result of providing access to our networks, revenue from services that are not fixed in amount and, instead, are based on usage are generally billed in arrears and recognized as the usage occurs.

Equipment

We sell wireless devices and accessories under the Verizon brand and other brands. Equipment revenue is generally recognized when the products are delivered to and accepted by the customer, as this is when control passes to the customer. In addition to offering the sale of equipment on a standalone basis, we have two primary offerings through which customers pay for a wireless device, in connection with a service contract: fixed-term plans and device payment plans.

Under a fixed-term plan, the customer is sold the wireless device without any upfront charge or at a discounted price in exchange for entering into a fixed-term service contract (typically for a term of 24 months or less).

Under a device payment plan, the customer is sold the wireless device in exchange for a non-interest-bearing installment note, which is repaid by the customer, typically over a 36-month term, and concurrently enters into a month-to-month contract for wireless service. We may offer certain promotions that provide billing credits applied over a specified term, contingent upon the customer maintaining service. The credits are included in the transaction price, which are allocated to the performance obligations based on their relative selling price and are recognized when earned.

A financing component exists in both our fixed-term plans and device payment plans because the timing of the payment for the device, which occurs over the contract term, differs from the satisfaction of the performance obligation, which occurs at contract inception upon transfer of the device to the customer. We periodically assess, at the contract level, the significance of the financing component inherent in our fixed-term and device payment plan receivable based on qualitative and quantitative considerations related to our customer classes. These considerations include assessing the commercial objective of our plans, the term and duration of financing provided, interest rates prevailing in the marketplace, and credit risks of our customer classes, all of which impact our selection of appropriate discount rates. Based on current facts and circumstances, we determined that the financing component in our existing wireless device payments and fixed-term contracts sold through the direct channel is not significant and therefore is not accounted for separately. See Note 8 for additional information on the interest on equipment financed on a device payment plan agreement when sold to the customer by an authorized agent in our indirect channel.

Wireless Contracts

For our wireless contracts, total contract revenue, which represents the transaction price for wireless service and wireless equipment, is allocated between service and equipment revenue based on their estimated standalone selling prices. We estimate the standalone selling price of the device or accessory to be its retail price excluding subsidies or conditional purchase discounts. We estimate the standalone selling price of wireless service to be the price that we offer to customers on month-to-month contracts that can be cancelled at any time without penalty (i.e., when there is no fixed-term for service) or when service is procured without the concurrent purchase of a wireless device. In addition, we also assess whether the service term is impacted by certain legally enforceable rights and obligations in our contract with customers, such as penalties that a customer would have to pay to early terminate a fixed-term contract or billing credits that would cease if the month-to-month wireless service is canceled. The assessment of these legally enforceable rights and obligations involves judgment and impacts our determination of the transaction price and related disclosures.

From time to time, we may offer certain promotions that provide our customers on device payment plans with the right to upgrade to a new device after paying a specified portion of their device payment plan agreement amount and trading in their device in good working order. We account for this trade-in right as a guarantee obligation. The full amount of the trade-in right's fair value

is recognized as a guarantee liability and results in a reduction to the revenue recognized upon the sale of the device. The guarantee liability was \$4 million and \$77 million at December 31, 2022 and 2021, respectively. The total transaction price is reduced by the guarantee, which is accounted for outside the scope of Topic 606, and the remaining transaction price is allocated between the performance obligations within the contract.

Our fixed-term plans generally include the sale of a wireless device at subsidized prices. This results in the creation of a contract asset at the time of sale, which represents the recognition of equipment revenue in excess of amounts billed.

For our device payment plans, billing credits are accounted for as consideration payable to a customer and are included in the determination of total transaction price, resulting in a contract liability.

We may provide a right of return on our products and services for a short time period after a sale. These rights are accounted for as variable consideration when determining the transaction price, and accordingly we recognize revenue based on the estimated amount to which we expect to be entitled after considering expected returns. Returns and credits are estimated at contract inception and updated at the end of each reporting period as additional information becomes available. We also may provide credits or incentives on our products and services for contracts with resellers, which are accounted for as variable consideration when estimating the amount of revenue to recognize.

Wireline Contracts

Total consideration for wireline services that are bundled in a single contract is allocated to each performance obligation based on our standalone selling price for each service. While many contracts include one or more service performance obligations, the revenue recognition pattern is generally not impacted by the allocation since the services are generally satisfied over the same period of time. We estimate the standalone selling price to be the price of the services when sold on a standalone basis without any promotional discount. In addition, we also assess whether the service term is impacted by certain legally enforceable rights and obligations in our contract with customers such as penalties that a customer would have to pay to early terminate a fixed-term contract. The assessment of these legally enforceable rights and obligations involves judgment and impacts our determination of transaction price and related disclosures.

We may provide performance-based credits or incentives on our products and services for contracts with our Business customers, which are accounted for as variable consideration when estimating the transaction price. Credits are estimated at contract inception and are updated at the end of each reporting period as additional information becomes available.

Wireless and Wireline Contracts

For offers that include third-party providers, we evaluate whether we are acting as the principal or as the agent with respect to the goods or services provided to the customer. This principal-versus-agent assessment involves judgment and focuses on whether the facts and circumstances of the arrangement indicate that the goods or services were controlled by us prior to transferring them to the customer. To evaluate if we have control, we consider various factors including whether we are primarily responsible for fulfillment, bear risk of loss and have discretion over pricing.

Other

Advertising revenues are generated through display advertising and search advertising. Display advertising revenue is generated by the display of graphical advertisements and other performance-based advertising. Search advertising revenue is generated when a consumer clicks on a text-based advertisement on the search results page. The divested Verizon Media Group (Verizon Media), primarily earned revenue through display advertising on Verizon Media properties, as well as on third-party properties through our advertising platforms, search advertising, and subscription arrangements. Revenue for display and search advertising contracts is recognized as ads are delivered, while subscription contracts are recognized over time. We are generally the principal in transactions carried out through our advertising platforms, and therefore report gross revenue based on the amount billed to our customers. The control and transfer of digital advertising inventory occurs in a rapid, real-time environment, where our proprietary technology enables us to identify, enhance, verify and solely control digital advertising inventory that we then sell to our customers. Our control is further supported by us being primarily responsible to our customers for fulfillment and the fact that we can exercise a level of discretion over pricing. We completed the sale of Verizon Media on September 1, 2021. See Note 3 for additional information on the sale of Verizon Media.

We offer telematics services including smart fleet management and optimization software. Telematics service revenue is generated primarily through subscription contracts. We recognize revenue over time for our subscription contracts.

We report taxes collected from customers on behalf of governmental authorities on revenue-producing transactions on a net basis.

Maintenance and Repairs

We charge the cost of maintenance and repairs, including the cost of replacing minor items not constituting substantial betterments, principally to Cost of services as these costs are incurred.

Advertising Costs

Costs for advertising products and services, as well as other promotional and sponsorship costs, are charged to Selling, general and administrative expense in the periods in which they are incurred. See Note 15 for additional information.

Earnings Per Common Share

Basic earnings per common share are based on the weighted-average number of shares outstanding during the period. Where appropriate, diluted earnings per common share include the dilutive effect of shares issuable under our stock-based compensation plans.

There were a total of approximately 1.9 million, 1.7 million and 2.0 million outstanding dilutive securities, primarily consisting of restricted stock units, included in the computation of diluted earnings per common share for the years ended December 31, 2022, 2021, and 2020, respectively.

Cash, Cash Equivalents and Restricted Cash

We consider all highly liquid investments with an original maturity of 90 days or less when purchased to be cash equivalents. Cash equivalents are stated at cost, which approximates quoted market value and includes amounts held in money market funds.

Cash collections on the receivables collateralizing our asset-backed debt securities are required at certain specified times to be placed into segregated accounts. Deposits to the segregated accounts are considered restricted cash and are included in Prepaid expenses and other and Other assets in our consolidated balance sheets.

Cash, cash equivalents and restricted cash are included in the following line items in the consolidated balance sheets:

		(dollars in millions)		
At December 31,		2022	2021	Increase / (Decrease)
Cash and cash equivalents	\$	2,605	\$ 2,921	\$ (316)
Restricted cash:				
Prepaid expenses and other		1,343	1,094	249
Other assets		163	146	17
Cash, cash equivalents and restricted cash	\$	4,111	\$ 4,161	\$ (50)

Investments in Debt and Equity Securities

Investments in equity securities that are not accounted for under equity method accounting or result in consolidation are to be measured at fair value. For investments in equity securities without readily determinable fair values, Verizon elects the measurement alternative permitted under GAAP to measure these investments at cost, less any impairment, plus or minus changes resulting from observable price changes in orderly transactions for an identical or similar investment of the same issuer. For investments in debt securities without quoted prices, Verizon uses an alternative matrix pricing method. Investments in equity securities that do not result in consolidation of the investee are included in Investments in unconsolidated businesses and debt securities are included in Other assets in our consolidated balance sheets.

Allowance for Credit Losses

Accounts receivable are recorded at amortized cost less an allowance for credit losses that are not expected to be recovered. The gross amount of accounts receivable and corresponding allowance for credit losses are presented separately in the consolidated balance sheets. We maintain allowances for credit losses resulting from the expected failure or inability of our customers to make required payments. We recognize the allowance for credit losses at inception and reassess quarterly based on management's expectation of the asset's collectability. The allowance is based on multiple factors including historical experience with bad debts, the credit quality of the customer base, the aging of such receivables and current macroeconomic conditions, such as the COVID-19 pandemic, as well as management's expectations of conditions in the future, as applicable. Our allowance for credit losses is based on management's assessment of the collectability of assets pooled together with similar risk characteristics.

We pool our device payment plan agreement receivables based on the credit quality indicators and shared risk characteristics of "new customers" and "existing customers." New customers are defined as customers who have been with Verizon for less than 210 days. Existing customers are defined as customers who have been with Verizon for 210 days or more. We record an allowance to reduce the receivables to the amount that is expected to be collectible. For device payment plan agreement receivables, we record bad debt expense based on a default and loss calculation using our proprietary loss model. The expected loss rate is determined based on customer credit scores and other qualitative factors as noted above. The loss rate is assigned individually on a customer by customer basis and the custom credit scores are then aggregated by vintage and used in our proprietary loss model to calculate the weighted-average loss rate used for determining the allowance balance.

We monitor the collectability of our wireless service receivables as one overall pool. Wireline service receivables are disaggregated and pooled by the following customer groups: consumer, small and medium business, global enterprise, public sector and wholesale. For wireless service receivables and wireline consumer and small and medium business receivables, the allowance is calculated based on a 12 month rolling average write-off balance multiplied by the average life-cycle of an account from billing to write-off. The risk of loss is assessed over the contractual life of the receivables and we adjust the historical loss amounts for current and future conditions based on management's qualitative considerations. For global enterprise, public sector and wholesale wireline receivables, the allowance for credit losses is based on historical write-off experience and individual customer credit risk, as applicable. We consider multiple factors in determining the allowance as discussed above.

Inventories

Inventory consists of wireless and wireline equipment held for sale, which is carried at the lower of cost (determined principally on either an average cost or first-in, first-out basis) or net realizable value.

Plant and Depreciation

We record property, plant and equipment at cost. Property, plant and equipment are generally depreciated on a straight-line basis.

Leasehold improvements are amortized over the shorter of the estimated life of the improvement or the remaining term of the related lease, calculated from the time the asset was placed in service.

When depreciable assets are retired or otherwise disposed of, the related cost and accumulated depreciation are deducted from the property, plant and equipment accounts and any gains or losses on disposition are recognized in Selling, general and administrative expense.

We capitalize and depreciate network software purchased or developed within property, plant and equipment assets. We also capitalize interest associated with the acquisition or construction of network-related assets. Capitalized interest is reported as a reduction in interest expense and depreciated as part of the cost of the network-related assets.

Computer Software and Cloud Computing Costs

We capitalize the cost of internal-use network and non-network software and defer the costs associated with cloud computing arrangements that have a useful life and term in excess of one year. Subsequent additions, modifications or upgrades to internal-use network and non-network software are capitalized only to the extent that they add significant new functionality. Planning, software maintenance and training costs for internal-use software and cloud computing arrangements are expensed in the period in which they are incurred. We capitalize interest associated with the development of internal-use network and non-network software. Capitalized non-network internal-use software costs are amortized using the straight-line method over a period of 7 years and are included in Other intangible assets, net in our consolidated balance sheets. Costs incurred in implementing a cloud computing arrangement are deferred during the application-development stage and recorded as Prepaid expense and other in our consolidated balance sheets. Once a project is substantially complete and ready for its intended use, we stop deferring the related cloud computing arrangement costs.

For a discussion of our impairment policy for capitalized non-network software costs, see "Goodwill and Other Intangible Assets" below. See Note 4 for additional information of internal-use non-network software reflected in our consolidated balance sheets. Similar to capitalized software costs, deferred costs associated with cloud computing arrangements are subject to impairment testing.

Goodwill and Other Intangible Assets

Goodwill

Goodwill is the excess of the acquisition cost of businesses over the fair value of the identifiable net assets acquired. Impairment testing for goodwill is performed annually in the fourth quarter or more frequently if impairment indicators are present.

To determine if goodwill is potentially impaired, we have the option to perform a qualitative assessment. However, we may elect to bypass the qualitative assessment and perform a quantitative impairment test even if no indications of a potential impairment exist. It is our policy to perform quantitative impairment assessment at least every three years.

Under the qualitative assessment, we consider several factors, including the business enterprise value of the reporting unit from the last quantitative test and the excess of fair value over carrying value from this test, macroeconomic conditions (including changes in interest rates and discount rates), industry and market considerations (including industry revenue and Earnings before interest, taxes, depreciation and amortization (EBITDA) margin results, projections and recent merger and acquisition activity), the recent and projected financial performance of the reporting unit, as well as other factors.

The quantitative impairment test for goodwill is performed at the reporting unit level and compares the fair value of the reporting unit (calculated using a combination of a market approach and a discounted cash flow method, as a form of the income

approach) to its carrying value. Estimated fair values of reporting units are Level 3 measures in the fair value hierarchy, see "Fair Value Measurements" discussion below for additional information. The market approach includes the use of comparative multiples of guideline companies to complement discounted cash flow results. The discounted cash flow method is based on the present value of two components, projected cash flows and a terminal value. The terminal value represents the expected normalized future cash flows of the reporting unit beyond the cash flows from the discrete projection period. The fair value of the reporting unit is calculated based on the sum of the present value of the cash flows from the discrete period and the present value of the terminal value. The discount rate represents our estimate of the weighted-average cost of capital, or expected return, that a marketplace participant would have required as of the valuation date. If the carrying value exceeds the fair value, an impairment charge is booked for the excess carrying value over fair value, limited to the total amount of goodwill of that reporting unit. During the fourth quarter each year, we update our five-year strategic planning review for each of our reporting units. Those plans consider current economic conditions and trends, estimated future operating results, our view of growth-rates and anticipated future economic and regulatory conditions.

See Note 4 for additional information regarding our goodwill impairment testing.

Intangible Assets Not Subject to Amortization

A significant portion of our intangible assets are wireless licenses that provide our wireless operations with the exclusive right to utilize designated radio frequency spectrum to provide wireless communication services. While licenses are issued for only a fixed time, generally ten to fifteen years, such licenses are subject to renewal by the Federal Communications Commission (FCC). License renewals have occurred routinely and at nominal cost. Moreover, we have determined that there are currently no legal, regulatory, contractual, competitive, economic or other factors that limit the useful life of our wireless licenses. As a result, we treat the wireless licenses as an indefinite-lived intangible asset. We re-evaluate the useful life determination for wireless licenses each year to determine whether events and circumstances continue to support an indefinite useful life. We aggregate our wireless licenses into one single unit of accounting, as we utilize our wireless licenses on an integrated basis as part of our nationwide wireless network.

We test our wireless licenses for potential impairment annually or more frequently if impairment indicators are present. We have the option to first perform a qualitative assessment to determine whether it is necessary to perform a quantitative impairment test. However, we may elect to bypass the qualitative assessment in any period and proceed directly to performing the quantitative impairment test. It is our policy to perform quantitative impairment assessment at least every three years.

As part of our qualitative assessment we consider several factors including the business enterprise value of our combined wireless business, macroeconomic conditions (including changes in interest rates and discount rates), industry and market considerations (including industry revenue and EBITDA margin results, projections and recent merger and acquisition activity), the recent and projected financial performance of our combined wireless business as a whole, as well as other factors including the result of our last quantitative assessment. See Note 4 for additional information regarding our impairment tests.

Our quantitative impairment assessment consists of comparing the estimated fair value of our aggregate wireless licenses to the aggregated carrying amount as of the test date. Under our quantitative assessment, we estimate the fair value of our wireless licenses using the Greenfield approach. The Greenfield approach is an income based valuation approach that values the wireless licenses by calculating the cash flow generating potential of a hypothetical start-up company that goes into business with no assets except the wireless licenses to be valued. A discounted cash flow analysis is used to estimate what a marketplace participant would be willing to pay to purchase the aggregated wireless licenses as of the valuation date. If the estimated fair value of the aggregated wireless licenses is less than the aggregated carrying amount of the wireless licenses, then an impairment charge is recognized.

Interest expense incurred while qualifying activities are performed to ready wireless licenses for their intended use is capitalized as part of wireless licenses. The capitalization period ends when the development is discontinued or substantially completed and the license is ready for its intended use.

Wireless licenses can be purchased through public auctions conducted by the FCC. Deposits required to participate in these auctions and purchase licenses are recorded within Other assets in our consolidated balance sheets until the corresponding licenses are received and within Net cash used in investing activities in our consolidated statements of cash flows.

Intangible Assets Subject to Amortization and Long-Lived Assets

Our intangible assets that do not have indefinite lives (primarily customer lists and non-network internal-use software) are amortized over their estimated useful lives. All of our intangible assets subject to amortization and other long-lived assets are reviewed for impairment whenever events or changes in circumstances indicate that the carrying amount of the asset may not be recoverable. If any indications of impairment are present, we would test for recoverability by comparing the carrying amount of the asset group to the net undiscounted cash flows expected to be generated from the asset group. If those net undiscounted cash flows do not exceed the carrying amount, we would perform the next step, which is to determine the fair value of the asset and record an impairment, if any. We re-evaluate the useful life determinations for these intangible assets each year to determine whether events and circumstances warrant a revision to their remaining useful lives.

See Note 4 for information related to the carrying amount of goodwill, wireless licenses and other intangible assets, as well as the major components and average useful lives of our other acquired intangible assets.

Leases

We lease network equipment including towers, distributed antenna systems, small cells, real estate, connectivity mediums which include dark fiber, equipment, and other various types of assets for use in our operations under both operating and finance leases. We assess whether an arrangement is a lease or contains a lease at inception. For arrangements considered leases or that contain a lease that is accounted for separately, we determine the classification and initial measurement of the right-of-use asset and lease liability at the lease commencement date, which is the date that the underlying asset becomes available for use.

For both operating and finance leases, we recognize a right-of-use asset, which represents our right to use the underlying asset for the lease term, and a lease liability, which represents the present value of our obligation to make payments arising over the lease term. The present value of the lease payments is calculated using the incremental borrowing rate for operating and finance leases. The incremental borrowing rate is determined using a portfolio approach based on the rate of interest that the Company would have to pay to borrow an amount equal to the lease payments on a collateralized basis over a similar term. Management uses the unsecured borrowing rate and risk-adjusts that rate to approximate a collateralized rate, which is updated on a quarterly basis.

In those circumstances where Verizon is the lessee, we account for non-lease components associated with our leases (e.g., common area maintenance costs) and lease components as a single lease component for substantially all of our asset classes. Additionally, in arrangements where we are the lessor, we have customer premise equipment for which we account for non-lease components (e.g., service revenue) and lease components as combined components under the revenue recognition guidance in Topic 606 as the service revenues are the predominant components in the arrangements.

Rent expense for operating leases is recognized on a straight-line basis over the term of the lease and is included in either Cost of services or Selling, general and administrative expense in our consolidated statements of income, based on the use of the facility or equipment on which rent is being paid. Variable rent payments related to both operating and finance leases are expensed in the period incurred. Our variable lease payments consist of payments dependent on various external indicators, including real estate taxes, common area maintenance charges and utility usage.

Operating leases with a term of 12 months or less are not recorded in our consolidated balance sheets; we recognize rent expense for these leases on a straight-line basis over the lease term.

We recognize the amortization of the right-of-use asset for our finance leases on a straight-line basis over the shorter of the lease term or the useful life of the right-of-use asset in Depreciation and amortization expense in our consolidated statements of income. The interest expense related to finance leases is recognized using the effective interest method based on the discount rate determined at lease commencement and is included within Interest expense in our consolidated statements of income.

See Note 6 for additional information related to leases, including disclosure required under ASU 2016-02, Leases (Topic 842).

Fair Value Measurements

Fair value of financial and non-financial assets and liabilities is defined as an exit price, representing the amount that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants. The three-tier hierarchy for inputs used in measuring fair value, which prioritizes the inputs used in the methodologies of measuring fair value for assets and liabilities, is as follows:

Level 1 — Quoted prices in active markets for identical assets or liabilities

Level 2 — Observable inputs other than quoted prices in active markets for identical assets and liabilities

Level 3 — Unobservable pricing inputs in the market

Financial assets and financial liabilities are classified in their entirety based on the lowest level of input that is significant to the fair value measurements. Our assessment of the significance of a particular input to the fair value measurements requires judgment and may affect the valuation of the assets and liabilities being measured and their categorization within the fair value hierarchy.

Income Taxes

Our effective tax rate is based on pre-tax income, statutory tax rates, tax laws and regulations and tax planning strategies available to us in the various jurisdictions in which we operate.

Deferred income taxes are provided for temporary differences in the basis between financial statement and income tax assets and liabilities. Deferred income taxes are recalculated annually at tax rates in effect for the years in which those tax assets and liabilities are expected to be realized or settled. We record valuation allowances to reduce our deferred tax assets to the amount that is more likely than not to be realized.

We use a two-step approach for recognizing and measuring tax benefits taken or expected to be taken in a tax return. The first step is recognition: we determine whether it is more likely than not that a tax position will be sustained upon examination, including resolution of any related appeals or litigation processes, based on the technical merits of the position. In evaluating whether a tax position has met the more-likely-than-not recognition threshold, we presume that the position will be examined by the appropriate taxing authority that has full knowledge of all relevant information. The second step is measurement: a tax position that meets the more-likely-than-not recognition threshold is measured to determine the amount of benefit to recognize in the financial statements. The tax position is measured at the largest amount of benefit that is greater than 50% likely of being realized upon ultimate settlement. Differences between tax positions taken in a tax return and amounts recognized in the financial statements will generally result in one or more of the following: an increase in a liability for income taxes payable, a reduction of an income tax refund receivable, a reduction in a deferred tax asset or an increase in a deferred tax liability.

Significant management judgment is required in evaluating our tax positions and in determining our effective tax rate.

Stock-Based Compensation

We measure and recognize compensation expense for all stock-based compensation awards made to employees and directors based on estimated fair values. See Note 10 for additional information.

Foreign Currency Translation and Transactions

The functional currency of our foreign operations is generally the local currency. For these foreign entities, we translate their financial statements into U.S. dollars using average exchange rates for the period for income statement amounts and using end-of-period exchange rates for assets and liabilities. We record these translation adjustments in Accumulated other comprehensive loss, a separate component of Equity, in our consolidated balance sheets. We record exchange gains and losses resulting from the conversion of transaction currency to functional currency as a component of Other income (expense), net.

Employee Benefit Plans

Pension and postretirement health care and life insurance benefits earned during the year, as well as interest on projected benefit obligations, are accrued. Prior service costs and credits resulting from changes in plan benefits are generally amortized over the average remaining service period of the employees expected to receive benefits. Expected return on plan assets is determined by applying the return on assets assumption to the actual fair value of plan assets. Actuarial gains and losses are recognized in Other income (expense), net in the year in which they occur. These gains and losses are measured annually as of December 31 or upon a remeasurement event. Verizon management employees no longer earn pension benefits or earn service towards the Company retiree medical subsidy. See Note 11 for additional information.

We recognize a pension or a postretirement plan's funded status as either an asset or liability in the consolidated balance sheets. Also, we measure any unrecognized prior service costs and credits that arise during the period as a component of Accumulated other comprehensive income (loss), net of applicable income tax.

Derivative Instruments

We enter into derivative transactions primarily to manage our exposure to fluctuations in foreign currency exchange rates and interest rates. We employ risk management strategies, which may include the use of a variety of derivatives including cross currency swaps, forward starting interest rate swaps, interest rate swaps, treasury rate locks, interest rate caps, swaptions and foreign exchange forwards. We do not hold derivatives for trading purposes.

We measure all derivatives at fair value and recognize them as either assets or liabilities in our consolidated balance sheets. Our derivative instruments are valued primarily using models based on readily observable market parameters for all substantial terms of our derivative contracts and thus are classified as Level 2. Changes in the fair values of derivative instruments applied as economic hedges are recognized in earnings in the current period. For fair value hedges, the change in the fair value of the derivative instruments is recognized in earnings, along with the change in the fair value of the hedged item. Unrealized gains or losses on excluded components of fair value hedges are recorded in Other comprehensive income (loss) and are recognized into earnings on a systematic and rational basis through the swap accrual over the life of the hedged item. For cash flow hedges, the change in the fair value of the derivative instruments is reported in Other comprehensive income (loss) and recognized in earnings when the hedged item is recognized in earnings. For net investment hedges of certain of our foreign operations, the change in the fair value of the hedging instruments is reported in Other comprehensive income (loss) as part of the cumulative translation adjustment and partially offsets the impact of foreign currency changes on the value of our net investment.

Cash flows from derivatives, which are designated as accounting hedges or applied as economic hedges, are presented consistently with the cash flow classification of the related hedged items. See Note 9 for additional information.

Variable Interest Entities

VEs are entities that lack sufficient equity to permit the entity to finance its activities without additional subordinated financial support from other parties, have equity investors that do not have the ability to make significant decisions relating to the entity's

operations through voting rights, do not have the obligation to absorb the expected losses, or do not have the right to receive the residual returns of the entity. We consolidate the assets and liabilities of VIEs when we are deemed to be the primary beneficiary. The primary beneficiary is the party that has the power to make the decisions that most significantly affect the economic performance of the VIE and has the obligation to absorb losses or the right to receive benefits that could potentially be significant to the VIE.

Recently Adopted Accounting Standards

On January 1, 2020, we adopted ASU 2016-13, Financial Instruments - Credit Losses (Topic 326) using the modified retrospective approach. The cumulative after-tax effect of the changes made to our consolidated financial statements for the adoption of Topic 326 was as follows:

(dollars in millions)	At December 31, 2019	Adjustments due to Topic 326	At January 1, 2020
Retained earnings	\$ 53,147	\$ (200)	\$ 52,947

See Note 8 for additional information related to credit losses, including disclosures required under Topic 326.

Note 2. Revenue and Contract Costs

We earn revenue from contracts with customers, primarily through the provision of telecommunications and other services and through the sale of wireless equipment.

Revenue by Category

We have two reportable segments that we operate and manage as strategic business units, Consumer and Business. Revenue is disaggregated by products and services within Consumer, and customer groups (Small and Medium Business, Global Enterprise, Public Sector and Other, and Wholesale) within Business. See Note 13 for additional information on revenue by segment. Corporate and other primarily includes insurance captive revenues as well as the historical results of divested businesses, including Verizon Media.

We also earn revenues that are not accounted for under Topic 606 from leasing arrangements (such as those for towers and equipment), captive reinsurance arrangements primarily related to wireless device insurance and the interest on equipment financed under a device payment plan agreement when sold to the customer by an authorized agent. As allowed by the practical expedient within Topic 842, we have elected to combine the lease and non-lease components for those arrangements of customer premise equipment where we are the lessor as components accounted for under Topic 606. Revenues from arrangements that were not accounted for under Topic 606 were approximately \$3.2 billion, \$3.1 billion and \$2.9 billion for the years ended December 31, 2022, 2021 and 2020, respectively.

Remaining Performance Obligations

When allocating the total contract transaction price to identified performance obligations, a portion of the total transaction price may relate to service performance obligations which were not satisfied or are partially satisfied as of the end of the reporting period. Below we disclose information relating to these unsatisfied performance obligations. We apply the practical expedient available under Topic 606 that provides the option to exclude the expected revenues arising from unsatisfied performance obligations related to contracts that have an original expected duration of one year or less. This situation primarily arises with respect to certain month-to-month service contracts. At December 31, 2022, month-to-month service contracts represented approximately 94% of our wireless postpaid contracts and 92% of our wireline Consumer and Small and Medium Business contracts, compared to December 31, 2021, for which month-to-month service contracts represented approximately 93% of our wireless postpaid contracts and 86% of our wireline Consumer and Small and Medium Business contracts.

Additionally, certain contracts provide customers the option to purchase additional services. The fees related to these additional services are recognized when the customer exercises the option (typically on a month-to-month basis).

Contracts for wireless services, with or without promotional credits that require maintenance of service, are generally either month-to-month and cancellable at any time, or considered to contain terms ranging from greater than one month to up to thirty-six months (typically under a device payment plan), or contain terms ranging from greater than one month to up to twenty-four months (typically under a fixed-term plan). Additionally, customers may incur charges based on usage or additional optional services purchased in conjunction with entering into a contract that can be cancelled at any time and therefore are not included in the transaction price. The transaction price allocated to service performance obligations, which are not satisfied or are partially satisfied as of the end of the reporting period, are generally related to contracts that are not accounted for as month-to-month contracts.

Our Consumer group customers also include traditional wholesale resellers that purchase and resell wireless service under their own brands to their respective customers. Reseller arrangements generally include a stated contract term, which typically

extends longer than two years and, in some cases, include a periodic minimum revenue commitment over the contract term for which revenues will be recognized in future periods.

Consumer customer contracts for wireline services are generally month-to-month; however, they may have a service term of two years or shorter than twelve months. Certain contracts with Business customers for wireline services extend into future periods, contain fixed monthly fees and usage-based fees, and can include annual commitments in each year of the contract or commitments over the entire specified contract term; however, a significant number of contracts for wireline services with our Business customers have a contract term that is twelve months or less.

Additionally, there are certain contracts with Business customers for wireline and telematics services that have a contractual minimum fee over the total contract term. We cannot predict the time period when revenue will be recognized related to those contracts; thus, they are excluded from the time bands below. These contracts have varying terms spanning over approximately nine years ending in April 2032 and have aggregate contract minimum payments totaling \$1.6 billion.

At December 31, 2022, the transaction price related to unsatisfied performance obligations that are expected to be recognized for 2023, 2024 and thereafter was \$22.1 billion, \$14.3 billion and \$8.1 billion, respectively. Remaining performance obligation estimates are subject to change and are affected by several factors, including terminations and changes in the timing and scope of contracts, arising from contract modifications.

Accounts Receivable and Contract Balances

The timing of revenue recognition may differ from the time of billing to our customers. Receivables presented in our consolidated balance sheets represent an unconditional right to consideration. Contract balances represent amounts from an arrangement when either Verizon has performed, by transferring goods or services to the customer in advance of receiving all or partial consideration for such goods and services from the customer, or the customer has made payment to Verizon in advance of obtaining control of the goods and/or services promised to the customer in the contract.

The following table presents information about receivables from contracts with customers:

(dollars in millions)	At December 31, 2022	At December 31, 2021	At January 1, 2021
Receivables ⁽¹⁾	\$ 11,274	\$ 10,758	\$ 12,029
Device payment plan agreement receivables ⁽²⁾	16,648	12,888	10,358

⁽¹⁾ Balances do not include receivables related to the following contracts: leasing arrangements (such as those for towers and equipment), captive reinsurance arrangements primarily related to wireless device insurance and the interest on equipment financed under a device payment plan agreement when sold to the customer by an authorized agent.

⁽²⁾ Included in device payment plan agreement receivables presented in Note 8. Receivables derived from the sale of equipment on a device payment plan through an authorized agent are excluded.

The following table presents information about contract balances:

(dollars in millions)	At December 31, 2022	At December 31, 2021	At January 1, 2021
Contract asset	\$ 863	\$ 934	\$ 937
Contract liability	8,234	7,229	5,598

Contract assets primarily relate to our rights to consideration for goods or services provided to customers but for which we do not have an unconditional right at the reporting date. Under a fixed-term plan, total contract revenue is allocated between wireless service and equipment revenues. In conjunction with these arrangements, a contract asset is created, which represents the difference between the amount of equipment revenue recognized upon sale and the amount of consideration received from the customer when the performance obligation related to the transfer of control of the equipment is satisfied. The contract asset is reclassified to accounts receivable as wireless services are provided and billed. We have the right to bill the customer as service is provided over time, which results in our right to the payment being unconditional. The contract asset balances are presented in our consolidated balance sheets as Prepaid expenses and other and Other assets. We recognize the allowance for credit losses at inception and reassess quarterly based on management's expectation of the asset's collectability.

Contract assets remained relatively flat during the years ended December 31, 2022 and 2021.

Contract liabilities arise when we bill our customers and receive consideration in advance of providing the goods or services promised in the contract. We typically bill service one month in advance, which is the primary component of the contract liability balance. Contract liabilities are recognized as revenue when services are provided to the customer. The contract liability balances are presented in our consolidated balance sheets as Other current liabilities and Other liabilities.

Contract liabilities increased \$1.0 billion during the year ended December 31, 2022. The change in contract liabilities was primarily due to increases in sales promotions recognized over time and upfront fees. Contract liabilities increased \$1.6 billion during the year ended December 31, 2021. The change in contract liabilities was primarily due to increases in sales promotions

recognized over time and upfront fees, increases in deferred revenue related to advanced billings, as well as the acquisition of TracFone, partially offset by the satisfaction of performance obligations related to wireless and Fios services, as well as the sale of Verizon Media.

Revenue recognized during both the years ended December 31, 2022 and 2021 related to contract liabilities existing at January 1, 2022 and 2021 were \$0 billion and \$4.3 billion, respectively, as performance obligations related to services were satisfied.

The balance of contract assets and contract liabilities recorded in our consolidated balance sheets were as follows:

(dollars in millions)	At December 31, 2022		At December 31, 2021	
Assets				
Prepaid expenses and other	\$	656	\$	739
Other assets		207		195
Total	\$	863	\$	934
Liabilities				
Other current liabilities	\$	6,583	\$	6,053
Other liabilities		1,651		1,176
Total	\$	8,234	\$	7,229

Contract Costs

As discussed in Note 1, Topic 606 requires the recognition of an asset for incremental costs to obtain a customer contract, which are then amortized to expense over the respective periods of expected benefit. We recognize an asset for incremental commission expenses paid to internal and external sales personnel and agents in conjunction with obtaining customer contracts. We only defer these costs when we have determined the commissions are incremental costs that would not have been incurred absent the customer contract and are expected to be recoverable. Costs to obtain a contract are amortized and recorded ratably as commission expense over the period representing the transfer of goods or services to which the assets relate. Costs to obtain wireless contracts are amortized over both of our Consumer and Business customers' estimated device upgrade cycles, as such costs are typically incurred each time a customer upgrades. Costs to obtain wireline contracts are amortized as expense over the estimated customer relationship period for our Consumer customers. Incremental costs to obtain wireline contracts for our Business customers are insignificant. Costs to obtain contracts are recorded in Selling, general and administrative expense.

We also defer costs incurred to fulfill contracts that: (1) relate directly to the contract; (2) are expected to generate resources that will be used to satisfy our performance obligation under the contract; and (3) are expected to be recovered through revenue generated under the contract. Contract fulfillment costs are expensed as we satisfy our performance obligations and recorded in Cost of services. These costs principally relate to direct costs that enhance our wireline business resources, such as costs incurred to install circuits.

We determine the amortization periods for our costs incurred to obtain or fulfill a customer contract at a portfolio level due to the similarities within these customer contract portfolios.

Other costs, such as general costs or costs related to past performance obligations, are expensed as incurred.

Collectively, costs to obtain a contract and costs to fulfill a contract are referred to as deferred contract costs, and amortized over two-to six-year period. Deferred contract costs are classified as current or non-current within Prepaid expenses and other and Other assets, respectively.

The balances of deferred contract costs included in our consolidated balance sheets were as follows:

(dollars in millions)	At December 31, 2022		At December 31, 2021	
Assets				
Prepaid expenses and other	\$	2,629	\$	2,432
Other assets		2,475		2,259
Total	\$	5,104	\$	4,691

For both the years ended December 31, 2022 and 2021, we recognized expense of \$0 billion associated with the amortization of deferred contract costs, primarily within Selling, general and administrative expense in our consolidated statements of income.

We assess our deferred contract costs for impairment on a quarterly basis. We recognize an impairment charge to the extent the carrying amount of a deferred cost exceeds the remaining amount of consideration we expect to receive in exchange for the goods and services related to the cost, less the expected costs related directly to providing those goods and services that have

not yet been recognized as expenses. There have been no impairment charges recognized for the years ended December 31, 2022 and 2021.

Note 3. Acquisitions and Divestitures

Spectrum License Transactions

In March 2020, the FCC's incentive auction, Auction 103, for spectrum licenses in the upper 37 GHz, 39 GHz, and 47 GHz bands concluded. Verizon participated in this incentive auction and was the high bidder on 4,940 licenses, which primarily consisted of 37 GHz and, to a lesser extent, 39 GHz spectrum. As an incumbent licensee, our 39 GHz licenses provided us with incentive payments that were applied towards the purchase price of spectrum in the auction. The value of the licenses won by Verizon amounted to \$3.4 billion, of which \$1.8 billion was settled with the relinquished 39 GHz licenses. The remaining balance was settled in cash of \$1.6 billion. In connection with the incentive auction, a pre-tax net loss of \$.2 billion (\$914 million after-tax) was recorded in Selling, general and administrative expense in the consolidated statement of income during 2020 because the exchange of the previously held licenses for new licenses had commercial substance.

In February 2021, the FCC concluded Auction 107 for C-Band wireless spectrum. Verizon paid \$45.5 billion for the licenses it won, of which \$44.6 billion was paid in the first quarter of 2021. In accordance with the rules applicable to the auction, Verizon is required to make additional payments to acquire the licenses. The payments are for our allocable share of clearing costs incurred by, and incentive payments due to, the incumbent license holders associated with the auction, which are estimated to be \$7.7 billion. During 2021, we made payments of \$.3 billion primarily related to certain obligations for projected clearing costs. During 2022, we made additional payments of \$1.6 billion for obligations related to accelerated clearing incentives and clearing costs, of which \$.4 billion were accrued as of December 31, 2021 in our consolidated balance sheet. We expect to continue to make payments related to clearing cost and incentive payment obligations through 2024, which we expect to be \$4.8 billion. These payments are dependent on the incumbent license holders accelerated clearing of the spectrum for Verizon's use and, therefore, the final timing and amounts could differ based on the incumbent holders' execution of their clearing process. In accordance with the FCC order, the clearing must be completed by December 2025. The carrying value of the wireless spectrum won in Auction 107 consists of all payments required to participate and purchase licenses in the auction, including Verizon's allocable share of clearing costs incurred by, and incentive payments due to, the incumbent license holders associated with the auction that we are obligated to pay in order to acquire the licenses, as well as capitalized interest to the extent qualifying activities have occurred.

In March 2022, Verizon signed agreements with satellite operators in which operators agreed to clear C-Band spectrum in certain markets and frequencies ahead of the previously expected December 2023 timeframe. During 2022, Verizon incurred costs associated with these agreements of approximately \$340 million, of which \$310 million was paid as of December 31, 2022. This early clearance accelerated Verizon's access to more spectrum in a number of key markets to support its fifth-generation (5G) network initiatives.

During 2022 and 2021, we entered into and completed various other wireless license acquisitions for cash consideration of an insignificant amount and \$5 million, respectively. During 2021, we recognized a pre-tax loss in connection with the sale of certain wireless licenses of \$223 million (\$167 million after tax).

Business Acquisitions and Divestitures

TracFone Wireless, Inc.

In September 2020, we entered into a purchase agreement (TracFone Purchase Agreement) with América Móvil to acquire TracFone, a leading provider of prepaid and value mobile services in the U.S. The transaction closed on November 23, 2021 (the Acquisition Date). The acquisition positions Verizon as the leading prepaid, value and premium wireless carrier by expanding Verizon's portfolio, bringing enhanced access of our wireless network and comprehensive suite of mobility products and services to a new customer base.

In accordance with the terms of the TracFone Purchase Agreement, Verizon acquired all of TracFone's outstanding stock in exchange for approximately \$3.5 billion in cash, net of cash acquired and working capital and other adjustments, 57,596,544 shares of our common stock valued at approximately \$3.0 billion, and up to an additional \$650 million in future cash contingent consideration related to the achievement of certain performance measures and other commercial arrangements. The fair value of the common stock was determined on the basis of its closing market price on the Acquisition Date. The estimated fair value of the contingent consideration as of the Acquisition Date was approximately \$560 million and represents a Level 3 measurement as defined in ASC 820, Fair Value Measurements and Disclosures. See Note 9 for additional information. The contingent consideration payable is based on the achievement of certain revenue and operational targets, measured over a two-year earn out period, as defined in the TracFone Purchase Agreement.

In May 2022, Verizon received net cash proceeds of \$248 million for the final settlement of working capital, which was included in our consideration as of the Acquisition Date. During 2022, Verizon made payments of \$188 million related to the contingent consideration, which is reflected in Cash flows from financing activities in our consolidated statement of cash flows for the year ended December 31, 2022. In January 2023, we made an additional payment of \$102 million related to the contingent consideration. Contingent consideration payments are expected to continue through 2024.

The TracFone acquisition was accounted for as a business combination. The purchase consideration was allocated to the assets acquired and liabilities assumed based on their fair values as of the Acquisition Date.

The following table summarizes the allocation of the consideration paid and payable to the identified assets acquired and liabilities assumed as of the Acquisition Date:

(dollars in millions)	November 23, 2021	Measurement Period Adjustments ⁽¹⁾	Adjusted Fair Value
Consideration:			
Cash, net of cash acquired and working capital and other adjustments	\$ 3,491	\$ 7	\$ 3,498
Fair value of Verizon common stock (57,596,544 shares)	2,981	—	2,981
Fair value of contingent consideration to be paid	542	18	560
Total consideration	\$ 7,014	\$ 25	\$ 7,039
Assets acquired:			
Current assets	\$ 1,370	\$ 9	\$ 1,379
Property, plant and equipment, net	96	(10)	86
Goodwill	3,723	100	3,823
Other intangible assets	4,374	(19)	4,355
Other assets	731	(58)	673
Total assets acquired	\$ 10,294	\$ 22	\$ 10,316
Liabilities assumed:			
Current liabilities	\$ 1,433	\$ 70	\$ 1,503
Deferred income taxes	1,007	37	1,044
Other liabilities	840	(110)	730
Total liabilities assumed	\$ 3,280	\$ (3)	\$ 3,277
Net assets acquired	\$ 7,014	\$ 25	\$ 7,039

⁽¹⁾ Adjustments to the fair value measurements reflect new information obtained about facts and circumstances that existed as of the Acquisition Date, that if known, would have affected the measurement of the amounts recognized as of that date. The most significant adjustments related to goodwill, deferred income taxes, deferred commission costs and other intangible assets.

During 2022, we recorded a measurement period adjustment resulting in changes to the fair value and amortization period of certain intangible assets. As such, we recorded an adjustment to the fair value of certain intangible assets with a corresponding adjustment to goodwill. As a result of the changes in the provisional fair value and amortization period of certain intangible assets during the one-year measurement period, amortization expense decreased \$110 million for the year ended December 31, 2022. Other intangible assets include \$2.4 billion related to customer relationships, with a weighted-average amortization period of 7 years, \$1.1 billion related to distribution relationships, with an amortization period of 5 years, \$740 million related to trade names with a weighted-average amortization period of 17 years and \$145 million related to acquired technology, with an amortization period of 10 years. The intangible assets were assigned fair values using an income approach. The valuations are considered Level 3 fair value measurements due to the use of significant inputs not observable in the market, which include the discount rate, royalty rate and amount and timing of future cash flows.

Goodwill is calculated as the difference between the Acquisition Date fair value of the consideration paid and payable and the fair value of the net assets acquired, representing future economic benefits that we expect to achieve as a result of the acquisition. The goodwill related to this acquisition is included within the Consumer segment.

Pursuant to the TracFone Purchase Agreement, América Móvil agreed to indemnify Verizon against certain pre-acquisition tax matters and other contingencies. We have recorded total contingent liabilities and offsetting indemnification assets of \$672 million for the expected reimbursement of these matters that had not been resolved as of the Acquisition Date. The liabilities are presented in Other liabilities and the indemnification assets are presented in Other assets, within our consolidated balance sheets. The amounts recognized represent reasonable estimates based on an evaluation of current facts and circumstances. Actual outcome may differ significantly from these estimates. We expect that any additional liabilities that may arise related to these indemnified matters would be indemnified and reimbursed by América Móvil.

Bluegrass Cellular

In October 2020, we entered into a definitive agreement to acquire certain assets of Bluegrass Cellular, a rural wireless operator serving central Kentucky. The transaction closed in March 2021. The aggregate cash consideration paid by Verizon at the closing of the transaction was approximately \$412 million, net of cash acquired.

Verizon Media Divestiture

On May 2, 2021, Verizon entered into a definitive agreement with an affiliate of Apollo Global Management Inc. (the Apollo Affiliate) pursuant to which we agreed to sell Verizon Media in return for consideration of \$4.3 billion in cash, subject to customary adjustments, \$750 million in non-convertible preferred limited partnership units of the Apollo Affiliate and 10% of the fully-diluted common limited partnership units of the Apollo Affiliate.

On September 1, 2021, we completed the sale of Verizon Media. As of the close of the transaction, cash proceeds, the fair value of the non-convertible preferred limited partnership units of the Apollo Affiliate and the fair value of 10% of the fully-diluted common limited partnership units of the Apollo Affiliate were \$4.3 billion, \$496 million, and \$124 million, respectively. We recorded a pre-tax gain on sale of approximately \$1.0 billion (after-tax \$1.0 billion) in Selling general and administrative expense in our consolidated statement of income for the year ended December 31, 2021. In addition, we incurred \$346 million of various costs associated with this disposition which are primarily recorded in Selling general and administrative expense in our consolidated statement of income for the year ended December 31, 2021.

During 2022 and 2021, the Apollo Affiliate redeemed \$37 million and \$100 million, respectively, of Verizon's preferred limited partnership interest reducing the carrying value of our preferred interest as of December 31, 2022 to \$359 million. The redemptions are reflected within Net cash used in investing activities in our consolidated statements of cash flows for the years ended December 31, 2022 and December 31, 2021, respectively. Verizon's 10% common interest in the Apollo Affiliate is accounted for as an equity method investment. The post-sale results of Verizon's common ownership interest in the Apollo Affiliate are recorded through the equity method of accounting, within Corporate and other.

In connection with the closing of the transaction, we entered into Transition Services Agreements with the Apollo Affiliate, under which Verizon will continue to provide and receive specified technical services to support operations for up to 18 months.

Under our ownership, Verizon Media generated revenues from contracts with customers under Topic 606 of approximately \$5.3 billion and \$7.0 billion for the years ended December 31, 2021 and 2020, respectively, reflected within our Corporate and Other segment.

Other

During 2022 and 2021, we completed various other acquisitions for cash consideration of an insignificant amount and approximately \$1 million, respectively.

In connection with a non-strategic divestiture and other activity we recorded a pre-tax net gain of an insignificant amount in our consolidated statement of income for the year ended December 31, 2022.

In December 2021, we completed the sale of an investment. In connection with this transaction, we recorded a pre-tax gain of \$31 million in Equity in earnings (losses) of unconsolidated businesses in our consolidated statement of income for the year ended December 31, 2021.

Note 4. Wireless Licenses, Goodwill and Other Intangible Assets

Wireless Licenses

The carrying amounts of Wireless licenses are as follows:

At December 31,	(dollars in millions)	
	2022	2021
Wireless licenses	\$ 149,796	\$ 147,619

At December 31, 2022 and 2021, approximately \$41.7 billion and \$54.9 billion, respectively, of wireless licenses were under development for commercial service for which we were capitalizing interest costs. We recorded approximately \$1.7 billion and \$1.6 billion of capitalized interest on wireless licenses for the years ended December 31, 2022 and 2021, respectively.

During 2022 and 2021, we renewed various wireless licenses in accordance with FCC regulations with an average renewal period of 5 years and 11 years, respectively. See Note 1 for additional information.

As discussed in Note 1, we test our wireless licenses for potential impairment annually or more frequently if impairment indicators are present. In 2022, we performed a qualitative impairment assessment, which indicated it was more likely than not that the fair value of our wireless licenses remained above their carrying amount and, therefore, did not result in an impairment. In 2021, we performed a quantitative impairment test, which compared the estimated fair value of our aggregate wireless licenses, estimated using the Greenfield approach, to the aggregated carrying amount of the licenses as of the test date. Our annual assessment in 2021 indicated that the fair value of our wireless licenses exceeded the carrying value and, therefore, did not result in an impairment.

Our strategy requires significant capital investments primarily to acquire wireless spectrum, put the spectrum into service, provide additional capacity for growth in our networks, invest in the fiber that supports our businesses, evolve and maintain our networks and develop and maintain significant advanced information technology systems and data system capabilities.

Goodwill

Changes in the carrying amount of Goodwill are as follows:

	(dollars in millions)			
	Consumer	Business	Other	Total
Balance at January 1, 2021 ⁽¹⁾	\$ 17,222	\$ 7,535	\$ 16	\$ 24,773
Acquisitions ⁽²⁾	3,818	—	34	3,852
Reclassifications, adjustments and other	2	(20)	(4)	(22)
Balance at December 31, 2021	21,042	7,515	46	28,603
Acquisitions ⁽³⁾	100	—	—	100
Reclassifications, adjustments and other ⁽⁴⁾	—	(13)	(19)	(32)
Balance at December 31, 2022	\$ 21,142	\$ 7,502	\$ 27	\$ 28,671

⁽¹⁾ Goodwill is net of accumulated impairment charges of \$4.8 billion, related to our historical Media reporting unit, which included Verizon Media. On September 1, 2021, we completed the sale of Verizon Media. See Note 3 for additional information.

⁽²⁾ Changes in goodwill due to acquisitions is related to TracFone, Bluegrass and other insignificant transactions. See Note 3 for additional information.

⁽³⁾ Changes in goodwill due to acquisitions is related to TracFone. See Note 3 for additional information.

⁽⁴⁾ Includes a goodwill impairment charge of \$16 million related to an early stage development company presented within Other, recorded in Selling, general and administrative expense in our consolidated statement of income for the year ended December 31, 2022.

We performed a qualitative impairment assessment for our Consumer reporting unit during the fourth quarter of 2022. Our qualitative impairment assessment indicated that it was more likely than not that the fair value of our Consumer reporting unit exceeded its carrying value and, therefore, did not result in an impairment.

We performed a quantitative impairment assessment for our Business reporting unit during the fourth quarter of 2022. We applied a combination of a market approach and a discounted cash flow method, as a form of the income approach, reflecting current assumptions and inputs, including our revised projections, discount rate and expected growth rates, which indicated that the fair value of our Business reporting unit exceeded its carrying value and, therefore, did not result in an impairment. We do not anticipate reasonable changes in significant estimates to change the outcome of the quantitative impairment assessment. However, as a result of our 2022 goodwill assessment, management believes there is an increasing risk that our Business reporting unit may be required to recognize an impairment charge in the future.

A projected sustained decline in the reporting unit's revenues and earnings could have a significant negative impact on its fair value and may result in impairment charges. Such a decline could be driven by, among other things: (1) decreases in sales volumes or long-term growth rate as a result of competitive pressures or other factors; or (2) the inability to achieve or delays in achieving the goals in strategic initiatives. Also, adverse changes to macroeconomic factors, such as increases in long-term interest rates, would also negatively impact the fair value of the reporting unit.

We performed quantitative impairment assessments for both our Consumer and Business reporting units in 2021. We applied a combination of a market approach and a discounted cash flow method, as a form of the income approach, which indicated that the fair values for our Consumer and Business reporting units exceeded their carrying values and, therefore, did not result in an impairment.

Other Intangible Assets

The following table displays the composition of Other intangible assets, net as well as the respective amortization period:

At December 31,	(dollars in millions)					
	2022			2021		
	Gross Amount	Accumulated Amortization	Net Amount	Gross ⁽¹⁾ Amount	Accumulated Amortization	Net Amount
Customer lists (5 to 13 years)	\$ 4,335	\$ (1,646)	\$ 2,689	\$ 4,201	\$ (1,126)	\$ 3,075
Non-network internal-use software (7 years)	23,421	(16,397)	7,024	21,310	(14,897)	6,413
Other (4 to 25 years)	2,806	(1,058)	1,748	2,974	(785)	2,189
Total	\$ 30,562	\$ (19,101)	\$ 11,461	\$ 28,485	\$ (16,808)	\$ 11,677

⁽¹⁾ Other intangible assets are net of assets disposed as a result of the closing of the Verizon Media sale on September 1, 2021 and includes assets acquired as a result of the acquisition of TracFone on November 23, 2021. See Note 3 for additional information.

The amortization expense for Other intangible assets was as follows:

Years	(dollars in millions)
2022	\$ 2,507
2021	2,087
2020	2,445

Estimated annual amortization expense for Other intangible assets is as follows:

Years	(dollars in millions)
2023	\$ 2,497
2024	2,256
2025	2,059
2026	1,813
2027	1,260

Note 5. Property, Plant and Equipment

The following table displays the details of Property, plant and equipment, which is stated at cost:

At December 31,	Lives (years)	(dollars in millions)	
		2022	2021
Land	-	\$ 747	\$ 673
Buildings and equipment	7 to 45	35,382	33,361
Central office and other network equipment	3 to 15	162,001	153,240
Antennas, cable, conduit, poles and towers	4 to 50	75,622	69,733
Leasehold improvements	5 to 20	10,159	9,587
Work in progress	-	12,889	13,057
Furniture, vehicles and other	3 to 20	10,889	10,246
		307,689	289,897
Less accumulated depreciation		200,255	190,201
Property, plant and equipment, net		\$ 107,434	\$ 99,696

Note 6. Leasing Arrangements

We enter into various lease arrangements for network equipment including towers, distributed antenna systems, small cells, real estate and connectivity mediums including dark fiber, equipment, and other various types of assets for use in our operations. Our leases have remaining lease terms ranging from 1 year to 30 years, some of which include options that we can elect to extend the leases term for up to 25 years, and some of which include options to terminate the leases. For the majority of leases entered into during the current period, we have concluded it is not reasonably certain that we would exercise the options to extend the lease or not terminate the lease. Therefore, as of the lease commencement date, our lease terms generally do not include these options. We include options to extend the lease when it is reasonably certain that we will exercise that option.

During March 2015, we completed a transaction with American Tower Corporation (American Tower) pursuant to which American Tower acquired the exclusive rights to lease and operate approximately 11,300 of our wireless towers for an upfront payment of

\$5.0 billion. We have subleased capacity on the towers from American Tower for a minimum of 10 years at current market rates in 2015, with options to renew. We continue to include the towers in Property, plant and equipment, net in our consolidated balance sheets and depreciate them accordingly. In addition to the rights to lease and operate the towers, American Tower assumed the interest in the underlying ground leases related to these towers. While American Tower can renegotiate the terms of and is responsible for paying the ground leases, we are still the primary obligor for these leases and accordingly, the present value of these ground leases are included in our operating lease right-of-use assets and operating lease liabilities. We do not expect to be required to make ground lease payments unless American Tower defaults, which we determined to be remote.

The components of net lease cost were as follows:

		(dollars in millions)		
Years Ended December 31,	Classification	2022	2021	2020
Operating lease cost ⁽¹⁾	Cost of services			
	Selling, general and administrative expense	\$ 5,345	\$ 5,248	\$ 5,016
Finance lease cost:				
Amortization of right-of-use assets	Depreciation and amortization expense	224	259	309
Interest on lease liabilities	Interest expense	36	34	39
Short-term lease cost ⁽¹⁾	Cost of services			
	Selling, general and administrative expense	23	21	22
Variable lease cost ⁽¹⁾	Cost of services			
	Selling, general and administrative expense	294	307	295
Sublease income	Service revenues and other	(199)	(193)	(167)
Total net lease cost		\$ 5,723	\$ 5,676	\$ 5,514

⁽¹⁾ All operating lease costs, including short-term and variable lease costs, are split between Cost of services and Selling, general and administrative expense in the consolidated statements of income based on the use of the facility or equipment that the rent is being paid on. See Note 1 for additional information. Variable lease costs represent payments that are dependent on a rate or index, or on usage of the asset.

Supplemental disclosure for the statements of cash flows related to operating and finance leases were as follows:

		(dollars in millions)		
Years Ended December 31,		2022	2021	2020
Cash Flows from Operating Activities				
Cash paid for amounts included in the measurement of lease liabilities				
Operating cash flows for operating leases		\$ (4,490)	\$ (4,658)	\$ (4,813)
Operating cash flows for finance leases		(36)	(34)	(39)
Cash Flows from Financing Activities				
Financing cash flows for finance leases		(449)	(394)	(394)
Supplemental lease cash flow disclosures				
Operating lease right-of-use assets obtained in exchange for new operating lease liabilities		2,392	9,778	3,800
Right-of-use assets obtained in exchange for new finance lease liabilities		832	461	562

Supplemental disclosures for the balance sheet related to finance leases were as follows:

		(dollars in millions)	
At December 31,		2022	2021
Assets			
Property, plant and equipment, net		\$ 1,138	\$ 1,046
Liabilities			
Debt maturing within one year		\$ 565	\$ 400
Long-term debt		1,167	925
Total Finance lease liabilities		\$ 1,732	\$ 1,325

The weighted-average remaining lease term and the weighted-average discount rate of our leases were as follows:

At December 31,	2022	2021
Weighted-average remaining lease term (years)		
Operating leases	8	9
Finance leases	4	4
Weighted-average discount rate		
Operating leases	3.2 %	3.1 %
Finance leases	2.5 %	2.2 %

The following table presents the maturity analysis of operating and finance lease liabilities as of December 31, 2022:

Years	(dollars in millions)	
	Operating Leases	Finance Leases
2023	\$ 4,591	\$ 586
2024	4,318	518
2025	3,929	388
2026	3,571	196
2027	3,253	90
Thereafter	10,160	67
Total lease payments	29,822	1,845
Less interest	4,130	113
Present value of lease liabilities	25,692	1,732
Less current obligation	4,134	565
Long-term obligation at December 31, 2022	\$ 21,558	\$ 1,167

As of December 31, 2022, we have contractually obligated lease payments amounting to \$2.0 billion primarily for office facility operating leases and small cell colocation and fiber operating leases that have not yet commenced. We have legally obligated lease payments for various other operating leases that have not yet commenced for which the total obligation was not significant. We have certain rights and obligations for these leases, but have not recognized an operating lease right-of-use asset or an operating lease liability since they have not yet commenced.

Note 7. Debt

Outstanding long-term debt obligations as of December 31, 2022 and 2021 are as follows:

(dollars in millions)

At December 31,	Maturities	Interest Rates %	2022	2021
Verizon Communications	< 5 Years	0.75 - 5.82	\$ 23,929	\$ 18,406
	5-10 Years	1.50 - 7.88	42,637	43,225
	> 10 Years	1.13 - 8.95	60,134	73,520
	< 5 Years	Floating ⁽¹⁾	2,992	4,086
	5-10 Years	Floating ⁽¹⁾	3,029	824
Alltel Corporation	5-10 Years	6.80 - 7.88	94	38
	> 10 Years	N/A	N/A	58
Operating telephone company subsidiaries—debentures	< 5 Years	N/A	N/A	141
	5-10 Years	6.00 - 8.75	475	375
	> 10 Years	5.13 - 7.38	139	250
Other subsidiaries—asset-backed debt	< 5 Years	0.41 - 5.72	9,767	9,620
	< 5 Years	Floating ⁽²⁾	10,271	4,610
Finance lease obligations (average rate of 2.5% and 2.2% in 2022 and 2021, respectively)			1,732	1,325
Unamortized discount, net of premium			(4,039)	(4,922)
Unamortized debt issuance costs			(671)	(688)
Total long-term debt, including current maturities			150,489	150,868
Less long-term debt maturing within one year			9,813	7,443
Total long-term debt			\$ 140,676	\$ 143,425
Long-term debt maturing within one year			\$ 9,813	\$ 7,443
Add commercial paper			150	—
Debt maturing within one year			9,963	7,443
Add long-term debt			140,676	143,425
Total debt			\$ 150,639	\$ 150,868

⁽¹⁾ The debt obligations bore interest at a floating rate based on the Compounded Secured Overnight Financing Rate (SOFR) for the interest period or the London Interbank Offered Rate (LIBOR) plus an applicable interest margin per annum, as applicable. Compounded SOFR is calculated using the SOFR Index published by the Federal Reserve Bank of New York in accordance with the formula set forth in the terms of the notes. The Compounded SOFR for the interest period ending in December 2022 was 3.450%. The one-month and three-month LIBOR at December 31, 2022 was 4.392% and 4.767%, respectively.

⁽²⁾ The debt obligations bore interest at floating rates, including floating rates associated with SOFR for the interest period, or LIBOR plus an applicable interest margin per annum, as applicable. Floating rates associated with SOFR for the interest period ending in December 2022 ranged from 3.807% to 4.372%.

Maturities of long-term debt (secured and unsecured) outstanding, including current maturities, excluding unamortized debt issuance costs, at December 31, 2022 are as follows:

Years	(dollars in millions)
2023	\$ 9,279
2024	16,252
2025	8,706
2026	8,304
2027	6,962
Thereafter	99,925

During 2022, we received \$17.8 billion of proceeds from long-term borrowings, which included \$10.7 billion of proceeds from asset-backed debt transactions. The net proceeds were primarily used for general corporate purposes including the repayment of debt and the funding of certain renewable energy projects. We used \$13.6 billion of cash to repay, redeem and repurchase long-term borrowings and finance lease obligations, including \$4.9 billion to prepay and repay asset-backed, long-term borrowings. The net proceeds of approximately \$1.0 billion from the green bond issued in 2022 are expected to be used to fund certain renewable energy projects.

During 2021, we received \$41.4 billion of proceeds from long-term borrowings, which included \$8.4 billion of proceeds from asset-backed debt transactions. The net proceeds were primarily used to finance the purchase of wireless licenses won in connection with the FCC's auction for C-Band wireless spectrum, Auction 107, and fund certain renewable energy projects. We

used \$18.9 billion of cash to repay, redeem and repurchase long-term borrowings and finance lease obligations, including \$4.8 billion to prepay and repay asset-backed, long-term borrowings. The net proceeds of approximately \$1.0 billion from the green bond issued in 2021 were used to fund certain renewable energy projects.

2022 Significant Debt Transactions

Debt or equity financing may be needed to fund additional investments or development activities or to maintain an appropriate capital structure to ensure our financial flexibility.

The following tables show the significant transactions involving the senior unsecured debt securities of Verizon and its subsidiaries that occurred during the year ended December 31, 2022.

Tender Offers

(dollars in millions)	Principal Amount Purchased		Cash Consideration ⁽¹⁾
Verizon and subsidiary 2.987% - 8.950% notes and debentures, due 2032 - 2056	\$	5,032	\$ 5,587

⁽¹⁾ The total cash consideration includes the tender offer consideration, plus any accrued and unpaid interest to the date of purchase.

Repayments, Redemptions and Repurchases

(dollars in millions)	Principal Repaid/ Redeemed/ Repurchased		Amount Paid ⁽¹⁾
Verizon floating rate (LIBOR + 1.000%) notes due 2022	\$	1,094	\$ 1,097
Subsidiary 8.000% debentures due 2022		103	108
Open market repurchases of various Verizon notes ⁽²⁾		896	759
Total	\$	2,093	\$ 1,964

⁽¹⁾ Represents amount paid to repay or repurchase, including any accrued interest

⁽²⁾ During 2022, we recorded net gains of \$119 million in connection with the open market repurchases, which were reflected within Other income (expense), net in our consolidated statements of income.

Issuances

(dollars in millions)	Principal Amount Issued		Net Proceeds ⁽¹⁾
Verizon 3.875% notes due 2052 ⁽²⁾	\$	1,000	\$ 982
Verizon 4.100% notes due 2055		655	650
Verizon 4.250% notes due 2030 ⁽³⁾		1,250	1,227
Verizon 4.750% notes due 2034 ⁽³⁾	€	1,250	1,231
Total			\$ 4,090

⁽¹⁾ Net proceeds were net of underwriting discounts and other issuance costs. In addition, for securities denominated in a currency other than the U.S. dollar, net proceeds are shown on a U.S. dollar equivalent basis.

⁽²⁾ An amount equal to the net proceeds from this green bond is expected to be used to fund, in whole or in part, certain renewable energy projects, including new and existing investments made by us during the period from December 1, 2021 through the maturity date of the green bond.

⁽³⁾ See Note 9 for additional information on cross currency swap transactions related to the issuances.

Short-term Borrowing and Commercial Paper Program

In March 2022, we entered into and fully drew from a \$1.0 billion short-term uncommitted credit facility. In December 2022, we fully repaid \$1.0 billion under the short-term uncommitted credit facility. As of December 31, 2022, we had no short-term borrowing outstanding.

In 2022, we issued \$27.4 billion in commercial paper and we repaid \$27.3 billion of commercial paper. As of December 31, 2022, we had \$150 million of commercial paper outstanding. These transactions are reflected within Cash flows from financing activities in our consolidated statements of cash flows.

Asset-Backed Debt

As of December 31, 2022, the carrying value of our asset-backed debt was \$20.0 billion. Our asset-backed debt includes Asset-Backed Notes (ABS Notes) issued to third-party investors (Investors) and loans (ABS Financing Facilities) received from banks and their conduit facilities (collectively, the Banks). Our consolidated asset-backed debt bankruptcy remote legal entities (each, an ABS Entity or collectively, the ABS Entities) issue the debt or are otherwise party to the transaction documentation in

connection with our asset-backed debt transactions. Under the terms of our asset-backed debt, Cellco Partnership (Cellco), a wholly-owned subsidiary of the Company, and certain other Company affiliates (collectively, the Originators) transfer device payment plan agreement receivables and certain other receivables (collectively referred to as certain receivables) to one of the ABS Entities, which in turn transfers such receivables to another ABS Entity that issues the debt. Verizon entities retain the equity interests and residual interests, as applicable, in the ABS Entities, which represent the rights to all funds not needed to make required payments on the asset-backed debt and other related payments and expenses.

Our asset-backed debt is secured by the transferred receivables and future collections on such receivables. These receivables transferred to the ABS Entities and related assets, consisting primarily of restricted cash, will only be available for payment of asset-backed debt and expenses related thereto, payments to the Originators in respect of additional transfers of certain receivables, and other obligations arising from our asset-backed debt transactions, and will not be available to pay other obligations or claims of Verizon's creditors until the associated asset-backed debt and other obligations are satisfied. The Investors or Banks, as applicable, which hold our asset-backed debt have legal recourse to the assets securing the debt, but do not have any recourse to Verizon with respect to the payment of principal and interest on the debt. Under a parent support agreement, the Company has agreed to guarantee certain of the payment obligations of Cellco and the Originators to the ABS Entities.

Cash collections on the receivables collateralizing our asset-backed debt securities are required at certain specified times to be placed into segregated accounts. Deposits to the segregated accounts are considered restricted cash and are included in Prepaid expenses and other and Other assets in our consolidated balance sheets.

Proceeds from our asset-backed debt transactions are reflected in Cash flows from financing activities in our consolidated statements of cash flows. The asset-backed debt issued is included in Debt maturing within one year and Long-term debt in our consolidated balance sheets.

ABS Notes

During the year ended December 31, 2022, we completed the following ABS Notes transactions:

(dollars in millions)	Interest Rates %	Expected Weighted-average Life to Maturity (in years)	Principal Amount Issued
January 2022			
Series 2022-1			
A Senior class notes	1.040	1.49	\$ 799
B Junior class notes	1.270	1.49	64
C Junior class notes	1.390	1.49	37
Series 2022-2			
A Senior class notes	1.530	2.99	710
B Junior class notes	1.830	2.99	57
C Junior class notes	2.010	2.99	33
January 2022 total			1,700
May 2022			
Series 2022-3			
A Senior class notes	3.010	1.49	400
B Junior class notes	3.250	1.49	—
C Junior class notes	3.500	1.49	16
Series 2022-4			
A Senior class notes	3.400	2.99	488
B Junior class notes	3.640	2.99	42
C Junior class notes	3.890	2.99	20
May 2022 total			966
August 2022			
Series 2022-5			
A-1a Senior fixed rate class notes	3.720	1.44	351
A-1b Senior floating rate class notes	Compounded SOFR + 0.620	1.44	57
B Junior class notes	3.960	1.44	—
C Junior class notes	4.210	1.44	—
Series 2022-6			
A Senior class notes	3.670	2.94	479
B Junior class notes	3.910	2.94	—
C Junior class notes	4.160	2.94	—
August 2022 total			887
November 2022			
Series 2022-7			
A-1a Senior fixed rate class notes	5.230	1.99	624
A-1b Senior floating rate class notes	Compounded SOFR + 0.850	1.99	267
B Junior class notes	5.480	1.99	—
C Junior class notes	5.720	1.99	41
November 2022 total			932
Total			\$ 4,485

Under the terms of each series of ABS Notes, there is a revolving period of 18 months, two years or up to three years, as applicable, during which we may transfer additional receivables to the ABS Entity. During the year ended December 31, 2022, we made aggregate principal repayments of \$4.3 billion on ABS Notes that have entered the amortization period, including principal payments made in connection with clean-up redemptions. During the year ended December 31, 2021, we made aggregate principal repayments of \$3.3 billion on ABS Notes that had entered the amortization period, including principal payments made in connection with clean-up redemptions.

In January 2023, we issued \$932 million aggregate principal amount of ABS Notes, with a blended interest rate of approximately 4.511%, through an ABS Entity.

ABS Financing Facilities

Under the two loan agreements in connection with the ABS Financing Facility entered into in December 2021 (2021 ABS Financing Facility), we prepaid an aggregate of \$515 million in January 2022, borrowed an additional \$1.9 billion in March 2022, prepaid an aggregate of \$30 million in May 2022 and borrowed an additional \$545 million in June 2022. In December 2022, we renewed the loan agreements in connection with the 2021 ABS Financing Facility which reset the revolving period of one or two years, as applicable, and we borrowed an additional \$1.9 billion. The aggregate outstanding balance under the 2021 ABS Financing Facility was \$7.9 billion as of December 31, 2022.

In October 2022, we entered into an ABS financing facility with a financial institution (2022 ABS Financing Facility). Under the terms of the 2022 ABS Financing Facility, the financial institution makes advances under asset-backed loans backed by certain wireless service accounts receivables of both Consumer customers and Business customers. The loan agreement entered into in connection with the 2022 ABS Financing Facility is outstanding with a final maturity date in June 2024 and the loan agreement bears interest at a floating rate. There is an 18 month revolving period, as set forth in the loan agreement, which may be extended with the approval of the financial institution. Under the loan agreement, we have the right to prepay all or a portion of the advances at any time without penalty, but in certain cases, with breakage costs. Subject to certain conditions, we may also remove receivables from the applicable ABS Entity. In October 2022, we borrowed \$2.0 billion under the loan agreement entered into in connection with the 2022 ABS Financing Facility. The aggregate outstanding balance under the 2022 ABS Financing Facility was \$2.0 billion as of December 31, 2022.

Variable Interest Entities

The ABS Entities meet the definition of a VIE for which we have determined that we are the primary beneficiary as we have both the power to direct the activities of the entity that most significantly impact the entity's performance and the obligation to absorb losses or the right to receive benefits of the entity. Therefore, the assets, liabilities and activities of the ABS Entities are consolidated in our financial results and are included in amounts presented on the face of our consolidated balance sheets.

The assets and liabilities related to our asset-backed debt arrangements included in our consolidated balance sheets were as follows:

(dollars in millions)	At December 31, 2022	At December 31, 2021
Assets		
Accounts receivable, net	\$ 13,906	\$ 10,705
Prepaid expenses and other	1,409	1,094
Other assets	9,894	5,455
Liabilities		
Accounts payable and accrued liabilities	22	10
Debt maturing within one year	6,809	5,024
Long-term debt	13,199	9,178

See Note 8 for additional information on certain receivables used to secure asset-backed debt.

Long-Term Credit Facilities

(dollars in millions)	Maturities	Facility Capacity	At December 31, 2022		
			Unused Capacity	Principal Amount Outstanding	
Verizon revolving credit facility ⁽¹⁾	2026	\$ 9,500	\$ 9,436	\$	—
Various export credit facilities ⁽²⁾	2024 - 2031	11,000	1,000		6,735
Total		\$ 20,500	\$ 10,436	\$	6,735

⁽¹⁾ The revolving credit facility does not require us to comply with financial covenants or maintain specified credit ratings, and it permits us to borrow even if our business has incurred a material adverse change. The revolving credit facility provides for the issuance of letters of credit. As of December 31, 2022, there have been no drawings against the \$9.5 billion revolving credit facility since its inception.

⁽²⁾ During 2022 and 2021, we drew down \$3.0 billion and \$1.0 billion, respectively, from these facilities. Borrowings under certain of these facilities are amortized semi-annually in equal installments up to the applicable maturity dates. Maturities reflect maturity dates of principal amounts outstanding. Any amounts borrowed under these facilities and subsequently repaid cannot be reborrowed.

In January 2023, we submitted an irrevocable request to borrow approximately \$515 million from funds available under our export credit facilities in March 2023.

Non-Cash Transactions

During the years ended December 31, 2022, 2021 and 2020, we financed, primarily through alternative financing arrangements, the purchase of approximately \$32 million, \$461 million, and \$1.7 billion, respectively, of long-lived assets consisting primarily of network equipment. As of December 31, 2022 and 2021, \$1.7 billion and \$1.3 billion, respectively, relating to these financing arrangements, including those entered into in prior years and liabilities assumed through acquisitions, remained outstanding. These purchases are non-cash financing activities and therefore are not reflected within Capital expenditures in our consolidated statements of cash flows.

Net Debt Extinguishment Losses

During the years ended December 31, 2022, 2021 and 2020, we recorded net debt extinguishment losses of \$1.1 billion, \$3.6 billion and \$121 million, respectively. The net losses are recorded in Other income (expense), net in our consolidated statements of income. The total losses are reflected within Other, net cash flow from operating activities and the portion of the losses representing cash payments are reflected within Other, net cash flow from financing activities in our consolidated statements of cash flows.

Guarantees

We guarantee the debentures of our operating telephone company subsidiaries. As of December 31, 2022, \$14 million aggregate principal amount of these obligations remained outstanding. Each guarantee will remain in place for the life of the obligation unless terminated pursuant to its terms, including the operating telephone company no longer being a wholly-owned subsidiary of the Company.

Debt Covenants

We and our consolidated subsidiaries are in compliance with all of our restrictive covenants in our debt agreements.

Note 8. Device Payment Plan Agreement and Wireless Service Receivables

The following table presents information about accounts receivable, net of allowances, recorded in our consolidated balance sheet:

(dollars in millions)	At December 31, 2022			
	Device payment plan agreement	Wireless service	Other receivables ⁽¹⁾	Total
Accounts receivable	\$ 13,374	\$ 5,140	\$ 6,818	\$ 25,332
Less Allowance for credit losses	445	143	238	826
Accounts receivable, net of allowance	\$ 12,929	\$ 4,997	\$ 6,580	\$ 24,506

⁽¹⁾ Other receivables primarily include wireline receivables and other receivables, the allowances for which are individually insignificant.

Included in Other assets and Accounts receivable, net at December 31, 2022 are net device payment plan agreement receivables and net wireless service receivables of \$23.6 billion, which have been transferred to ABS Entities and continue to be

reported in our consolidated balance sheet. Included in Other assets and Accounts receivable, net at December 31, 2021, are net device payment plan agreement receivables of \$16.0 billion, which have been transferred to ABS Entities and continue to be reported in our consolidated balance sheet. See Note 7 for additional information. We believe the carrying value of these receivables approximate their fair value using a Level 3 expected cash flow model.

Under the Verizon device payment program, our eligible wireless customers purchase wireless devices under a device payment plan agreement. Customers that activate service on devices purchased under the device payment program pay lower service fees as compared to those under our fixed-term service plans, and their device payment plan charge is included on their wireless monthly bill. We no longer offer Consumer customers new fixed-term, subsidized service plans for devices; however, we continue to offer subsidized plans to our Business customers. We also continue to service existing plans for customers who have not yet purchased and activated devices under the Verizon device payment program.

Wireless Device Payment Plan Agreement Receivables

The following table displays device payment plan agreement receivables, net, recognized in our consolidated balance sheets:

		(dollars in millions)	
At December 31,		2022	2021
Device payment plan agreement receivables, gross	\$	26,188	\$ 21,303
Unamortized imputed interest		(479)	(358)
Device payment plan agreement receivables, at amortized cost		25,709	20,945
Allowance ⁽¹⁾		(881)	(759)
Device payment plan agreement receivables, net	\$	24,828	\$ 20,186
Classified in our consolidated balance sheets:			
Accounts receivable, net	\$	12,929	\$ 12,783
Other assets		11,899	7,403
Device payment plan agreement receivables, net	\$	24,828	\$ 20,186

⁽¹⁾ Includes allowance for both short-term and long-term device payment plan agreement receivables.

For indirect channel wireless contracts with customers, we impute risk adjusted interest on the device payment plan agreement receivables. We record the imputed interest as a reduction to the related accounts receivable. Interest income, which is included within Service revenues and other in our consolidated statements of income, is recognized over the financed device payment term.

Promotions

In connection with certain device payment plan agreements, we may offer a promotion to allow our customers to upgrade to a new device after paying down a certain specified portion of the required device payment plan agreement amount as well as trading in their device in good working order. When a customer enters into a device payment plan agreement with the right to upgrade to a new device, we account for this trade-in right as a guarantee obligation. We recognize a liability measured at fair value for the customer's right to trade in the device which is determined by considering several factors, including the weighted-average selling prices obtained in recent resales of similar devices eligible for trade-in. At December 31, 2022 and December 31, 2021, the amount of the guarantee liability was \$54 million and \$77 million, respectively.

We may offer certain promotions that allow a customer to trade in their owned device in connection with the purchase of a new device. Under these types of promotions, the customer receives a credit for the value of the trade-in device. At December 31, 2022 and December 31, 2021, the amount of trade-in liability was \$562 million and \$366 million, respectively.

In addition, we may provide the customer with additional future billing credits that will be applied against the customer's monthly bill as long as service is maintained. These future billing credits are accounted for as consideration payable to a customer and are included in the determination of total transaction price, resulting in a contract liability.

Device payment plan agreement receivables, net, does not reflect the trade-in liability, additional future credits or the guarantee liability.

Origination of Device Payment Plan Agreements

When originating device payment plan agreements, we use internal and external data sources to create a credit risk score to measure the credit quality of a customer and to determine eligibility for the device payment program. Verizon's experience has been that the payment attributes of longer tenured customers are highly predictive for estimating their reliability to make future payments. Customers with longer tenures tend to exhibit similar risk characteristics to other customers with longer tenures, and receivables due from customers with longer tenures tend to perform better than receivables from customers that have not previously been Verizon customers. As a result of this experience, we make initial lending decisions based upon whether the customers are "established customers" or "short-tenured customers." If a Consumer customer has been a customer for 45 days

or more, or if a Business customer has been a customer for 12 months or more, the customer is considered an "established customer." For established customers, the credit decision and ongoing credit monitoring processes rely on a combination of internal and external data sources. If a Consumer customer has been a customer less than 45 days, or a Business customer has been a customer for less than 12 months, the customer is considered a "short-tenured customer." For short-tenured customers, the credit decision and credit monitoring processes rely more heavily on external data sources.

Available external credit data from credit reporting agencies along with internal data are used to create custom credit risk scores for Consumer customers. The custom credit risk score is generated automatically from the applicant's credit data using proprietary custom credit models. The credit risk score measures the likelihood that the potential customer will become severely delinquent and be disconnected for non-payment. For a small portion of short-tenured customer applications, a traditional credit report is not available from one of the national credit reporting agencies because the potential customer does not have sufficient credit history. In those instances, alternative credit data is used for the risk assessment. For Business customers, we also verify the existence of the business with external data sources.

Based on the custom credit risk score, we assign each customer a credit class, each of which has specified offers of credit. This includes an account level spending limit and a maximum amount of credit allowed per device for Consumer customers or a required down payment percentage for Business customers.

Credit Quality Information

Subsequent to origination, we assess indicators for the quality of our wireless device payment plan agreement portfolio using two models, one for new customers and one for existing customers. The model for new customers pools all Consumer and Business wireless customers based on less than 210 days as "new customers." The model for existing customers pools all Consumer and Business wireless customers based on 210 days or more as "existing customers."

The following table presents device payment plan agreement receivables, at amortized cost, as of December 31, 2022, by credit quality indicator and year of origination:

(dollars in millions)	Year of Origination ⁽¹⁾				Total
	2022	2021	Prior to 2021		
New customers	\$ 2,761	\$ 904	\$ 24	\$	3,689
Existing customers	16,312	5,544	164		22,020
Total	\$ 19,073	\$ 6,448	\$ 188	\$	25,709

⁽¹⁾ Includes accounts that have been suspended at a point in time.

The data presented in the table above was last updated on December 31, 2022.

We assess indicators for the quality of our wireless service receivables portfolio as one overall pool. As of December 31, 2022, wireless service receivables, at amortized cost, originating in 2022 and 2021 were \$5.1 billion and \$58 million, respectively.

Allowance for Credit Losses

The credit quality indicators are used in determining the estimated amount and the timing of expected credit losses for the device payment plan agreement and wireless service receivables portfolios.

For device payment plan agreement receivables, we record bad debt expense based on a default and loss calculation using our proprietary loss model. The expected loss rate is determined based on customer credit scores and other qualitative factors as noted above. The loss rate is assigned individually on a customer by customer basis and the custom credit scores are then aggregated by vintage and used in our proprietary loss model to calculate the weighted-average loss rate used for determining the allowance balance.

We monitor the collectability of our wireless service receivables as one overall pool. Wireline service receivables are disaggregated and pooled by the following customer groups: consumer, small and medium business, global enterprise, public sector and wholesale. For wireless service receivables and wireline consumer and small and medium business receivables, the allowance is calculated based on a 12 month rolling average write-off balance multiplied by the average life-cycle of an account from billing to write-off. The risk of loss is assessed over the contractual life of the receivables and is adjusted based on the historical loss amounts for current and future conditions based on management's qualitative considerations. For global enterprise, public sector and wholesale wireline receivables, the allowance for credit losses is based on historical write-off experience and individual customer credit risk, if applicable.

Activity in the allowance for credit losses by portfolio segment of receivables was as follows:

(dollars in millions)	Device Payment Plan Agreement Receivables ⁽¹⁾	Wireless Service Plan Receivables
Balance at January 1, 2022	\$ 759	\$ 130
Current period provision for expected credit losses	958	428
Write-offs charged against the allowance	(869)	(463)
Recoveries collected	33	48
Balance at December 31, 2022	\$ 881	\$ 143

⁽¹⁾ Includes allowance for both short-term and long-term device payment plan agreement receivables.

We monitor delinquency and write-off experience based on the quality of our device payment plan agreement and wireless service receivables portfolios. The extent of our collection efforts with respect to a particular customer are based on the results of our proprietary custom internal scoring models that analyze the customer's past performance to predict the likelihood of the customer falling further delinquent. These custom scoring models assess a number of variables, including origination characteristics, customer account history and payment patterns. Since our customers' behaviors may be impacted by general economic conditions, we analyzed whether changes in macroeconomic conditions impact our credit loss experience and have concluded that our credit loss estimates are generally not materially impacted by reasonable and supportable forecasts of future economic conditions. Based on the score derived from these models, accounts are grouped by risk category to determine the collection strategy to be applied to such accounts. For device payment plan agreement receivables and wireless service receivables, we consider an account to be delinquent and in default status if there are unpaid charges remaining on the account on the day after the bill's due date. The risk class determines the speed and severity of the collections effort including initiatives taken to facilitate customer payment.

The balance and aging of the device payment plan agreement receivables, at amortized cost, were as follows:

(dollars in millions)	At December 31, 2022
Unbilled	\$ 24,480
Billed:	
Current	998
Past due	231
Device payment plan agreement receivables, at amortized cost	\$ 25,709

Note 9. Fair Value Measurements and Financial Instruments

Recurring Fair Value Measurements

The following table presents the balances of assets and liabilities measured at fair value on a recurring basis as of December 31, 2022:

(dollars in millions)

	Level 1 ⁽¹⁾	Level 2 ⁽²⁾	Level 3 ⁽³⁾	Total
Assets:				
Prepaid expenses and other:				
Fixed income securities	\$ —	\$ 37	\$ —	\$ 37
Cross currency swaps	—	42	—	42
Foreign exchange forwards	—	6	—	6
Interest rate caps	—	63	—	63
Other assets:				
Fixed income securities	—	349	—	349
Cross currency swaps	—	263	—	263
Interest rate caps	—	30	—	30
Total	\$ —	\$ 790	\$ —	\$ 790
Liabilities:				
Other current liabilities:				
Interest rate swaps	\$ —	\$ 731	\$ —	\$ 731
Cross currency swaps	—	346	—	346
Interest rate caps	—	63	—	63
Foreign exchange forwards	—	1	—	1
Contingent consideration	—	—	274	274
Other liabilities:				
Interest rate swaps	—	3,902	—	3,902
Cross currency swaps	—	3,295	—	3,295
Interest rate caps	—	30	—	30
Contingent consideration	—	—	43	43
Total	\$ —	\$ 8,368	\$ 317	\$ 8,685

⁽¹⁾ Quoted prices in active markets for identical assets or liabilities.

⁽²⁾ Observable inputs other than quoted prices in active markets for identical assets and liabilities.

⁽³⁾ Unobservable pricing inputs in the market.

The following table presents the balances of assets and liabilities measured at fair value on a recurring basis as of December 31, 2021:

(dollars in millions)

	Level 1 ⁽¹⁾	Level 2 ⁽²⁾	Level 3 ⁽³⁾	Total
Assets:				
Prepaid expenses and other:				
Fixed income securities	\$ —	\$ 18	\$ —	\$ 18
Interest rate swaps	—	188	—	188
Cross currency swaps	—	9	—	9
Foreign exchange forwards	—	12	—	12
Other assets:				
Fixed income securities	—	391	—	391
Interest rate swaps	—	285	—	285
Cross currency swaps	—	580	—	580
Interest rate caps	—	44	—	44
Total	\$ —	\$ 1,527	\$ —	\$ 1,527
Liabilities:				
Other current liabilities:				
Interest rate swaps	\$ —	\$ 1	\$ —	\$ 1
Forward starting interest rate swaps	—	302	—	302
Cross currency swaps	—	218	—	218
Contingent consideration	—	—	231	231
Other liabilities:				
Interest rate swaps	—	665	—	665
Cross currency swaps	—	1,406	—	1,406
Interest rate caps	—	44	—	44
Contingent consideration	—	—	313	313
Total	\$ —	\$ 2,636	\$ 544	\$ 3,180

⁽¹⁾ Quoted prices in active markets for identical assets or liabilities.

⁽²⁾ Observable inputs other than quoted prices in active markets for identical assets and liabilities.

⁽³⁾ Unobservable pricing inputs in the market.

Certain of our equity investments do not have readily determinable fair values and are excluded from the tables above. Such investments are measured at cost, less any impairment, plus or minus changes resulting from observable price changes in orderly transactions for an identical or similar investment of the same issuer and are included in Investments in unconsolidated businesses in our consolidated balance sheets. As of December 31, 2022 and December 31, 2021, the carrying amount of our investments without readily determinable fair values was \$804 million and \$808 million, respectively. During 2022, there were insignificant adjustments due to observable price changes and insignificant impairment charges. Cumulative adjustments due to observable price changes and impairment charges were approximately \$176 million and \$82 million, respectively.

Verizon has a liability for contingent consideration related to its acquisition of TracFone, completed in November 2021. The fair value is calculated using a probability-weighted discounted cash flow model and represents a Level 3 measurement. Level 3 instruments include valuation based on unobservable inputs reflecting our own assumptions, consistent with reasonably available assumptions made by other market participants. Subsequent to the Acquisition Date, at each reporting date, the contingent consideration liability is remeasured to fair value. During 2022, we recorded gains of \$57 million related to fair value adjustments for the contingent consideration within Selling, general and administrative expense in our consolidated statement of income. During 2022, we made payments of \$88 million related to the contingent consideration. In January 2023, we made an additional payment of \$102 million related to the contingent consideration. See Note 3 for additional information.

Fixed income securities consist primarily of investments in municipal bonds. The valuation of the fixed income securities is based on the quoted prices for similar assets in active markets or identical assets in inactive markets or models that apply inputs from observable market data. The valuation determines that these securities are classified as Level 2.

Derivative contracts are valued using models based on readily observable market parameters for all substantial terms of our derivative contracts and thus are classified within Level 2. We use mid-market pricing for fair value measurements of our derivative instruments. Our derivative instruments are recorded on a gross basis.

We recognize transfers between levels of the fair value hierarchy as of the end of the reporting period.

Fair Value of Short-term and Long-term Debt

The fair value of our debt is determined using various methods, including quoted prices for identical debt instruments, which is a Level 1 measurement, as well as quoted prices for similar debt instruments with comparable terms and maturities, which is a Level 2 measurement.

The fair value of our short-term and long-term debt, excluding finance leases, was as follows:

(dollars in millions)	Carrying Amount	Fair Value				Total
		Level 1	Level 2	Level 3		
At December 31, 2021	\$ 149,543	\$ 106,599	\$ 62,606	\$ —	\$	169,205
At December 31, 2022	148,906	84,385	54,656	—		139,041

Derivative Instruments

We enter into derivative transactions primarily to manage our exposure to fluctuations in foreign currency exchange rates and interest rates. We employ risk management strategies, which may include the use of a variety of derivatives including interest rate swaps, cross currency swaps, forward starting interest rate swaps, treasury rate locks, interest rate caps, swaptions and foreign exchange forwards. We do not hold derivatives for trading purposes.

The following table sets forth the notional amounts of our outstanding derivative instruments:

At December 31,	2022		2021	
Interest rate swaps	\$	26,071	\$	19,779
Cross currency swaps		34,976		32,502
Forward starting interest rate swaps		—		1,000
Foreign exchange forwards		920		932

The following tables summarize the activities of our designated derivatives:

Years Ended December 31,	2022		2021	
Interest Rate Swaps:				
Notional value entered into	\$	7,155	\$	6,050
Notional value settled		863		4,018
Pre-tax gain recognized in Interest expense		2		2
Cross Currency Swaps:				
Notional value entered into		2,474		6,214
Notional value settled		—		—
Pre-tax loss recognized in Other comprehensive loss ⁽¹⁾		(430)		(2,285)
Pre-tax loss on cross currency swaps recognized in Interest expense		(1,373)		N/A
Pre-tax gain on hedged debt recognized in Interest expense		1,373		N/A
Excluded components recognized in Other comprehensive loss		(498)		N/A
Initial value of the excluded component amortized into Interest expense		81		N/A
Forward Starting Interest Rate Swaps:				
Notional value entered into		—		—
Notional value settled		1,000		1,000
Pre-tax gain recognized in Other comprehensive loss		196		258
Treasury Rate Locks:				
Notional value entered into		—		4,650
Notional value settled		—		4,650
Pre-tax gain recognized in Other comprehensive loss		—		251

N/A - not applicable

⁽¹⁾ Represents amounts recorded under the cash flow hedge model. These instruments were re-designated as fair value hedges on March 31, 2022.

	(dollars in millions)	
Years Ended December 31,	2022	2021
Other, net Cash Flows from Operating Activities:		
Cash received for settlement of interest rate swaps	\$ 40	\$ 107
Cash paid for settlement of forward starting interest rate swaps	(107)	(237)
Cash received for settlement of treasury rate locks	—	251

The following table displays the amounts recorded in Long-term debt in our consolidated balance sheets related to cumulative basis adjustments for fair value hedges. The cumulative amounts exclude cumulative basis adjustments related to foreign exchange risk.

	(dollars in millions)	
At December 31,	2022	2021
Carrying amount of hedged liabilities	\$ 21,741	\$ 20,027
Cumulative amount of fair value hedging adjustment included in the carrying amount of the hedged liabilities	(4,512)	(113)
Cumulative amount of fair value hedging adjustment remaining for which hedge accounting has been discontinued	488	575

Interest Rate Swaps

We enter into interest rate swaps to achieve a targeted mix of fixed and variable rate debt. We principally receive fixed rates and pay variable rates, resulting in a net increase or decrease to Interest expense. These swaps are designated as fair value hedges and hedge against interest rate risk exposure of designated debt issuances. We record the interest rate swaps at fair value in our consolidated balance sheets as assets and liabilities. Changes in the fair value of the interest rate swaps are recorded to Interest expense, which are primarily offset by changes in the fair value of the hedged debt due to changes in interest rates.

Cross Currency Swaps

We have entered into cross currency swaps previously designated as cash flow hedges through March 31, 2022 to exchange our British Pound Sterling, Euro, Swiss Franc, Canadian Dollar and Australian Dollar-denominated cash flows into U.S. dollars and to fix our cash payments in U.S. dollars, as well as to mitigate the impact of foreign currency transaction gains or losses. A portion of the loss recognized in Other comprehensive income (loss) was reclassified to Interest expense to offset the related pre-tax foreign currency transaction gain or loss on the underlying hedged item.

On March 31, 2022, we elected to de-designate our cross currency swaps as cash flow hedges and re-designated these swaps as fair value hedges. For these hedges, we have elected to exclude the change in fair value of the cross currency swaps related to both time value and cross currency basis spread from the assessment of hedge effectiveness (the excluded components). The initial value of the excluded components of \$1.0 billion as of March 31, 2022 will continue to be amortized into Interest expense over the remaining life of the hedging instruments. We estimate that \$109 million will be amortized into Interest expense within the next 12 months.

In addition to the previously mentioned cross currency swaps, we have executed additional cross currency swaps to exchange Euro-denominated cash flows into U.S. dollars to fix our cash payments in U.S. dollars. These swaps are designated as fair value hedges. We record the cross currency swaps at fair value in our consolidated balance sheets as assets and liabilities. Changes in the fair value of the cross currency swaps attributable to changes in the spot rate of the hedged item and changes in the recorded value of the hedged debt due to changes in spot rates are recorded in the same income statement line item. We present exchange gains and losses from the conversion of foreign currency denominated debt as a part of Interest expense. During the year ended December 31, 2022, these amounts completely offset each other and no net gain or loss was recorded.

Changes in the fair value of cross currency swaps attributable to time value and cross currency basis spread are initially recorded to Other comprehensive income (loss). Unrealized gains or losses on excluded components are recorded in Other comprehensive income (loss) and are recognized into Interest expense on a systematic and rational basis through the swap accrual over the life of the hedging instrument. The amount remaining in Accumulated other comprehensive loss related to cash flow hedges on the date of transition will be reclassified to earnings when the hedged item is recognized in earnings or when it becomes probable that the forecasted transactions will not occur. During the year ended December 31, 2022, the amortization of the initial value of the excluded component completely offset the amortization related to the amount remaining in Other comprehensive income (loss) related to cash flow hedges. See Note 14 for additional information.

Forward Starting Interest Rate Swaps

We have entered into forward starting interest rate swaps designated as cash flow hedges in order to manage our exposure to interest rate changes on future forecasted transactions. We hedge our exposure to the variability in future cash flows based on the expected maturities of the related forecasted debt issuance. We recognize gains and losses resulting from interest rate movements in Other comprehensive income (loss).

Treasury Rate Locks

We have entered into treasury rate locks designated as cash flow hedges to mitigate our interest rate risk on future transactions. We recognize gains and losses resulting from interest rate movements in Other comprehensive income (loss).

Net Investment Hedges

We have designated certain foreign currency debt instruments as net investment hedges to mitigate foreign exchange exposure related to non-U.S. dollar net investments in certain foreign subsidiaries against changes in foreign exchange rates. The notional amount of Euro-denominated debt designated as a net investment hedge was €750 million as of both December 31, 2022 and 2021.

Undesignated Derivatives

We also have the following derivative contracts which we use as economic hedges but for which we have elected not to apply hedge accounting.

The following table summarizes the activity of our derivatives not designated in hedging relationships:

		(dollars in millions)	
Years Ended December 31,		2022	2021
Foreign Exchange Forwards:			
Notional value entered into	\$	10,689	\$ 12,604
Notional value settled		10,701	13,077
Pre-tax loss recognized in Other income (expense), net		(97)	(62)
Swaptions:			
Notional value sold		1,000	2,000
Notional value settled		1,000	2,000
Pre-tax gain (loss) recognized in Interest expense		(33)	11

Foreign Exchange Forwards

We enter into British Pound Sterling and Euro foreign exchange forwards to mitigate our foreign exchange rate risk related to non-functional currency denominated monetary assets and liabilities of international subsidiaries.

Swaptions

We enter into swaptions to achieve a targeted mix of fixed and variable rate debt.

Concentrations of Credit Risk

Financial instruments that subject us to concentrations of credit risk consist primarily of temporary cash investments, short-term and long-term investments, trade receivables, including device payment plan agreement receivables, certain notes receivable, including lease receivables, and derivative contracts.

Counterparties to our derivative contracts are major financial institutions with whom we have negotiated derivatives agreements (ISDA master agreements) and credit support annex (CSA) agreements which provide rules for collateral exchange. The CSA agreements contain rating based thresholds such that we or our counterparties may be required to hold or post collateral based upon changes in outstanding positions as compared to established thresholds and changes in credit ratings. We do not offset fair value amounts recognized for derivative instruments and fair value amounts recognized for the right to reclaim cash collateral or the obligation to return cash collateral arising from derivative instruments recognized at fair value. At December 31, 2022, we did not hold any collateral. At December 31, 2022, we posted \$2.3 billion of collateral related to derivative contracts under collateral exchange agreements, which was recorded as Prepaid expenses and other in our consolidated balance sheet. At December 31, 2021, we held and posted \$0.1 billion and an insignificant amount, respectively, of collateral related to derivative contracts under collateral exchange arrangements, which were recorded as Other current liabilities and Prepaid expenses and other, respectively, in our consolidated balance sheet. While we may be exposed to credit losses due to the nonperformance of our counterparties, we consider the risk remote and do not expect that any such nonperformance would result in a significant effect on our results of operations or financial condition due to our diversified pool of counterparties.

Note 10. Stock-Based Compensation

Verizon Long-Term Incentive Plan

In May 2017, our shareholders approved the 2017 Long-Term Incentive Plan (the 2017 Plan) and terminated the Company's authority to grant new awards under the Verizon 2009 Long-Term Incentive Plan (the 2009 Plan). The 2017 Plan provides for broad-based equity grants to employees, including executive officers, and permits the granting of stock options, stock appreciation rights, restricted stock, restricted stock units, performance shares, performance stock units and other awards. Upon

approval of the 2017 Plan, we reserved for issuance under the 2017 Plan the number of shares that were remaining but not issued under the 2009 Plan. Shares subject to outstanding awards under the 2009 Plan that expire, are canceled or otherwise terminated will also be available for awards under the 2017 Plan. As of December 31, 2022, 69 million shares are reserved for future issuance under the 2017 Plan.

Restricted Stock Units

Restricted Stock Units (RSUs) granted under the 2017 Plan generally vest in three equal installments on each anniversary of the grant date. The RSUs that are paid in stock upon vesting and are thus classified as equity awards are measured using the grant date fair value of Verizon common stock and are not remeasured at the end of each reporting period. The RSUs that are settled in cash are classified as liability awards and the liability is measured at its fair value at the end of each reporting period. All RSUs granted under the 2017 Plan have dividend equivalent units (DEUs), which will be paid to participants if, and only to the extent the applicable RSU award vests, and is paid at the time the RSU award is paid, and in the same proportion as the RSU award. In 2020, Verizon announced a broad-based program that provides for the annual award of cash-settled RSUs under the 2017 Plan to all full-time and part-time employees who meet eligibility requirements.

We estimate forfeitures at the time of grant and revise those estimates in subsequent periods if actual forfeitures differ from those estimates. We use historical data to estimate forfeitures and recognize that estimated compensation cost of restricted stock units, net of estimated forfeitures, on a straight-line basis over the vesting period.

Performance Stock Units

The 2017 Plan also provides for grants of Performance Stock Units (PSUs) that generally vest at the end of the third year after the grant. As defined by the 2017 Plan, the Human Resources Committee of the Board of Directors determines the number of PSUs a participant earns based on the extent to which the corresponding performance goals have been achieved over the three-year performance cycle. The PSUs that are paid in stock upon vesting and are classified as equity awards are measured using the grant date fair value of Verizon common stock and are not remeasured at the end of each reporting period. The PSUs that are settled in cash and are classified as liability awards are measured at its fair value at the end of each reporting period and, therefore, will fluctuate based on the price of Verizon common stock as well as performance relative to the targets. All PSUs granted under the 2017 Plan have DEUs, which will be paid to participants if, and only to the extent the applicable PSU award vests, and is paid at the time that PSU award is paid, and in the same proportion as the PSU award. The granted and cancelled activity for the PSU award includes adjustments for the performance goals achieved.

The following table summarizes Verizon's Restricted Stock Unit and Performance Stock Unit activity:

(shares in thousands)	Restricted Stock Units		Performance Stock Units	
	Equity Awards	Liability Awards	Equity Awards	Liability Awards
Outstanding January 1, 2020	7,259	14,713	—	15,551
Granted	3,638	15,161	4,358	1,389
Payments	(3,814)	(9,311)	—	(7,160)
Cancelled/Forfeited	(182)	(1,004)	(116)	(143)
Outstanding December 31, 2020	6,901	19,559	4,242	9,637
Granted	4,079	16,845	5,353	1,692
Payments	(3,417)	(10,797)	—	(6,718)
Cancelled/Forfeited	(784)	(8,317)	(955)	(146)
Outstanding December 31, 2021	6,779	17,290	8,640	4,465
Granted	4,149	11,309	5,752	197
Payments	(3,313)	(6,363)	—	(2,075)
Cancelled/Forfeited	(362)	(1,627)	(567)	(2,171)
Outstanding December 31, 2022	7,253	20,609	13,825	416

As of December 31, 2022, unrecognized compensation expense related to the unvested portion of Verizon's RSUs and PSUs was approximately \$604 million and is expected to be recognized over approximately 2 years.

The equity awards granted in 2022, 2021 and 2020 have weighted-average grant date fair values of \$3.26, \$55.39 and \$57.38 per unit, respectively. During 2022, 2021 and 2020, we paid \$433 million, \$986 million and \$961 million, respectively, to settle RSUs and PSUs classified as liability awards.

Stock-Based Compensation Expense

After-tax compensation expense for stock-based compensation related to RSUs and PSUs described above included in Net income attributable to Verizon was \$609 million, \$625 million and \$780 million for 2022, 2021 and 2020, respectively.

Note 11. Employee Benefits

We maintain non-contributory defined benefit pension plans for certain employees. In addition, we maintain postretirement health care and life insurance plans for certain retirees and their dependents, which are both contributory and non-contributory, and include a limit on our share of the cost for certain current and future retirees. In accordance with our accounting policy for pension and other postretirement benefits, operating expenses include service costs associated with pension and other postretirement benefits while other credits and/or charges based on actuarial assumptions, including projected discount rates, an estimated return on plan assets, and impact from health care trend rates are reported in Other income (expense), net. These estimates are updated in the fourth quarter or upon a remeasurement event, to reflect actual return on plan assets and updated actuarial assumptions. The adjustment is recognized in the income statement during the fourth quarter or upon a remeasurement event pursuant to our accounting policy for the recognition of actuarial gains and losses.

Pension and Other Postretirement Benefits

Pension and other postretirement benefits for certain employees are subject to collective bargaining agreements. Modifications in benefits have been bargained from time to time, and we may also periodically amend the benefits in the management plans. The following tables summarize benefit costs, as well as the benefit obligations, plan assets, funded status and rate assumptions associated with pension and postretirement health care and life insurance benefit plans.

Obligations and Funded Status

At December 31,	Pension		Health Care and Life	
	2022	2021	2022	2021
Change in Benefit Obligations				
Beginning of year	\$ 20,167	\$ 22,236	\$ 14,710	\$ 16,168
Service cost	246	282	94	112
Interest cost	544	394	332	289
Plan amendments	427	—	4	—
Actuarial gain, net	(3,865)	(605)	(3,297)	(930)
Benefits paid	(782)	(816)	(736)	(929)
Curtailment and termination benefits	2	6	—	—
Settlements paid	(1,370)	(1,330)	—	—
End of year	15,369	20,167	11,107	14,710
Change in Plan Assets				
Beginning of year	20,087	20,128	581	572
Actual return on plan assets	(4,249)	2,049	(87)	53
Company contributions	53	56	692	885
Benefits paid	(782)	(816)	(736)	(929)
Settlements paid	(1,370)	(1,330)	—	—
End of year	13,739	20,087	450	581
Funded Status - End of year	\$ (1,630)	\$ (80)	\$ (10,657)	\$ (14,129)

At December 31,	Pension		Health Care and Life	
	2022	2021	2022	2021
Amounts recognized in the balance sheets				
Non-current assets	\$ 4	\$ 376	\$ —	\$ —
Current liabilities	(48)	(55)	(718)	(748)
Non-current liabilities	(1,586)	(401)	(9,939)	(13,381)
Total	\$ (1,630)	\$ (80)	\$ (10,657)	\$ (14,129)
Amounts recognized in Accumulated other comprehensive loss (pre-tax)				
Prior service cost (benefit)	\$ 747	\$ 402	\$ (1,355)	\$ (1,889)
Total	\$ 747	\$ 402	\$ (1,355)	\$ (1,889)

The accumulated benefit obligation for all defined benefit pension plans was \$5.3 billion and \$20.1 billion at December 31, 2022 and 2021, respectively.

Actuarial Gain, Net

The net actuarial gain in 2022 is primarily the result of a \$7.0 billion gain (\$4.1 billion gain in the pension plan and \$2.9 billion gain in our postretirement benefit plans) due to an increase in our discount rate assumption used to determine the current year liabilities of our pension plans and postretirement benefit plans from a weighted-average of 2.9% at December 31, 2021 to a weighted-average of 5.2% at December 31, 2022.

The net actuarial gain in 2021 is primarily the result of a \$1.1 billion gain due to an increase in our discount rate assumption used to determine the current year liabilities of our pension plans and postretirement benefit plans from a weighted-average of 2.6% at December 31, 2020 to a weighted-average of 2.9% at December 31, 2021.

Plan Amendments

In July 2022, union members ratified the extension of our collective bargaining agreements with the Communications Workers of America and the International Brotherhood of Electrical Workers for three years, from August 5, 2023 until August 1, 2026. Amendments triggered by the collective bargaining negotiations were made to certain pension plans for certain union-represented employees. The impact of the plan amendments was an increase in our defined benefit pension plans plan obligations of approximately \$427 million, which has been recorded as a net decrease to Accumulated other comprehensive income (loss) of \$17 million (net of taxes of \$110 million). The annual impact of the amount recorded in Accumulated other comprehensive income (loss) that will be reclassified to net periodic benefit cost is minimal.

The reclassifications from the amounts recorded in Accumulated other comprehensive income (loss) as a result of collective bargaining agreements and plan amendments made in 2016, 2017 and 2018 resulted in a net decrease to net periodic benefit cost and net increase to pre-tax income of approximately \$390 million during 2022 and \$708 million during 2021 and 2020.

Information for pension plans with an accumulated benefit obligation in excess of plan assets follows:

		(dollars in millions)	
At December 31,		2022	2021
Accumulated benefit obligation	\$	15,286	\$ 456
Fair value of plan assets		13,694	—

Information for pension plans with a projected benefit obligation in excess of plan assets follows:

		(dollars in millions)	
At December 31,		2022	2021
Projected benefit obligation	\$	15,328	\$ 456
Fair value of plan assets		13,694	—

Net Periodic Benefit Cost (Income)

The following table summarizes the components of net periodic benefit cost (income) related to our pension and postretirement health care and life insurance plans:

		(dollars in millions)					
		Pension			Health Care and Life		
Years Ended December 31,		2022	2021	2020	2022	2021	2020
Service cost - Cost of services	\$	216	\$ 247	\$ 245	\$ 79	\$ 94	\$ 89
Service cost - Selling, general and administrative expense		30	35	60	15	18	21
Service cost		246	282	305	94	112	110
Amortization of prior service cost (credit)		82	61	61	(530)	(894)	(966)
Expected return on plan assets		(1,119)	(1,234)	(1,186)	(27)	(22)	(28)
Interest cost		544	394	505	332	289	429
Remeasurement loss (gain), net		1,505	(1,419)	744	(3,182)	(960)	866
Curtailment and termination benefits		2	—	—	—	—	—
Other components		1,014	(2,198)	124	(3,407)	(1,587)	301
Total	\$	1,260	\$ (1,916)	\$ 429	\$ (3,313)	\$ (1,475)	\$ 411

The service cost component of net periodic benefit cost (income) is recorded in Cost of services and Selling, general and administrative expense in the consolidated statements of income while the other components, including mark-to-market adjustments, if any, are recorded in Other income (expense), net.

Other pre-tax changes in plan assets and benefit obligations recognized in Other comprehensive (income) loss are as follows:

At December 31,	Pension			Health Care and Life		
	2022	2021	2020	2022	2021	2020
Reversal of amortization items						
Prior service cost (benefit)	\$ (82)	\$ (61)	\$ (61)	\$ 530	\$ 894	\$ 966
Total recognized in Other comprehensive loss (income) (pre-tax)	\$ (82)	\$ (61)	\$ (61)	\$ 530	\$ 894	\$ 966

Assumptions

The weighted-average assumptions used in determining benefit obligations follow:

At December 31,	Pension		Health Care and Life	
	2022	2021	2022	2021
Discount Rate	5.20 %	3.00 %	5.20 %	2.90 %
Rate of compensation increases	3.00 %	3.00 %	N/A	N/A

N/A - not applicable

The weighted-average assumptions used in determining net periodic cost follow:

At December 31,	Pension			Health Care and Life		
	2022	2021	2020	2022	2021	2020
Discount rate in effect for determining service cost	3.80 %	3.20 %	3.30 %	3.20 %	3.00 %	3.50 %
Discount rate in effect for determining interest cost	3.20	1.90	2.40	2.30	1.80	2.80
Expected return on plan assets	6.70	6.50	6.50	4.90	4.20	4.50
Rate of compensation increases	3.00	3.00	3.00	N/A	N/A	N/A

N/A - not applicable

In determining our pension and other postretirement benefit obligations, we used a weighted-average discount rate of 5.2% in 2022. The rates were selected to approximate the composite interest rates available on a selection of high-quality bonds available in the market at December 31, 2022. The bonds selected had maturities that coincided with the time periods during which benefits payments are expected to occur, were non-callable and available in sufficient quantities to ensure marketability (at least \$300 million par outstanding).

In order to project the long-term target investment return for the total portfolio, estimates are prepared for the total return of each major asset class over the subsequent 10-year period. Those estimates are based on a combination of factors including the current market interest rates and valuation levels, consensus earnings expectations and historical long-term risk premiums. To determine the aggregate return for the pension trust, the projected return of each individual asset class is then weighted according to the allocation to that investment area in the trust's long-term asset allocation policy.

The assumed health care cost trend rates are as follows:

At December 31,	Health Care and Life		
	2022	2021	2020
Weighted-average healthcare cost trend rate assumed for next year	6.60 %	6.20 %	6.20 %
Rate to which cost trend rate gradually declines	4.50	4.50	4.50
Year the rate reaches the level it is assumed to remain thereafter	2031	2029	2029

Plan Assets

The Company's overall investment strategy is to achieve a mix of assets that allows us to meet projected benefit payments while taking into consideration risk and return. While target allocation percentages will vary over time, the current target allocation for plan assets is designed so that 41% to 51% of the assets have the objective of achieving a return in excess of the growth in liabilities (comprised of public equities, private equities, real estate, hedge funds, high yield bonds and emerging market debt) and 55% to 65% of the assets are invested as liability hedging assets (where interest rate sensitivity of the liability hedging assets better match the interest rate sensitivity of the liability) and a maximum of 10% is in cash. This allocation will shift as funded status improves to a higher allocation of liability hedging assets. Target policies will be revisited periodically to ensure they are in line with fund objectives. Both active and passive management approaches are used depending on perceived market efficiencies and various other factors. Due to our diversification and risk control processes, there are no significant concentrations of risk, in terms of sector, industry, geography or company names.

Pension and healthcare and life plans assets do not include significant amounts of Verizon bonds or common stock.

Pension Plans

The fair values for the pension plans by asset category at December 31, 2022 are as follows:

(dollars in millions)				
Asset Category	Total	Level 1	Level 2	Level 3
Cash and cash equivalents	\$ 817	\$ 779	\$ 38	\$ —
Equity securities	332	318	14	—
Fixed income securities				
U.S. Treasuries and agencies	1,541	1,312	229	—
Corporate bonds	2,413	13	2,400	—
International bonds	528	10	518	—
Other	711	4	707	—
Real estate	1,002	—	—	1,002
Other				
Private equity	569	—	—	569
Hedge funds	88	—	36	52
Total investments at fair value	8,001	2,436	3,942	1,623
Investments measured at NAV	5,738			
Total	\$ 13,739	\$ 2,436	\$ 3,942	\$ 1,623

The fair values for the pension plans by asset category at December 31, 2021 are as follows:

(dollars in millions)				
Asset Category	Total	Level 1	Level 2	Level 3
Cash and cash equivalents	\$ 1,221	\$ 1,208	\$ 13	\$ —
Equity securities	2,482	2,463	19	—
Fixed income securities				
U.S. Treasuries and agencies	1,785	1,652	133	—
Corporate bonds	4,046	123	3,923	—
International bonds	1,407	23	1,384	—
Other	695	—	695	—
Real estate	972	—	—	972
Other				
Private equity	569	—	—	569
Hedge funds	224	—	114	110
Total investments at fair value	13,401	5,469	6,281	1,651
Investments measured at NAV	6,686			
Total	\$ 20,087	\$ 5,469	\$ 6,281	\$ 1,651

The following is a reconciliation of the beginning and ending balance of pension plan assets that are measured at fair value using significant unobservable inputs:

(dollars in millions)									
	Equity Securities	Corporate Bonds	International Bonds	Real Estate	Private Equity	Hedge Funds	Total		
Balance at January 1, 2021	\$ 2	\$ 176	\$ 20	\$ 757	\$ 414	\$ 138	\$ 1,507		
Actual gain (loss) on plan assets	(1)	(5)	—	(21)	(19)	1	(45)		
Purchases (sales)	(1)	1	(4)	197	147	81	421		
Transfers out	—	(172)	(16)	39	27	(110)	(232)		
Balance at December 31, 2021	—	—	—	972	569	110	1,651		
Actual gain on plan assets	—	—	—	19	30	19	68		
Purchases (sales)	—	—	—	14	(11)	6	9		
Transfers out	—	—	—	(3)	(19)	(83)	(105)		
Balance at December 31, 2022	\$ —	\$ —	\$ —	\$ 1,002	\$ 569	\$ 52	\$ 1,623		

Health Care and Life Plans

The fair values for the other postretirement benefit plans by asset category at December 31, 2022 are as follows:

(dollars in millions)				
Asset Category	Total	Level 1	Level 2	Level 3
Cash and cash equivalents	\$ 30	\$ 1	\$ 29	\$ —
Equity securities	252	252	—	—
Fixed income securities				
U.S. Treasuries and agencies	101	82	19	—
Corporate bonds	35	25	10	—
International bonds	12	9	3	—
Other	11	—	11	—
Total investments at fair value	441	369	72	—
Investments measured at NAV	9			
Total	\$ 450	\$ 369	\$ 72	\$ —

The fair values for the other postretirement benefit plans by asset category at December 31, 2021 are as follows:

(dollars in millions)				
Asset Category	Total	Level 1	Level 2	Level 3
Cash and cash equivalents	\$ 36	\$ —	\$ 36	\$ —
Equity securities	284	284	—	—
Fixed income securities				
U.S. Treasuries and agencies	160	150	10	—
Corporate bonds	64	50	14	—
International bonds	14	10	4	—
Other	10	—	10	—
Total investments at fair value	568	494	74	—
Investments measured at NAV	13			
Total	\$ 581	\$ 494	\$ 74	\$ —

The following are general descriptions of asset categories, as well as the valuation methodologies and inputs used to determine the fair value of each major category of assets.

Cash and cash equivalents include short-term investment funds (less than 90 days to maturity), primarily in diversified portfolios of investment grade money market instruments and are valued using quoted market prices or other valuation methods. The carrying value of cash equivalents approximates fair value due to the short-term nature of these investments.

Investments in securities traded on national and foreign securities exchanges are valued by the trustee at the last reported sale prices on the last business day of the year or, if no sales were reported on that date, at the last reported bid prices. Government obligations, corporate bonds, international bonds and asset-backed debt are valued using matrix prices with input from independent third-party valuation sources. Over-the-counter securities are valued at the bid prices or the average of the bid and ask prices on the last business day of the year from published sources or, if not available, from other sources considered reliable such as multiple broker quotes.

Commingled funds not traded on national exchanges are priced by the custodian or fund's administrator at their net asset value (NAV). Commingled funds held by third-party custodians appointed by the fund managers provide the fund managers with a NAV. The fund managers have the responsibility for providing this information to the custodian of the respective plan.

The investment manager of the entity values venture capital, corporate finance and natural resource limited partnership investments. Real estate investments are valued at amounts based upon appraisal reports prepared by either independent real estate appraisers or the investment manager using discounted cash flows or market comparable data. Loans secured by mortgages are carried at the lesser of the unpaid balance or appraised value of the underlying properties. The values assigned to these investments are based upon available and current market information and do not necessarily represent amounts that might ultimately be realized. Because of the inherent uncertainty of valuation, estimated fair values might differ significantly from the values that would have been used had a ready market for the securities existed. These differences could be material.

Forward currency contracts, futures, and options are valued by the trustee at the exchange rates and market prices prevailing on the last business day of the year. Both exchange rates and market prices are readily available from published sources. These securities are classified by the asset class of the underlying holdings.

Hedge funds are valued by the custodian at NAV based on statements received from the investment manager. These funds are valued in accordance with the terms of their corresponding offering or private placement memoranda.

Commingled funds, hedge funds, venture capital, corporate finance, natural resource and real estate limited partnership investments for which fair value is measured using the NAV per share as a practical expedient are not leveled within the fair value hierarchy but are included in total investments.

Employer Contributions

In 2022, we made no discretionary contribution to our qualified pension plans, \$53 million of contributions to our nonqualified pension plans and \$692 million of contributions to our other postretirement benefit plans. No mandatory qualified pension plans contributions are expected or required to be made in 2023. In 2023, we expect discretionary qualified pension plan contributions to be approximately \$200 million, nonqualified pension plan contributions are estimated to be insignificant and contributions to our other postretirement benefit plans are estimated to be approximately \$820 million.

Estimated Future Benefit Payments

The benefit payments to retirees are expected to be paid as follows:

Year			(dollars in millions)	
	Pension Benefits		Health Care and Life	
2023	\$	1,679	\$	867
2024		1,583		863
2025		1,578		854
2026		1,553		845
2027		937		837
2028 to 2032		4,628		4,082

Savings Plan and Employee Stock Ownership Plans

We maintain four leveraged employee stock ownership plans (ESOP). We match a certain percentage of eligible employee contributions to certain savings plans with shares of our common stock from this ESOP. At December 31, 2022, the number of allocated shares of common stock in this ESOP was 43 million. There were no unallocated shares of common stock in this ESOP at December 31, 2022. All leveraged ESOP shares are included in earnings per share computations.

Total savings plan costs were \$620 million in 2022, \$690 million in 2021 and \$730 million in 2020.

Severance Benefits

The following table provides an analysis of our severance liability:

(dollars in millions)

Year	Beginning of Year		Charged to Expense		Payments		Other		End of Year
2020	\$	565	\$	309	\$	(248)	\$	(24)	\$ 602
2021		602		233		(258)		(29)	548
2022		548		319		(214)		—	653

Severance, Pension and Benefits (Credits) Charges

During 2022, in accordance with our accounting policy to recognize actuarial gains and losses in the period in which they occur, we recorded net pre-tax pension and benefits credits of \$1.7 billion in our pension and postretirement benefit plans. The credits were recorded in Other income (expense), net in our consolidated statement of income and were primarily driven by a credit of \$7.0 billion due to an increase in our discount rate assumption used to determine the current year liabilities of our pension plans (\$4.1 billion) and postretirement benefit plans (\$2.9 billion) from a weighted-average of 2.9% at December 31, 2021 to a weighted-average of 5.2% at December 31, 2022, a charge of \$5.5 billion due to the difference between our estimated and our actual return on assets and a credit of \$06 million due to other actuarial assumption adjustments. During 2022, we also recorded net pre-tax severance charges of \$319 million in Selling, general and administrative expense in our consolidated statements of income.

During 2021, we recorded net pre-tax pension and benefits credits of \$2.4 billion in our pension and postretirement benefit plans. The credits were recorded in Other income (expense), net in our consolidated statement of income and were primarily driven by a credit of \$1.1 billion due to an increase in our discount rate assumption used to determine the current year liabilities of our pension plans and postretirement benefit plans from a weighted-average of 2.6% at December 31, 2020 to a weighted-average of 2.9% at December 31, 2021, a credit of \$847 million due to the difference between our estimated and our actual return on assets and a credit of \$453 million due to other actuarial assumption adjustments. During 2021, we also recorded net pre-tax severance charges of \$33 million in Selling, general and administrative expense in our consolidated statements of income.

During 2020, we recorded net pre-tax pension and benefits charges of \$.6 billion in our pension and postretirement benefit plans. The charges were recorded in Other income (expense), net in our consolidated statement of income and were primarily driven by a charge of \$3.2 billion due to a decrease in our discount rate assumption used to determine the current year liabilities of our pension plans and postretirement benefit plans from a weighted-average of 3.3% at December 31, 2019 to a weighted-average of 2.6% at December 31, 2020, partially offset by a credit of \$.6 billion due to the difference between our estimated and our actual return on assets. During 2020, we also recorded net pre-tax severance charges of \$309 million in Selling, general and administrative expense in our consolidated statements of income.

Note 12. Taxes

The components of income before provision for income taxes are as follows:

Years Ended December 31,	(dollars in millions)		
	2022	2021	2020
Domestic	\$ 26,822	\$ 27,607	\$ 22,844
Foreign	1,449	1,813	1,123
Total	\$ 28,271	\$ 29,420	\$ 23,967

The components of the provision for income taxes are as follows:

Years Ended December 31,	(dollars in millions)		
	2022	2021	2020
Current			
Federal	\$ 2,411	\$ 1,876	\$ 2,826
Foreign	201	248	159
State and Local	938	414	1,081
Total	3,550	2,538	4,066
Deferred			
Federal	2,529	3,354	1,432
Foreign	(22)	(97)	1
State and Local	466	1,007	120
Total	2,973	4,264	1,553
Total income tax provision	\$ 6,523	\$ 6,802	\$ 5,619

The following table shows the principal reasons for the difference between the effective income tax rate and the statutory federal income tax rate:

Years Ended December 31,	2022	2021	2020
Statutory federal income tax rate	21.0 %	21.0 %	21.0 %
State and local income tax rate, net of federal tax benefits	3.9	3.8	3.9
Noncontrolling interest	(0.4)	(0.4)	(0.5)
Divestitures	—	(0.6)	0.1
Tax credits	(0.5)	(0.5)	(0.8)
Other, net	(0.9)	(0.2)	(0.3)
Effective income tax rate	23.1 %	23.1 %	23.4 %

The effective income tax rate for 2022 and 2021 was 23.1%. The effective income tax rate for the twelve months ended December 31, 2022 was comparable to the similar period in 2021. The decrease in the provision for income taxes was primarily due to the decrease in income before income taxes in the current period.

The effective income tax rate for 2021 was 23.1% compared to 23.4% for 2020. The decrease in the effective income tax rate was primarily due to the sale of Verizon Media in 2021, partially offset by the non-recurring tax benefit recognized in 2020 from a series of legal entity restructurings. The increase in the provision for income taxes was primarily due to the increase in income before income taxes in 2021.

The amounts of cash taxes paid by Verizon are as follows:

Years Ended December 31,	(dollars in millions)		
	2022	2021	2020
Income taxes, net of amounts refunded	\$ 2,736	\$ 3,040	\$ 2,725
Employment taxes	1,245	1,225	618
Property and other taxes	1,959	1,756	2,093
Total	\$ 5,940	\$ 6,021	\$ 5,436

Deferred Tax Assets and Liabilities

Deferred taxes arise because of differences in the book and tax bases of certain assets and liabilities. Significant components of deferred tax assets and liabilities are as follows:

At December 31,	(dollars in millions)	
	2022	2021
Deferred Tax Assets		
Employee benefits	\$ 3,888	\$ 4,388
Tax loss, credit, and other carry forwards	1,940	2,224
Other - assets	6,986	7,314
	12,814	13,926
Valuation allowances	(1,347)	(1,574)
Deferred tax assets	11,467	12,352
Deferred Tax Liabilities		
Spectrum and other intangible amortization	25,851	24,935
Depreciation	21,388	19,893
Other - liabilities	7,496	8,041
Deferred tax liabilities	54,735	52,869
Net deferred tax liability	\$ 43,268	\$ 40,517

At December 31, 2022, undistributed earnings of our foreign subsidiaries indefinitely invested outside the U.S. amounted to approximately \$2.5 billion. The majority of Verizon's cash flow is generated from domestic operations and we are not dependent on foreign cash or earnings to meet our funding requirements, nor do we intend to repatriate these undistributed foreign earnings to fund U.S. operations. Furthermore, a portion of these undistributed earnings represents amounts that legally must be kept in reserve in accordance with certain foreign jurisdictional requirements and are unavailable for distribution or repatriation. As a result, we have not provided U.S. deferred taxes on these undistributed earnings because we intend that they will remain indefinitely reinvested outside of the U.S. and, therefore unavailable for use in funding U.S. operations. Determination of the amount of unrecognized deferred taxes related to these undistributed earnings is not practicable.

At December 31, 2022, we had net after-tax loss, credit, and other carry forwards for income tax purposes of approximately \$1.9 billion that relate to federal, state and foreign taxes. Of these net after-tax loss, credit, and other carry forwards, approximately \$1.2 billion will expire between 2023 and 2042 and approximately \$732 million may be carried forward indefinitely.

During 2022, the valuation allowance decreased approximately \$227 million. The \$1.3 billion valuation allowance at December 31, 2022, as well as the 2022 activity is primarily related to state and foreign taxes.

Unrecognized Tax Benefits

A reconciliation of the beginning and ending balance of unrecognized tax benefits is as follows:

	(dollars in millions)		
	2022	2021	2020
Balance at January 1,	\$ 3,134	\$ 2,944	\$ 2,870
Additions based on tax positions related to the current year	123	150	160
Additions for tax positions of prior years	122	621	258
Reductions for tax positions of prior years	(419)	(330)	(166)
Settlements	(92)	(163)	(46)
Lapses of statutes of limitations	(56)	(88)	(132)
Balance at December 31,	\$ 2,812	\$ 3,134	\$ 2,944

Included in the total unrecognized tax benefits at December 31, 2022, 2021 and 2020 is \$2.5 billion, \$2.8 billion and \$2.5 billion, respectively, that if recognized, would favorably affect the effective income tax rate.

We recognized the following net after-tax expenses (benefit) related to interest and penalties in the provision for income taxes:

Years Ended December 31,	(dollars in millions)	
2022	\$	35
2021		(21)
2020		5

The after-tax accruals for the payment of interest and penalties in the consolidated balance sheets are as follows:

At December 31,		(dollars in millions)
2022	\$	544
2021		551

The decrease in unrecognized tax benefits was primarily related to the resolution of issues with the Internal Revenue Service (IRS) involving tax years 2015-2016 as well as final purchase accounting adjustments made in connection with the 2021 acquisition of TracFone. The uncertain tax benefits related to the acquisition of TracFone are the subject of an indemnity from América Móvil for which there is a corresponding indemnity asset.

Verizon and/or its subsidiaries file income tax returns in the U.S. federal jurisdiction, and various state, local and foreign jurisdictions. As a large taxpayer, we are under audit by the IRS and multiple state and foreign jurisdictions for various open tax years. The IRS is currently examining the Company's U.S. income tax returns for tax years 2017-2018 and Cellco's U.S. income tax return for tax years 2017-2019. Tax controversies are ongoing for tax years as early as 2009 in certain states and as early as 2000 outside the U.S. The amount of the liability for unrecognized tax benefits will change in the next twelve months due to the expiration of the statute of limitations in various jurisdictions and it is reasonably possible that various current tax examinations will conclude or require reevaluations of the Company's tax positions during this period. An estimate of the range of the possible change cannot be made until these tax matters are further developed or resolved.

Note 13. Segment Information

Reportable Segments

We have two reportable segments that we operate and manage as strategic business units - Consumer and Business. We measure and evaluate our reportable segments based on segment operating income, consistent with the chief operating decision maker's assessment of segment performance.

Our segments and their principal activities consist of the following:

Segment	Description
Verizon Consumer Group	Our Consumer segment provides consumer-focused wireless and wireline communications services and products. Our wireless services are provided across one of the most extensive wireless networks in the U.S. under the Verizon brand, TracFone brands and through wholesale and other arrangements. We also provide FWA broadband through our wireless networks. Our wireline services are provided in nine states in the Mid-Atlantic and Northeastern U.S., as well as Washington D.C., over our 100% fiber-optic network through our Verizon Fios product portfolio and over a traditional copper-based network to customers who are not served by Fios.
Verizon Business Group	Our Business segment provides wireless and wireline communications services and products, including data, video and conferencing services, corporate networking solutions, security and managed network services, local and long distance voice services and network access to deliver various IoT services and products. We also provide FWA broadband through our wireless networks. We provide these products and services to businesses, government customers and wireless and wireline carriers across the U.S. and select products and services to customers around the world.

Our Consumer segment's wireless and wireline products and services are available to our retail customers, as well as resellers that purchase wireless network access from us on a wholesale basis. Our Business segment's wireless and wireline products and services are organized by the primary customer groups targeted by these offerings: Small and Medium Business, Global Enterprise, Public Sector and Other, and Wholesale.

Corporate and other primarily includes insurance captives, investments in unconsolidated businesses and development stage businesses that support our strategic initiatives, as well as unallocated corporate expenses, certain pension and other employee benefit related costs and interest and financing expenses. Corporate and other also includes the historical results of divested businesses including Verizon Media, and other adjustments and gains and losses that are not allocated in assessing segment performance due to their nature. Although such transactions are excluded from the business segment results, they are included in reported consolidated earnings. Gains and losses from these transactions that are not individually significant are included in segment results as these items are included in the chief operating decision maker's assessment of segment performance.

We completed the sale of Verizon Media on September 1, 2021. See Note 3 for additional information on the sale of Verizon Media.

The following tables provides operating financial information for our two reportable segments:

(dollars in millions)

2022	Consumer	Business	Total Reportable Segments
External Operating Revenues			
Service	\$ 73,139	\$ —	\$ 73,139
Wireless equipment	23,168	—	23,168
Other ⁽¹⁾	6,996	—	6,996
Small and Medium Business	—	12,595	12,595
Global Enterprise	—	9,733	9,733
Public Sector and Other	—	6,117	6,117
Wholesale	—	2,584	2,584
Intersegment revenues	203	43	246
Total Operating Revenues⁽²⁾	103,506	31,072	134,578
Cost of services	17,746	10,483	28,229
Cost of wireless equipment	25,134	5,362	30,496
Selling, general and administrative expense	19,064	8,284	27,348
Depreciation and amortization expense	12,716	4,312	17,028
Total Operating Expenses	74,660	28,441	103,101
Operating Income	\$ 28,846	\$ 2,631	\$ 31,477

⁽¹⁾ Other revenue includes fees that partially recover the direct and indirect costs of complying with regulatory and industry obligations and programs, revenues associated with certain products included in our device protection offerings, leasing and interest when equipment is sold to the customer by an authorized agent under a device payment plan agreement.

⁽²⁾ Service and other revenues and Wireless equipment revenues included in our Business segment amounted to approximately \$7.0 billion and \$4.0 billion, respectively, for the year ended December 31, 2022.

(dollars in millions)

2021	Consumer	Business	Total Reportable Segments
External Operating Revenues			
Service	\$ 67,723	\$ —	\$ 67,723
Wireless equipment	19,781	—	19,781
Other ⁽¹⁾	7,568	—	7,568
Small and Medium Business	—	11,751	11,751
Global Enterprise	—	10,218	10,218
Public Sector and Other	—	6,324	6,324
Wholesale	—	2,680	2,680
Intersegment revenues	228	69	297
Total Operating Revenues⁽²⁾	95,300	31,042	126,342
Cost of services	16,581	10,653	27,234
Cost of wireless equipment	20,523	4,544	25,067
Selling, general and administrative expense	16,562	8,324	24,886
Depreciation and amortization expense	11,679	4,084	15,763
Total Operating Expenses	65,345	27,605	92,950
Operating Income	\$ 29,955	\$ 3,437	\$ 33,392

⁽¹⁾ Other revenue includes fees that partially recover the direct and indirect costs of complying with regulatory and industry obligations and programs, revenues associated with certain products included in our device protection offerings, leasing and interest when equipment is sold to the customer by an authorized agent under a device payment plan agreement.

⁽²⁾ Service and other revenues and Wireless equipment revenues included in our Business segment amounted to approximately \$7.7 billion and \$3.4 billion, respectively, for the year ended December 31, 2021.

	(dollars in millions)		
2020	Consumer	Business	Total Reportable Segments
External Operating Revenues			
Service	\$ 64,884	\$ —	\$ 64,884
Wireless equipment	15,492	—	15,492
Other ⁽¹⁾	7,916	—	7,916
Small and Medium Business	—	11,112	11,112
Global Enterprise	—	10,405	10,405
Public Sector and Other	—	6,362	6,362
Wholesale	—	3,013	3,013
Intersegment revenues	241	70	311
Total Operating Revenues⁽²⁾	88,533	30,962	119,495
Cost of services	15,610	10,659	26,269
Cost of wireless equipment	15,736	4,064	19,800
Selling, general and administrative expense	16,936	8,380	25,316
Depreciation and amortization expense	11,395	4,086	15,481
Total Operating Expenses	59,677	27,189	86,866
Operating Income	\$ 28,856	\$ 3,773	\$ 32,629

⁽¹⁾ Other revenue includes fees that partially recover the direct and indirect costs of complying with regulatory and industry obligations and programs, revenues associated with certain products included in our device protection offerings, leasing and interest when equipment is sold to the customer by an authorized agent under a device payment plan agreement.

⁽²⁾ Service and other revenues and Wireless equipment revenues included in our Business segment amounted to approximately \$8.1 billion and \$2.9 billion, respectively, for the year ended December 31, 2020.

The following table provides Fios revenues for our two reportable segments:

Years Ended December 31,	(dollars in millions)		
	2022	2021	2020
Consumer	\$ 11,622	\$ 11,558	\$ 11,082
Business	1,201	1,136	1,057
Total Fios revenue	\$ 12,823	\$ 12,694	\$ 12,139

The following table provides Wireless service revenue for our reportable segments and includes intersegment activity:

Years Ended December 31,	(dollars in millions)		
	2022	2021	2020
Consumer	\$ 61,509	\$ 56,103	\$ 53,605
Business	12,845	12,366	11,805
Total Wireless service revenue	\$ 74,354	\$ 68,469	\$ 65,410

Reconciliation to Consolidated Financial Information

The reconciliation of segment operating revenues and expenses to consolidated operating revenues and operating income below includes the effects of special items that the chief operating decision maker does not consider in assessing segment performance, primarily because of their nature.

A reconciliation of the reportable segment operating revenues to consolidated operating revenues is as follows:

Years Ended December 31,	(dollars in millions)		
	2022	2021	2020
Operating Revenues			
Total reportable segments	\$ 134,578	\$ 126,342	\$ 119,495
Corporate and other	2,510	7,722	9,334
Reconciling items:			
Eliminations	(253)	(451)	(537)
Consolidated Operating Revenues	\$ 136,835	\$ 133,613	\$ 128,292

A reconciliation of the total reportable segments' operating income to consolidated income before provision for income taxes is as follows:

		(dollars in millions)		
Years Ended December 31,		2022	2021	2020
Operating Income				
Total reportable segments	\$	31,477	\$ 33,392	\$ 32,629
Corporate and other		(319)	(449)	(1,472)
Reconciling items:				
Severance charges		(304)	(209)	(221)
Other components of net periodic pension and benefit charges (Note 11)		(387)	(769)	(817)
Loss on spectrum licenses		—	(223)	(1,195)
Net gain (loss) from disposition of business		—	706	(126)
Consolidated operating income		30,467	32,448	28,798
Equity in earnings (losses) of unconsolidated businesses		44	145	(45)
Other income (expense), net		1,373	312	(539)
Interest expense		(3,613)	(3,485)	(4,247)
Income Before Provision For Income Taxes	\$	28,271	\$ 29,420	\$ 23,967

No single customer accounted for more than 10% of our total operating revenues during the years ended December 31, 2022, 2021 and 2020. International operating revenues were not significant during the years ended December 31, 2022, 2021 and 2020. As of December 31, 2022 and 2021, international long-lived assets were not significant.

The chief operating decision maker does not review disaggregated assets on a segment basis; therefore, such information is not presented. Depreciation and amortization included in the measure of segment profitability is primarily allocated based on proportional usage, and is included within Total reportable segment operating income.

Note 14. Equity and Comprehensive Income (Loss)

Equity

Common Stock

In February 2020, the Board of Directors of the Company authorized a share buyback program to repurchase up to 100 million shares of our common stock. The program will terminate when the aggregate number of shares purchased reaches 100 million or a new share repurchase plan superseding the current plan is authorized, whichever is sooner. During the years ended December 31, 2022, 2021, and 2020, we did not repurchase any shares of our common stock under our authorized share buyback program. At December 31, 2022, the maximum number of shares that could be purchased by or on behalf of Verizon under our share buyback program was 100 million.

Common stock has been used from time to time to satisfy some of the funding requirements of employee and shareholder plans. During the years ended December 31, 2022, 2021, and 2020, we issued 2.1 million, 2.1 million and 2.3 million shares of common stock from treasury stock, which had aggregate values of \$91 million, \$91 million and \$101 million, respectively.

In connection with our acquisition of TracFone in November 2021, we issued approximately 57.6 million shares of our common stock from treasury stock valued at approximately \$3.0 billion. See Note 3 for additional information.

Noncontrolling Interests

During the year ended December 31, 2022, Verizon entered into and completed agreements to acquire additional interests in certain controlled wireless partnerships for cash consideration of \$508 million. Verizon continues to retain controlling financial interest within these partnerships; therefore, the changes in ownership interest were accounted for as equity transactions. This resulted in a reduction of additional paid-in capital of \$450 million, reflected in Other, and a reduction of noncontrolling interest of \$58 million, reflected in Distributions and other, both within our consolidated statement of changes in equity for the year ended December 31, 2022. These transactions were recorded within Other, net cash flow from financing activities in our consolidated statement of cash flows for the year ended December 31, 2022.

Accumulated Other Comprehensive Income (Loss)

Comprehensive income consists of net income and other gains and losses affecting equity that, under U.S. GAAP, are excluded from net income. Significant changes in the components of Other comprehensive income (loss), net of provision for income taxes are described below.

The changes in the balances of Accumulated other comprehensive income (loss) by component are as follows:

(dollars in millions)	Foreign currency translation adjustments	Unrealized gain (loss) on cash flow hedges	Unrealized loss on fair value hedges	Unrealized gain (loss) on marketable securities	Defined benefit pension and postretirement plans	Total
Balance at January 1, 2020	\$ (584)	\$ (816)	\$ —	\$ 27	\$ 2,371	\$ 998
Other comprehensive income	180	953	—	7	—	1,140
Amounts reclassified to net income	—	(1,524)	—	(9)	(676)	(2,209)
Net other comprehensive income (loss)	180	(571)	—	(2)	(676)	(1,069)
Balance at December 31, 2020	(404)	(1,387)	—	25	1,695	(71)
Other comprehensive loss	(141)	(1,318)	—	(8)	—	(1,467)
Amounts reclassified to net income	—	1,233	—	(1)	(621)	611
Net other comprehensive income (loss)	(141)	(85)	—	(9)	(621)	(856)
Balance at December 31, 2021	(545)	(1,472)	—	16	1,074	(927)
Excluded components recognized in other comprehensive income	—	—	(371)	—	—	(371)
Other comprehensive loss	(153)	(174)	—	(25)	(317)	(669)
Amounts reclassified to net income	—	496	(60)	—	(334)	102
Net other comprehensive income (loss)	(153)	322	(431)	(25)	(651)	(938)
Balance at December 31, 2022	\$ (698)	\$ (1,150)	\$ (431)	\$ (9)	\$ 423	\$ (1,865)

The amounts presented above in Net other comprehensive income (loss) are net of taxes. The amounts reclassified to net income related to unrealized gain (loss) on cash flow hedges and unrealized loss on fair value hedges in the table above are included in Other income (expense), net and Interest expense in our consolidated statements of income. See Note 9 for additional information. The amounts reclassified to net income related to unrealized gain (loss) on marketable securities in the table above are included in Other income (expense), net in our consolidated statements of income. The amounts reclassified to net income related to defined benefit pension and postretirement plans in the table above are included in Other income (expense), net in our consolidated statements of income. See Note 11 for additional information.

Note 15. Additional Financial Information

The following tables provide additional financial information related to our consolidated financial statements:

Income Statement Information

Years Ended December 31,	(dollars in millions)		
	2022	2021	2020
Depreciation expense	\$ 14,592	\$ 14,119	\$ 14,275
Interest costs on debt balances	5,429	5,148	4,632
Net amortization of debt discount	214	178	170
Capitalized interest costs	(2,030)	(1,841)	(555)
Advertising expense	3,556	3,394	3,107

Years Ended December 31,	(dollars in millions)		
	2022	2021	2020
Other income (expense), net			
Interest income	\$ 146	\$ 48	\$ 65
Other components of net periodic benefit (cost) income	2,386	3,785	(425)
Net debt extinguishment losses	(1,077)	(3,541)	(129)
Other, net	(82)	20	(50)
	\$ 1,373	\$ 312	\$ (539)

Balance Sheet Information

At December 31,	(dollars in millions)		
	2022	2021	
Prepaid expenses and other			
Prepaid taxes	\$ 167	\$ 1,093	
Deferred contract costs	2,629	2,432	
Collateral payments related to derivative contracts	2,286	21	
Restricted cash	1,343	1,094	
Other prepaid expense and other	1,933	2,266	
	\$ 8,358	\$ 6,906	
Accounts payable and accrued liabilities			
Accounts payable	\$ 8,750	\$ 8,040	
Accrued expenses	7,824	9,123	
Accrued vacation, salaries and wages	3,950	4,485	
Interest payable	1,577	1,561	
Taxes payable	1,876	1,624	
	\$ 23,977	\$ 24,833	
Other current liabilities			
Dividends payable	\$ 2,764	\$ 2,709	
Contract liability	6,583	6,053	
Other	2,750	2,263	
	\$ 12,097	\$ 11,025	

As of December 31, 2022 and 2021, Property, plant and equipment includes approximately \$6.0 billion and \$5.9 billion of additions that have not yet been paid.

Cash Flow Information

		(dollars in millions)		
Years Ended December 31,	2022	2021	2020	
Cash Paid				
Interest, net of amounts capitalized	\$ 3,316	\$ 3,435	\$ 4,420	
Income taxes, net of amounts refunded	2,736	3,040	2,725	
Other, net Cash Flows from Operating Activities				
Changes in device payment plan agreement non-current receivables	\$ (4,919)	\$ (2,438)	\$ 558	
Net debt extinguishment losses	1,077	3,541	129	
Loss on spectrum licenses	—	223	1,195	
Gain on disposition of Media business	—	(1,051)	—	
Other, net	64	(368)	898	
	\$ (3,778)	\$ (93)	\$ 2,780	
Other, net Cash Flows from Financing Activities				
Net debt related costs ⁽¹⁾	\$ (366)	\$ (2,309)	\$ (1,055)	
Other, net	(1,706)	(1,523)	(1,648)	
	\$ (2,072)	\$ (3,832)	\$ (2,703)	

⁽¹⁾ These costs include the premiums paid for the early extinguishment of debt, fees paid in connection with exchange and tender offers, and settlements of associated instruments.

Note 16. Commitments and Contingencies

In the ordinary course of business, Verizon is involved in various commercial litigation and regulatory proceedings at the state and federal level. Where it is determined, in consultation with counsel based on litigation and settlement risks, that a loss is probable and estimable in a given matter, Verizon establishes an accrual. In none of the currently pending matters is the amount of accrual material. An estimate of the reasonably possible loss or range of loss in excess of the amounts already accrued cannot be made at this time due to various factors typical in contested proceedings, including: (1) uncertain damage theories and demands; (2) a less than complete factual record; (3) uncertainty concerning legal theories and their resolution by courts or regulators; and (4) the unpredictable nature of the opposing party and its demands. We continuously monitor these proceedings as they develop and adjust any accrual or disclosure as needed. We do not expect that the ultimate resolution of any pending regulatory or legal matter in future periods will have a material effect on our financial condition, but it could have a material effect on our results of operations for a given reporting period.

Verizon is currently involved in approximately 25 federal district court actions alleging that Verizon is infringing various patents. Most of these cases are brought by non-practicing entities and effectively seek only monetary damages; a small number are brought by companies that have sold products and could seek injunctive relief as well. These cases have progressed to various stages and a small number may go to trial in the coming 12 months if they are not otherwise resolved.

In connection with the execution of agreements for the sales of businesses and investments, Verizon ordinarily provides representations and warranties to the purchasers pertaining to a variety of nonfinancial matters, such as ownership of the securities being sold, as well as indemnity from certain financial losses. From time to time, counterparties may make claims under these provisions, and Verizon will seek to defend against those claims and resolve them in the ordinary course of business.

As of December 31, 2022, letters of credit totaling approximately \$34 million, which were executed in the normal course of business and support several financing arrangements and payment obligations to third parties, were outstanding.

During 2022, Verizon entered into four renewable energy purchase agreements (REPAs) with third parties in addition to 20 signed in previous years. Each of the REPAs is based on the expected operation of a renewable energy-generating facility and has a fixed price term of 12 to 18 years from the commencement of the facility's entry into commercial operation. Seven of the facilities have entered into commercial operation, and the remainder are under development. The REPAs generally are expected to be financially settled based on the prevailing market price as energy is generated by the facilities.

We have various unconditional purchase obligations, which represent agreements to purchase goods or services that are enforceable and legally binding. We estimate that these unconditional purchase obligations, for contracts with terms in excess of one year, total \$24.5 billion, and primarily represent commitments to purchase network equipment, software and services, content, marketing services and other items which will be used or sold in the ordinary course of business from a variety of suppliers. Of this total amount, \$9.4 billion is attributable to 2023, \$6.9 billion is attributable to 2024, \$5.5 billion is attributable to 2025, \$1.4 billion is attributable to 2026, \$772 million is attributable to 2027 and \$524 million is attributable to years thereafter. These amounts do not represent our entire anticipated purchases in the future, but represent only those items that are the

subject of contractual obligations. Our commitments are generally determined based on the noncancelable quantities to which we are contractually obliged. Since the commitments to purchase programming services from television networks and broadcast stations have no minimum volume requirement, we estimated our obligation based on number of subscribers at December 31, 2022, and applicable rates stipulated in the contracts in effect at that time. We also purchase products and services as needed with no firm commitment.

Item 9. Changes in and Disagreements with Accountants on Accounting and Financial Disclosure

None.

Item 9A. Controls and Procedures

Evaluation of Disclosure Controls and Procedures

Our Chief Executive Officer and Chief Financial Officer have evaluated the effectiveness of the registrant's disclosure controls and procedures (as defined in Rules 13a-15(e) and 15d-15(e) of the Securities Exchange Act of 1934), as of the end of the period covered by this Annual Report, that ensure that information relating to the registrant which is required to be disclosed in this report is recorded, processed, summarized and reported within required time periods using the criteria for effective internal control established in Internal Control—Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission in 2013. Based on this evaluation, our Chief Executive Officer and Chief Financial Officer have concluded that the registrant's disclosure controls and procedures were effective as of December 31, 2022.

Changes in Internal Control over Financial Reporting

In the ordinary course of business, we routinely review our system of internal control over financial reporting and make changes to our systems and processes that are intended to ensure an effective internal control environment. In the third quarter of 2020, we began a multi-year implementation of a new global enterprise resource planning (ERP) system, which will replace many of our existing core financial systems. The new ERP system is designed to enhance the flow of financial information, facilitate data analysis and accelerate information reporting. The implementation is expected to occur in phases over the next several years.

As the phased implementation of the new ERP system continues, we could have changes to our processes and procedures which, in turn, could result in changes to our internal controls over financial reporting. As such changes occur, we will evaluate quarterly whether such changes materially affect our internal control over financial reporting.

There were no changes in Verizon's internal control over financial reporting during the fourth quarter 2022 that have materially affected, or are reasonably likely to materially affect, our internal control over financial reporting.

Management's Annual Report on Internal Control over Financial Reporting

The management of Verizon Communications Inc. is responsible for establishing and maintaining adequate internal control over financial reporting of Verizon. Management has evaluated internal control over financial reporting of Verizon using the criteria for effective internal control established in Internal Control—Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission in 2013.

Management has assessed the effectiveness of Verizon's internal control over financial reporting as of December 31, 2022. Based on this assessment, management believes that the internal control over financial reporting of Verizon is effective as of December 31, 2022. In connection with this assessment, there were no material weaknesses in Verizon's internal control over financial reporting identified by management. The Company's independent registered public accounting firm, Ernst & Young LLP, has provided an attestation report on Verizon's internal control over financial reporting and is included in Item 8 of this Annual Report.

Item 9B. Other Information

None.

Item 9C. Disclosure Regarding Foreign Jurisdictions that Prevent Inspections

Not applicable.

PART III

Item 10. Directors, Executive Officers and Corporate Governance

Set forth below is information with respect to our current executive officers.

Name	Age	Office	Held Since
Hans Vestberg	57	Chairman and Chief Executive Officer	2019
Matthew D. Ellis	51	Executive Vice President and Chief Financial Officer	2016
Samantha Hammock	44	Executive Vice President and Chief Human Resources Officer	2021
Kyle Malady	55	Executive Vice President and President - Global Networks & Technology	2022
Rima Qureshi	58	Executive Vice President and Chief Strategy Officer	2017
Sowmyanarayan Sampath	46	Executive Vice President and Group CEO - Verizon Business	2022
Craig L. Silliman	55	Executive Vice President and President - Verizon Global Services	2023
Anthony T. Skiadas	54	Senior Vice President and Controller	2013
Vandana Venkatesh	51	Executive Vice President and General Counsel	2022

Each of the above officers has held the indicated office or other high-level managerial positions with the Company or one of its subsidiaries for at least five years, with the exception of Samantha Hammock, who has been with the Company since 2020. Officers are not elected for a fixed term of office and may be removed from office at any time at the discretion of the Board of Directors.

Samantha Hammock is the Executive Vice President and Chief Human Resources Officer of the Company. Ms. Hammock joined the Company in December 2020 as Senior Vice President of Global Talent and began serving in her current role in December 2021. Prior to joining Verizon, Ms. Hammock spent 14 years at the American Express Company, a globally integrated payments company and provider of credit and charge cards to consumers and businesses around the world, where she served as Head of Talent and Learning from April 2020 to December 2020, Chief Learning Officer from 2017 to April 2020, and Vice President, Leadership Strategy, from 2016 to April 2020.

For other information required by this item, see the sections entitled "Governance — Item 1: Election of Directors — Nominees for election and — Election process, — Our governance framework — Where to find more information, — Board committees — Audit Committee and — Other risk-related matters — Business conduct and ethics" in our definitive Proxy Statement to be filed with the Securities and Exchange Commission and delivered to shareholders in connection with our 2023 Annual Meeting of Shareholders, which are incorporated herein by reference.

Item 11. Executive Compensation

For information with respect to executive compensation, see the sections entitled "Governance — Non-employee Director compensation" and "Executive Compensation — Compensation discussion and analysis, — Compensation Committee Report and — Compensation tables" in our definitive Proxy Statement to be filed with the Securities and Exchange Commission and delivered to shareholders in connection with our 2023 Annual Meeting of Shareholders, which are incorporated by reference herein. There were no relationships to be disclosed under paragraph (e)(4) of Item 407 of Regulation S-K.

Item 12. Security Ownership of Certain Beneficial Owners and Management and Related Stockholder Matters

For information with respect to the security ownership of certain beneficial owners, the directors and executive officers, see the section entitled "Stock ownership — Security ownership of certain beneficial owners and management" in our definitive Proxy Statement to be filed with the Securities and Exchange Commission and delivered to shareholders in connection with our 2023 Annual Meeting of Shareholders, which is incorporated herein by reference.

The following table provides information as of December 31, 2022 for (i) all equity compensation plans previously approved by the Company's shareholders, and (ii) all equity compensation plans not previously approved by the Company's shareholders. Since May 4, 2017, the Company has only issued awards under the 2017 Verizon Communications, Inc. Long-Term Incentive Plan (2017 LTIP), which provides for awards of stock options, restricted stock, restricted stock units, performance stock units and other equity-based hypothetical stock units to employees of Verizon. No new awards are permitted to be issued under any other equity compensation plan. In accordance with SEC rules, the table does not include outstanding awards that are payable solely in cash by the terms of the award, and such awards do not reduce the number of shares remaining for issuance under the 2017 LTIP.

Plan category	Number of securities to be issued upon exercise of outstanding options, warrants and rights (a)	Weighted-average exercise price of outstanding options, warrants and rights (b)	Number of securities remaining available for future issuance under equity compensation plans (excluding securities reflected in column (a)) (c)
Equity compensation plans approved by security holders	21,078,199 ⁽¹⁾	\$ — ⁽²⁾	68,502,052 ⁽³⁾
Equity compensation plans not approved by security holders	92,978 ⁽⁴⁾	—	—
Total	21,171,177	\$ —	68,502,052

⁽¹⁾ This amount includes: 21,078,199 shares of common stock subject to outstanding restricted stock units and performance stock units, including dividend equivalents accrued on such awards through December 31, 2022. This does not include performance stock units, deferred stock units and deferred share equivalents payable solely in cash.

⁽²⁾ The Company's outstanding restricted stock units, performance stock units and deferred stock units do not have exercise prices associated with the settlement of these awards.

⁽³⁾ This number reflects the number of shares of common stock that remained available for future issuance under the 2017 LTIP.

⁽⁴⁾ This number reflects shares subject to deferred stock units credited to the Verizon Income Deferral Plan, which were awarded in 2002 under the Verizon Communications Broad-Based Incentive Plan. No new awards are permitted to be issued under this plan.

Item 13. Certain Relationships and Related Transactions, and Director Independence

For information with respect to certain relationships and related transactions and director independence, see the sections entitled "Governance — Our governance framework — Other risk-related matters — Related person transactions and — Item 1: Election of Directors — Our Board's independence" in our definitive Proxy Statement to be filed with the Securities and Exchange Commission and delivered to shareholders in connection with our 2023 Annual Meeting of Shareholders, which are incorporated herein by reference.

Item 14. Principal Accounting Fees and Services

Our independent registered public accounting firm is Ernst & Young LLP, New York, NY, Auditor Firm ID:42.

For information with respect to principal accounting fees and services, see the section entitled "Audit matters — Item 4: Ratification of appointment of independent registered public accounting firm" in our definitive Proxy Statement to be filed with the Securities and Exchange Commission and delivered to shareholders in connection with our 2023 Annual Meeting of Shareholders, which is incorporated herein by reference.

Item 15. Exhibits and Financial Statement Schedules

(a) Documents filed as part of this report:

	Page
(1) Financial Statements	
Report of Independent Registered Public Accounting Firm on Internal Control Over Financial Reporting	51
Report of Independent Registered Public Accounting Firm on Financial Statements	52
Financial Statements covered by Report of Independent Registered Public Accounting Firm:	
Consolidated Statements of Income	54
Consolidated Statements of Comprehensive Income	55
Consolidated Balance Sheets	56
Consolidated Statements of Cash Flows	57
Consolidated Statements of Changes in Equity	58
Notes to Consolidated Financial Statements	59
(2) Financial Statement Schedule	
II – Valuation and Qualifying Accounts	113
(3) Exhibits	

Exhibits identified in parentheses below, on file with the SEC, are incorporated herein by reference as exhibits hereto. Unless otherwise indicated, all exhibits so incorporated are from File No. 1-8606.

Pursuant to Regulation S-K, Item 601(b)(4)(iii)(A), certain instruments which define the rights of holders of long-term debt of Verizon and its consolidated subsidiaries are not filed herewith, and Verizon hereby agrees to furnish a copy of any such instrument to the SEC upon request.

Exhibit Number	Description
3a	Restated Certificate of Incorporation of Verizon Communications Inc. (filed as Exhibit 3a to Form 10-Q for the period ended June 30, 2014 and incorporated herein by reference).
3b	Bylaws of Verizon Communications Inc., as amended and restated, effective as of September 30, 2022 (filed as Exhibit 3b to Form 8-K filed on September 30, 2022 and incorporated herein by reference).
4a	Indenture between Verizon Communications Inc., both individually and as successor in interest to Verizon Global Funding Corp., and U.S. Bank National Association, as successor trustee to Wachovia Bank, National Association, formerly known as First Union National Bank, as Trustee, dated as of December 1, 2000 (filed as Exhibit 4.1 to Verizon Global Funding Corp.'s Registration Statement on Form S-4, Registration No. 333-64792, and incorporated herein by reference).
4b	First Supplemental Indenture between Verizon Communications Inc., both individually and as successor in interest to Verizon Global Funding Corp., and U.S. Bank National Association, as successor trustee to Wachovia Bank, National Association, formerly known as First Union National Bank, as Trustee, dated as of May 15, 2001 (filed as Exhibit 4.2 to Verizon Global Funding Corp.'s Registration Statement on Form S-3, Registration No. 333-67412, and incorporated herein by reference).
4c	Second Supplemental Indenture between Verizon Communications Inc., both individually and as successor in interest to Verizon Global Funding Corp., and U.S. Bank National Association, as successor trustee to Wachovia Bank, National Association, formerly known as First Union National Bank, as Trustee, dated as of September 29, 2004 (filed as Exhibit 4.1 to Form 8-K filed on February 9, 2006, and incorporated herein by reference).
4d	Third Supplemental Indenture between Verizon Communications Inc., both individually and as successor in interest to Verizon Global Funding Corp., and U.S. Bank National Association, as successor trustee to Wachovia Bank, National Association, formerly known as First Union National Bank, as Trustee, dated as of February 1, 2006 (filed as Exhibit 4.2 to Form 8-K filed on February 9, 2006, and incorporated herein by reference).
4e	Fourth Supplemental Indenture between Verizon Communications Inc., both individually and as successor in interest to Verizon Global Funding Corp., and U.S. Bank National Association, as successor trustee to Wachovia Bank, National Association, formerly known as First Union National Bank, as Trustee, dated as of April 4, 2016 (filed as Exhibit 4.5 to Verizon Communications Inc.'s Registration Statement on Form S-4, Registration No. 333-212307, and incorporated herein by reference).
4f	Fifth Supplemental Indenture between Verizon Communications Inc., both individually and as successor in interest to Verizon Global Funding Corp., and U.S. Bank National Association, as successor trustee to Wachovia Bank, National Association, formerly known as First Union National Bank, as Trustee, dated as of May 15, 2020 (filed as Exhibit 4.1 to Form 8-K filed on May 15, 2020, and incorporated herein by reference).
	Pursuant to Regulation S-K, Item 601(b)(4)(iii)(A), certain instruments which define the rights of holders of long-term debt of Verizon Communications Inc. and its consolidated subsidiaries are not filed herewith, and the Company hereby agrees to furnish a copy of any such instrument to the SEC upon request.
4g	Description of Verizon's Securities Registered Pursuant to Section 12 of the Securities and Exchange Act of 1934, filed herewith.
10a	2017 Verizon Communications Inc. Long-Term Incentive Plan (incorporated by reference to Appendix B of the Registrant's Proxy Statement included in Schedule 14A filed on March 20, 2017). ^{**}
10a(i)	2018 Special Performance Stock Unit Agreement pursuant to the 2017 Verizon Communications Inc. Long-Term Incentive Plan for H. Vestberg (filed as Exhibit 10 to Form 10-Q for the period ended September 30, 2018 and incorporated herein by reference). ^{**}
10a(ii)	Form of 2020 Performance Stock Unit Agreement pursuant to the 2017 Verizon Communications Inc. Long-Term Incentive Plan (filed as Exhibit 10a to Form 10-Q for the period ended March 31, 2020 and incorporated herein by reference). ^{**}
10a(iii)	Form of 2020 Restricted Stock Unit Agreement pursuant to the 2017 Verizon Communications Inc. Long-Term Incentive Plan (filed as Exhibit 10b to Form 10-Q for the period ended March 31, 2020 and incorporated herein by reference). ^{**}
10a(iv)	Form of 2021 Performance Stock Unit Agreement pursuant to the 2017 Verizon Communications Inc. Long-Term Incentive Plan (filed as Exhibit 10a to Form 10-Q for the period ended March 31, 2021 and incorporated herein by reference). ^{**}
10a(v)	Form of 2021 Restricted Stock Unit Agreement pursuant to the 2017 Verizon Communications Inc. Long-Term Incentive Plan (filed as Exhibit 10b to Form 10-Q for the period ended March 31, 2021 and incorporated herein by reference). ^{**}
10a(vi)	Form of 2022 Performance Stock Unit Agreement pursuant to the 2017 Verizon Communications Inc. Long-Term Incentive Plan (filed as Exhibit 10a to Form 10-Q for the period ended March 31, 2022 and incorporated herein by reference). ^{**}

	<u>10a(vii)</u>	Form of 2022 Restricted Stock Unit Agreement pursuant to the 2017 Verizon Communications Inc. Long-Term Incentive Plan (filed as Exhibit 10b to Form 10-Q for the period ended March 31, 2022 and incorporated herein by reference).**
<u>10b</u>		Verizon Communications Inc. Short-Term Incentive Plan (filed as Exhibit 10a to Form 10-Q for the period ended March 31, 2019 and incorporated herein by reference).**
<u>10c</u>		Verizon Executive Deferral Plan (filed as Exhibit 10e to Form 10-K for the period ended December 31, 2017 and incorporated herein by reference).**
<u>10d</u>		Verizon Communications Inc. Income Deferral Plan (filed as Exhibit 10f to Form 10-Q for the period ended June 30, 2002 and incorporated herein by reference).**
	<u>10d(i)</u>	Description of Amendment to Verizon Communications Inc. Income Deferral Plan (filed as Exhibit 10o(i) to Form 10-K for the year ended December 31, 2004 and incorporated herein by reference).**
<u>10e</u>		Verizon Excess Pension Plan (filed as Exhibit 10p to Form 10-K for the year ended December 31, 2004 and incorporated herein by reference).**
	<u>10e(i)</u>	First Amendment to Verizon Excess Pension Plan (filed as Exhibit 10p(i) to Form 10-K for the year ended December 31, 2004 and incorporated herein by reference).**
<u>10f</u>		Bell Atlantic Senior Management Long-Term Disability and Survivor Protection Plan, as amended (filed as Exhibit 10h to Form SE filed on March 27, 1986 and Exhibit 10b(ii) to Form 10-K for the year ended December 31, 1997 and incorporated herein by reference).**
<u>10g</u>		Verizon Executive Life Insurance Plan, As Amended and Restated September 2009 (filed as Exhibit 10s to Form 10-K for the year ended December 31, 2010 and incorporated herein by reference).**
<u>10h</u>		Form of Aircraft Time Sharing Agreement (filed as Exhibit 10i to Form 10-K for the year ended December 31, 2020 and incorporated herein by reference).**
<u>10i</u>		Verizon Senior Manager Severance Plan (filed as Exhibit 10d to Form 10-Q for the period ended March 31, 2010 and incorporated herein by reference).**
<u>21</u>		List of principal subsidiaries of Verizon Communications Inc., filed herewith.
<u>23</u>		Consent of Ernst & Young LLP, filed herewith.
<u>24</u>		Powers of Attorney, filed herewith.
<u>31.1</u>		Certification of Chief Executive Officer pursuant to Section 302 of the Sarbanes-Oxley Act of 2002, filed herewith.
<u>31.2</u>		Certification of Chief Financial Officer pursuant to Section 302 of the Sarbanes-Oxley Act of 2002, filed herewith.
<u>32.1</u>		Certification of Chief Executive Officer pursuant to Section 906 of the Sarbanes-Oxley Act of 2002, filed herewith.
<u>32.2</u>		Certification of Chief Financial Officer pursuant to Section 906 of the Sarbanes-Oxley Act of 2002, filed herewith.
101.INS		XBRL Instance Document - the instance document does not appear in the interactive data file because its XBRL tags are embedded within the inline XBRL document.
101.SCH		XBRL Taxonomy Extension Schema Document.
101.PRE		XBRL Taxonomy Presentation Linkbase Document.
101.CAL		XBRL Taxonomy Calculation Linkbase Document.
101.LAB		XBRL Taxonomy Label Linkbase Document.
101.DEF		XBRL Taxonomy Extension Definition Linkbase Document.
104		Cover Page Interactive Data File (formatted as inline XBRL with applicable taxonomy extension information contained in Exhibits 101).
**		Indicates management contract or compensatory plan or arrangement.

Schedule II - Valuation and Qualifying Accounts

Verizon Communications Inc. and Subsidiaries

For the Years Ended December 31, 2022, 2021 and 2020

(dollars in millions)

(dollars in millions)							
Description	Balance at Beginning of Period	Additions			Deductions ^(b)	Balance at End of Period ^(c)	
		Charged to Expenses		Charged to Other Accounts ^(a)			
Allowance for credit losses deducted from accounts receivable:							
Year 2022	\$ 1,151	\$ 1,531	\$	69	\$ 1,490	\$ 1,261	
Year 2021	1,507	743		139	1,238	1,151	
Year 2020	1,125 ^(d)	1,390		165	1,173	1,507	

Description	Balance at Beginning of Period	Additions				Deductions ^(f)	Balance at End of Period
		Charged to Expenses	Other	Charged to Accounts ^(e)			
Valuation allowance for deferred tax assets:							
Year 2022	\$ 1,574	\$ 41	\$ —	\$ —	\$ 268	\$ 1,347	
Year 2021	2,183	339	—	—	948	1,574	
Year 2020	2,260	202	84	—	363	2,183	

^(a) Charged to Other Accounts primarily includes amounts previously written off which were credited directly to this account when recovered.

^(b) Deductions primarily include amounts written off as uncollectible or transferred to other accounts or utilized.

^(c) Allowance for credit losses includes approximately \$436 million, \$255 million, and \$254 million at December 31, 2022, 2021, and 2020, respectively, related to long-term device payment receivables.

^(d) Includes opening balance sheet adjustment related to the adoption of Topic 326.

^(e) Charged to Other Accounts includes current year increase to valuation allowance charged to equity and reclassifications from other balance sheet accounts.

^(f) Reductions to valuation allowances related to deferred tax assets.

Item 16. Form 10-K Summary

None.

Signatures

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

VERIZON COMMUNICATIONS INC.

By: /s/ Anthony T. Skiadas Date: February 10, 2023
Anthony T. Skiadas
Senior Vice President and Controller

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacities and on the dates indicated.

Principal Executive Officer:

/s/ Hans E. Vestberg February 10, 2023
Hans E. Vestberg
Chairman and Chief Executive Officer

Principal Financial Officer:

/s/ Matthew D. Ellis February 10, 2023
Matthew D. Ellis
Executive Vice President and Chief Financial Officer

Principal Accounting Officer:

/s/ Anthony T. Skiadas February 10, 2023
Anthony T. Skiadas
Senior Vice President and Controller

	*	Director	February 10, 2023
Hans E. Vestberg			
	*	Director	February 10, 2023
Shellye L. Archambeau			
	*	Director	February 10, 2023
Roxanne S. Austin			
	*	Director	February 10, 2023
Mark T. Bertolini			
	*	Director	February 10, 2023
Vittorio Colao			
	*	Director	February 10, 2023
Melanie L. Healey			
	*	Director	February 10, 2023
Laxman Narasimhan			
	*	Director	February 10, 2023
Clarence Otis, Jr.			
	*	Director	February 10, 2023
Daniel H. Schulman			
	*	Director	February 10, 2023
Rodney E. Slater			
	*	Director	February 10, 2023
Carol B. Tomé			
	*	Director	February 10, 2023
Gregory G. Weaver			
* By: /s/ Anthony T. Skiadas			
Anthony T. Skiadas			
(as attorney-in-fact)			